

PSoC™ Automotive Multitouch touchscreen controller

Module design best practices

About this document

Scope and purpose

This document describes Infineon's hardware design recommendations when using the PSoC™ Automotive Multitouch touchscreen controllers. Guidelines are detailed for all aspects of the touchscreen design: stack-up, electrode pattern, trace routing, shielding, and buttons. Guidelines are also given for the FPC and PCB. Requirements for hover and multichip designs are also covered.

Intended audience

This document is intended for Infineon customers, touchscreen vendors, panel manufacturers, FPC/PCB manufacturers, and touchscreen module designers. This document assumes that the reader is familiar with the features and operation of the Infineon touch controller used in the design as detailed in its Technical Reference Manual.

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1 Introduction

The objective of this document is to provide design guidelines for building touchscreen modules using Infineon touch controllers. A touchscreen module consists of the following parts:

- A touch sensor made of either glass or film with indium tin oxide (ITO), metal mesh, or other transparent conductive sensor electrodes.
- A cover lens.
- Metal or ITO traces that route all the sensor electrodes to the flexible printed circuit (FPC) connector.
- The FPC.
- (Optional) A PCB if the controller is not mounted on the FPC.
- (Optional) A display mounted underneath the panel.
- An Infineon touchscreen controller.

Some CYAT817x devices support force sensing, in which the device can detect and measure the amount of force that is being applied to the touchscreen by the user's finger. The best practices for force sensing are very dependent on the overall electrical and mechanical design of the system, and it is not practical to provide a set of best practices guidelines. Therefore, force sensing is not covered in this document. For designs requiring force sensing, you should work directly with Infineon.

1.1 Acronyms

Table 1 Acronyms

Acronym	Description	Definition
ACF	anisotropic conductive film	A Z-axis conductive material used to bond FPC pads onto exposed ITO pads on a panel.
Cm	mutual capacitance	Mutual capacitance between a transmitter and receiver electrode.
COB	chip on board	A circuit design with the touchscreen controller mounted on the main system board instead of on the FPC connected to the touchscreen.
Cp	parasitic capacitance	Self/parasitic capacitance between an electrode and ground.
CYATx	Cypress, an Infineon company, automotive	Prefix (and shorthand) for generation 6 and 7 touch controllers (also see TMAx).
dCm	delta Cm	Change in mutual capacitance due to a finger touch.
dCp	delta Cp	Change in self capacitance due to a finger touch.
DSD	dual solid diamond	A single-layer (with bridges) ITO pattern with four diamonds per unit cell.
EMC	electromagnetic compatibility	
EMI	electromagnetic interference	
EPAD	exposed pad	An electrical contact (usually V _{SS}) on a silicon chip package.
ESD	electrostatic discharge	

Introduction

Acronym	Description	Definition
EUT	equipment under test	
FPC	flexible printed circuit	
ITO	indium tin oxide	A transparent, conductive ceramic used in touchscreens.
HBM	human body model	ESD model emulating a discharge of a human through their skin.
HMM	human metal model	ESD model emulating a discharge of a human through a small piece of metal (see IEC 61000-4-2).
OCA	optically clear adhesive	
PCB	printed circuit board	
PET	polyethylene terephthalate	A transparent plastic, often used as a touch sensor substrate. May be described as “film”.
PMMA	polymethyl methacrylate	A transparent plastic, often used as a touch sensor cover lens.
SLIM	single layer independent multi-touch	A single-layer ITO pattern that supports multi-touch.
SNR	signal-to-noise ratio	A standard metric for signal quality in a system.
SOL	sensor-on-lens	A touch panel stack-up construction that reduced thickness and cost.
SSD	single solid diamond	A signal-layer (with bridges) ITO pattern.
TMAx	touch multitouch all-point	Prefix (and shorthand) for generation 3, 4, and 5 touch controllers (also see CYATx).
TTHE	touch tuning host emulator	Software tool used to tune and debug touch solutions.
TVS	transient voltage suppressor	A semiconductor device used to protect circuits from external voltage transients.

2 Touch panels

Touchscreen systems are made of a transparent conducting material, generally indium tin oxide (ITO), which is patterned onto plastic film (PET) or glass substrates. These sensors are connected to the sensing circuitry in a Infineon touch controller. A human touch in the vicinity of a sensor electrode changes its projected capacitance. The touch controller measures the projected capacitance of the sensor array and analyzes the changes in capacitance to resolve finger positions and gestures.

This section describes the touch panel portion of the system. The touch panel is the assembly that senses the touch. It consists of several components:

- A stack-up of materials, including an array of touch-sensitive electrodes, a cover lens or glass, various protective films, polarizers, etc.
- An array of ITO or metal mesh electrodes, arranged in a pattern of repeating unit cells. The electrode array is deposited onto glass or film substrate which is part of the stack-up.
- Routing traces from the electrodes to bonding pads in an area where a separate flexible printed circuit (FPC) is bonded.
- Some designs may include capacitive buttons as separate patterns on the panel.
- Some designs may include slider structures.
- Some designs may include a shield layer to prevent display noise from coupling into the electrodes and degrading performance.
- Some designs will laminate the stack-up directly to a display. In other designs, there is an air gap between the display and the rest of the stack-up.
- Design elements to help protect the system against electrostatic discharge (ESD).

This section provides design guidance for each component in the touch panel: the stack-up, pattern, routing, capacitive buttons, shield, display, and ESD protection.

Infineon provides comprehensive sensor-design services to all qualified PSoC™ Automotive Multitouch customers and ITO partners. This service ensures that each touch application has an optimized touch panel for the selected touch controller, panel process technology, and dimensional constraints of the end product.

2.1 Key parameters

This section introduces the key electrical parameters for the overall system:

- **Mutual Capacitance (C_M)** - This is the capacitance between a transmit pin and receive pin of the touch controller. When a finger touches the panel, C_M changes, and this change is what is used to determine the location of the finger.

In a well-designed system, the mutual capacitance is determined by the panel pattern. The PCB, FPC, and panel routing (outside the active area of the panel) should have only a minimal impact on C_M , and many of the best practices described in this document are designed to minimize these effects.

For the panel to work properly, there must be a sufficient change in C_M when a finger (or glove) touches the panel. The change in C_M is usually determined by simulation, and Infineon strongly recommends panel simulation to validate the suitability of the panel.

- **Parasitic Capacitance (C_P)** - This is the “parasitic” capacitance from a TX or RX electrode to ground, that is, the capacitance without a finger touch. When a finger or glove touches the panel (or hovers above the panel), C_P changes. C_P is used as part of the detection algorithm.

C_P must be small and balanced between electrodes for the system to work well. This document gives guidance for the panel, FPC, and PCB to help ensure successful operation.

- **Resistance** - The ITO used in touch panels is inherently resistive. When combined with C_P , the resistance creates a low-pass filter. For proper operation, the RC time constant of the panel needs to be sufficiently small, so that the TX electrodes can be driven at sufficiently high frequencies. The slower the panel is scanned, the smaller the number of samples in the allotted timeframe, reducing SNR. Also, noise from the display may interfere with the scans at lower frequencies, further reducing SNR.

2.2 Stack-up

The stack-up is a critical design element for the system. The arrangement, thickness, and material selection of the stack-up determines the amount of change in capacitance (dCm, change in mutual capacitance) as a result of the touch. The stack-up design determines how small a finger can be detected, the thickness of glove that can be detected, how much noise the system can tolerate, etc.

Infineon strongly recommends close cooperation with customers during the stack-up definition phase. Infineon provides simulation services that will give a good prediction of how well the system will work, what features can be supported, etc.

2.2.1 Cover lens

The cover lens can be characterized in terms of material and thickness. There are two prevailing material types: glass and plastic. Cost generally determines the type of cover lens. The thickness of the cover lens (plus any coatings and bonding layers) critically affects the touchscreen performance. [Table 8](#) provides constraints on cover lens thickness associated with each sensor pattern.

2.2.2 Bezel

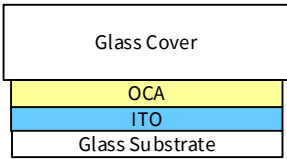
The bezel of a touch panel is the opaque area around the perimeter of the display. This bezel area is used to mask the traces between the touchscreen controller and the row/column electrodes. A non-transparent color is typically printed on the second surface (bottom-side) of the cover lens to hide the traces. However, the materials used to make different colors can create different electrical performance. For example, carbon black and chrome are conductive and interact with the e-field of projected capacitance sensing. Therefore, if a product with a touchscreen is offered with multiple bezel colors, then all supported colors must be tested across the temperature range supported by the product.

For electrostatic discharge (ESD) protection, it is desirable to use a conductive bezel and connect it to system ground, provided it is not close enough to the routing traces to interfere with performance.

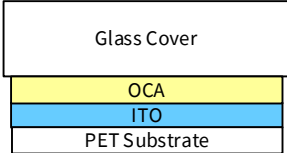
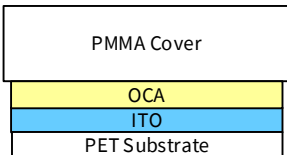
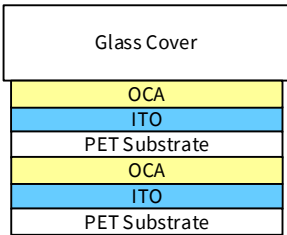
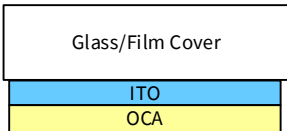
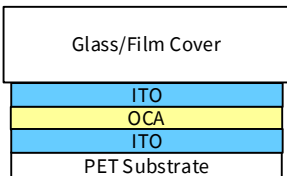
2.2.3 Common industry stack-ups

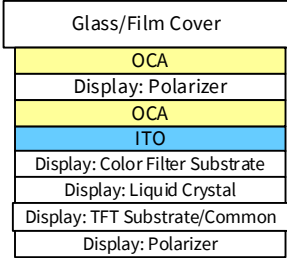
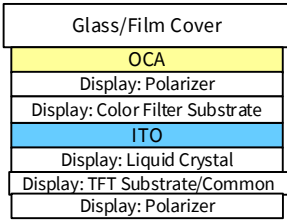
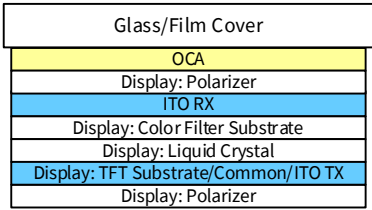
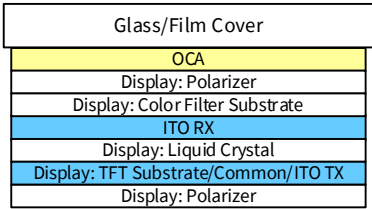
[Table 2](#) shows common industry stack-ups that are used in the capacitive touchscreen market. They are not drawn to scale. The cover lens is the top-side layer of the stack-up that provides most of the mechanical strength. Unless there are additional surface coatings present (such as hard coat, anti-reflection, anti-smudge, and hydrophobic), the top side of the cover lens is what a user's finger physically touches to operate the touchscreen. "OCA" is an optically clear adhesive layer that attaches one layer to another.

Table 2 Industry stack-ups

Stack-up	Comments
Glass/Glass (GG) 	Conventional stack-up used for diamond patterns "Glass Cover" can be other material such as PMMA.

Touch panels

Stack-up	Comments
Glass/Film (GF) 	
Film/Film (FF) 	
Glass/Film/Film (GFF) 	Traditionally used for MH3 patterns Configurations can be further described as “Face Up” or “Face Down” in reference to ITO being positioned on the top side (face up) of the PET Film or bottom-side (face down) of the PET Film. “Glass Cover” can be other material such as PMMA.
Sensor on Lens (SOL) 	RX electrodes typically on cover lens Also used for in-cell processes.
Hybrid SOL (G1F) 	RX electrodes typically on cover lens TX electrodes on PET Film

Stack-up	Comments
Sensor On-Cell (SOC)  The diagram shows a stack-up for Sensor On-Cell (SOC). From top to bottom: Glass/Film Cover, OCA (yellow), Display: Polarizer, OCA (yellow), ITO (blue), Display: Color Filter Substrate, Display: Liquid Crystal, Display: TFT Substrate/Common, and Display: Polarizer.	
1-Layer Sensor In-Cell (SIC)  The diagram shows a stack-up for 1-Layer Sensor In-Cell (SIC). From top to bottom: Glass/Film Cover, OCA (yellow), Display: Polarizer, Display: Color Filter Substrate, ITO (blue), Display: Liquid Crystal, Display: TFT Substrate/Common, and Display: Polarizer.	
Hybrid Sensor In-Cell (SIC)  The diagram shows a stack-up for Hybrid Sensor In-Cell (SIC). From top to bottom: Glass/Film Cover, OCA (yellow), Display: Polarizer, ITO RX (blue), Display: Color Filter Substrate, Display: Liquid Crystal, Display: TFT Substrate/Common/ITO TX (blue), and Display: Polarizer.	
True Sensor In-Cell (SIC)  The diagram shows a stack-up for True Sensor In-Cell (SIC). From top to bottom: Glass/Film Cover, OCA (yellow), Display: Polarizer, Display: Color Filter Substrate, ITO RX (blue), Display: Liquid Crystal, Display: TFT Substrate/Common/ITO TX (blue), and Display: Polarizer.	

2.2.4 Air gap

Some designs will have a small air gap, typically less than 1 mm, between the panel stack-up and the display. Although not generally required, an air gap will generally give better performance due to less display noise being coupled into the system. Air gap recommendations are given in [Table 8](#).

If an air gap is included in the design, care should be taken to minimize panel bending when force is applied. Even minor panel bending can significantly distort the electric fields.

2.2.5 Glove touch guidelines

Successful glove touch detection requires a panel with sufficient sensitivity. In most cases, a panel that generates a mutual capacitance change (dCm) of 100 fF when touched with a 9 mm diameter finger will adequately detect a 5 mm thick gloved finger. For low-noise systems (such as those with AMOLED displays, air gaps between display and touch panel, and/or low-resistance ITO shields), good glove performance can be achieved with smaller dCm values - as low as 50 fF. For higher-noise systems, a dCm value of 100 fF may not be enough.

For new designs, the following steps should be taken to ensure good glove performance:

- Work with Infineon during the panel design, to simulate the dCm value of the panel.
- Use a low-noise display. See section [High, medium, and low-noise displays](#) for what constitutes a “low noise” display. If a low-noise display is not possible, use a shield layer. If a shield layer is not possible, use an air gap ≥ 0.2 mm.
- Provide access to the horizontal and vertical synch signals of the display. These signals can be used by the touch controller to synchronize the scanning when the display is generating less noise.
- Do not use split routing for the panel design. Double route the longest sensors if possible.

2.2.6 Signal disparity guidelines

Signal Disparity (SD) is the ratio of touch signal between a grounded device and an ungrounded device. For example, if a grounded device provides a touch signal of 200, but only provides a touch signal of 100 when ungrounded, $SD = 200 / 100 = 2$. For most designs, an $SD \leq 1.5$ is acceptable.

Note that SD is not usually a concern in automotive designs, where a well designed module is strongly grounded to the chassis ground.

[Table 8](#) contains a recommended minimum cover lens thickness range to provide acceptable levels of SD.

In general, MH3 sensors have the least amount of SD, while SSD has the most SD. DSD has SD in between MH3 and SSD. Sensor design is covered in section [Electrode patterns](#).

2.2.7 Additional layers

Some panel designs include anti-reflective or anti-glare films. These films can be conductive, or partially conductive. If any ungrounded conductive film is applied to the panel, it should be designed such that the film does not extend to the panel edge. Otherwise, ESD events near the panel edge will be coupled to the film, and then to the sensor electrodes, degrading ESD performance. The film design should leave the largest possible gap that does not degrade the visual appearance of the display.

2.3 Electrode patterns

This section describes the common patterns used in the touch panel.

2.3.1 Diamond patterns

A summary of diamond pattern recommendations is given in [Table 9](#).

Diamond patterns are suitable for where a single-layer ITO process is required. Infineon supports two types of diamond patterns: single-solid diamond (SSD) and dual-solid diamond (DSD). [Figure 1](#) and [Figure 2](#) show graphical representations of these two patterns.

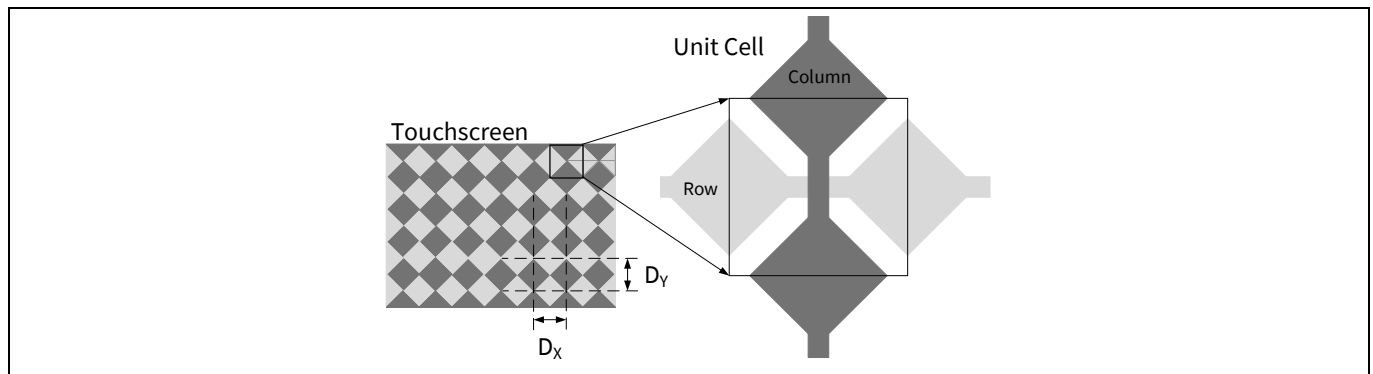


Figure 1 Single solid diamond pattern

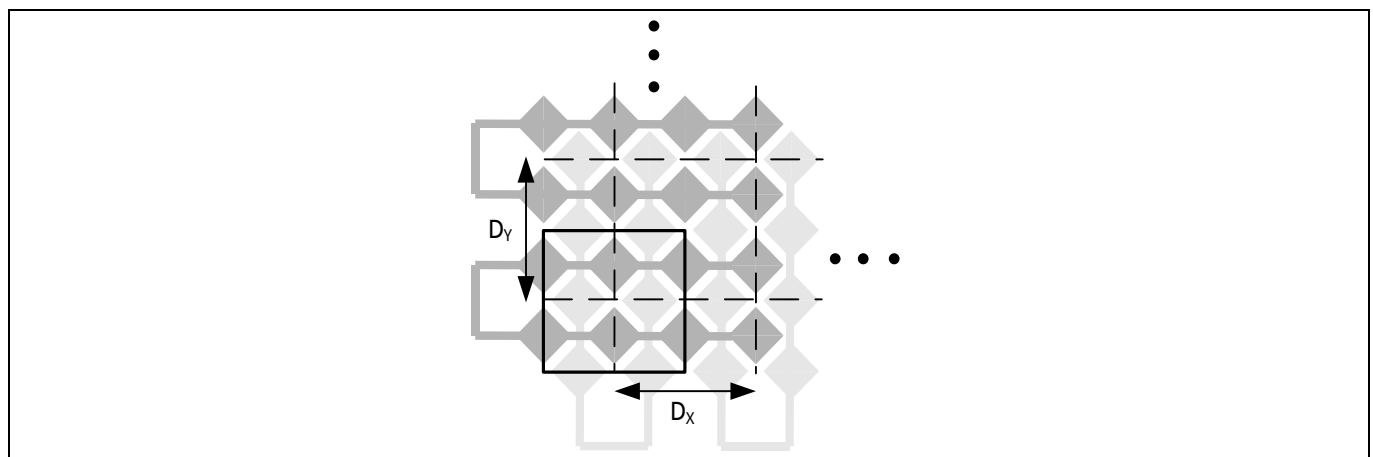


Figure 2 Dual solid diamond pattern

Sensors based on diamond patterns are described by three different dimensional parameters: electrode pitch (D), gap (G), and bridge width (W).

2.3.1.1 Electrode pitch (DX, DY)

Electrode pitch refers to the centerline spacing between adjacent rows or columns of electrodes, and that spacing can differ in the X and Y directions. The electrode pitch affects the accuracy of the reported touches on a touch panel; in general, the smaller the pitch, the higher the accuracy. A typical electrode pitch is 5 mm. However, the tradeoffs of a smaller pitch are higher I/O count and longer RC time constant, which for large panels may lead to a slower maximum scan rate.

For CYATx devices, the number of RX electrodes should not exceed the number of RX channels (see [Table 6](#)). For older TMAx devices, minimize the number of scans (see [Equation 3-3](#) and [Equation 3-4](#)).

For CYATx devices, the number of Tx electrodes is recommended to be a multiple of 7, 9, or 11 (or one Tx less). For older TMAx devices, the number of Tx electrodes should be a multiple of 4 (or one Tx less). This for the multi-TX scanning algorithm to achieve the best panel scan time.

2.3.1.2 Gap (G)

Gap refers to the distance between the edges of adjacent diamonds. The gap between the row and column electrodes helps to determine the optical quality of a touch panel. This sensor parameter is typically constrained by the process technology. For instance, for a glass substrate, the gap typically ranges between 30 μm and 100 μm ; while for a film technology the gap typically ranges between 100 μm and 300 μm .

2.3.1.3 Bridge width (W)

Bridge width refers to the width of the interconnecting bridges for row and column electrodes. This width can be different in the X and Y directions. Depending on the technology, the bridges connecting the rows and columns can be either ITO or metal. For ITO bridges, the bridge-width should be sized to meet the resistance and parasitic capacitance requirements. Metal bridges have lower resistance (including the contact resistances) than ITO bridges and should be used for electrodes that span the longer axis of the touch panel.

2.3.1.4 Guidelines for diamond patterns

Avoid large unconnected islands of ITO (known as “dummies”) that can be patterned within a unit cell to achieve optical uniformity. These dummies increase parasitic capacitance, which degrades touchscreen performance. If dummies are used, a good compromise is to split them into two or three slivers, as shown on the right side of [Figure 3](#).

Noise from the display couples into the sensor electrodes of a diamond pattern. Diamond patterns require an air gap or a shield layer unless used with a low-noise display, such as an active matrix organic LED (AMOLED).

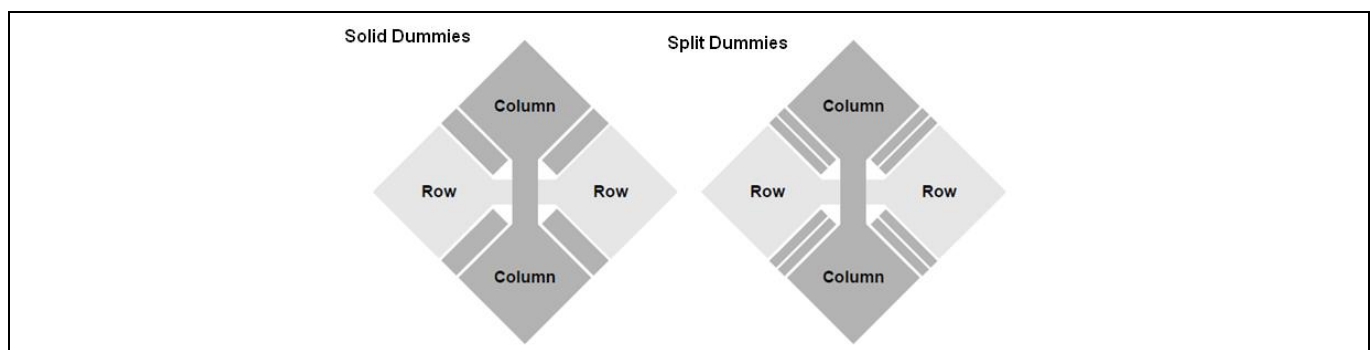


Figure 3 Diamonds with dummies

2.3.2 Manhattan patterns

A summary of Manhattan pattern recommendations is given in [Table 10](#).

Manhattan (MH) patterns are for designs with dual-layer PET, double-sided single glass, or single-sided dual-layer glass processes. The left side of [Figure 4](#) shows an example of a panel with a Manhattan pattern. The right side of the figure shows a unit cell for the pattern known as MH3, with a wide stripe on the bottom-side layer (TX electrode) and three narrow stripes on the top-side layer (RX electrode). In [Figure 4](#) the wide TX electrodes span the short axis of the panel while the RX electrodes span the long axis of the panel. Whether the TX electrodes span the short or long axis of the panel depends on multiple factors, including the number of available RX channels, balancing of RX vs. TX electrode resistances and capacitances, and system-level ESD handling. See section [Pin assignment](#) for more information. Each of the three stripes that form an RX electrode are connected at both ends to reduce their total resistance and so that they work as a single electrode.

Two advantages of Manhattan patterns are:

- Inherent shielding of display noise by the TX layer (Manhattan-patterned sensors may not require a separate shield layer or an air gap for noisy displays).
- Less susceptible to signal disparity.

The disadvantage of Manhattan patterns is that they require at least two ITO layers, which may increase cost.

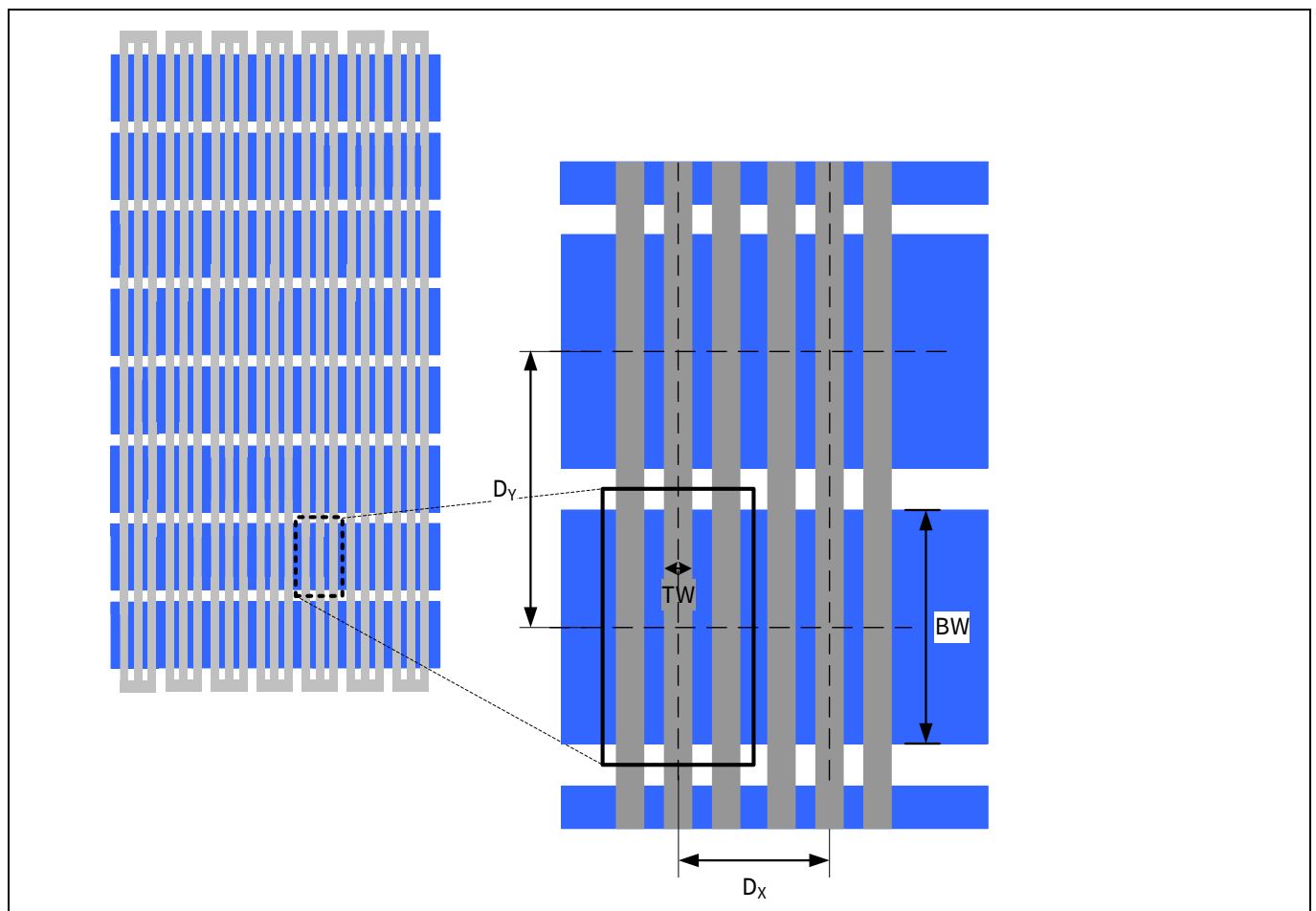


Figure 4 Manhattan pattern

Touch panels

Sensors based on Manhattan patterns are described by four different dimensional parameters: electrode pitch (D), bottom-side width (BW), number of stripes, and top-side stripe width (TW).

2.3.2.1 Electrode pitch (DX, DY)

Electrode pitch refers to the centerline spacing between adjacent rows or columns of electrodes, and that spacing can differ in the X and Y directions. The electrode pitch affects the accuracy of the reported touches on a touch panel; in general, the smaller the pitch, the higher the accuracy. A typical electrode pitch is 5 mm. However, the tradeoffs of a smaller pitch are higher I/O count and longer RC time constant, which for large panels may lead to a slower maximum scan rate.

For CYATx devices, the number of RX electrodes should not exceed the number of RX channels (see [Table 6](#)). For older TMAx devices, minimize the number of scans (see [Equation 3-3](#) and [Equation 3-4](#)).

For CYATx devices, the number of Tx electrodes is recommended to be a multiple of 7, 9, or 11 (or one Tx less). For older TMAx devices, the number of Tx electrodes should be a multiple of 4 (or one Tx less). This is for the multi-TX scanning algorithm to achieve the best panel scan time.

2.3.2.2 Bottom-side width (BW)

Bottom-side width refers to the width of the bottom-side layer TX electrode, as shown in blue in [Figure 4](#).

2.3.2.3 Number of stripes

Number of stripes refers to the stripes per RX electrode on the top-side ITO layer. Infineon recommends using three stripes, as shown in [Figure 4](#).

2.3.2.4 Top-side width (TW)

Top-side width refers to the width of each stripe in an RX electrode on the top-side layer.

2.3.2.5 Guidelines for Manhattan patterns

- Place TX electrodes as the bottom-side layer and RX electrodes as the top-side layer.
- Make the width of the bottom-side layer electrodes as wide as possible (within the process technology's tolerance).
- Connect all stripes within the same RX electrode together on both ends, even if the panel is single-routed.
- Route a ground line in the bezel area to prevent parasitic TX/RX coupling when a finger is placed near the edge of the panel. See [Figure 5](#). This may be either the system ground, or the VSS ground of the touch controller.

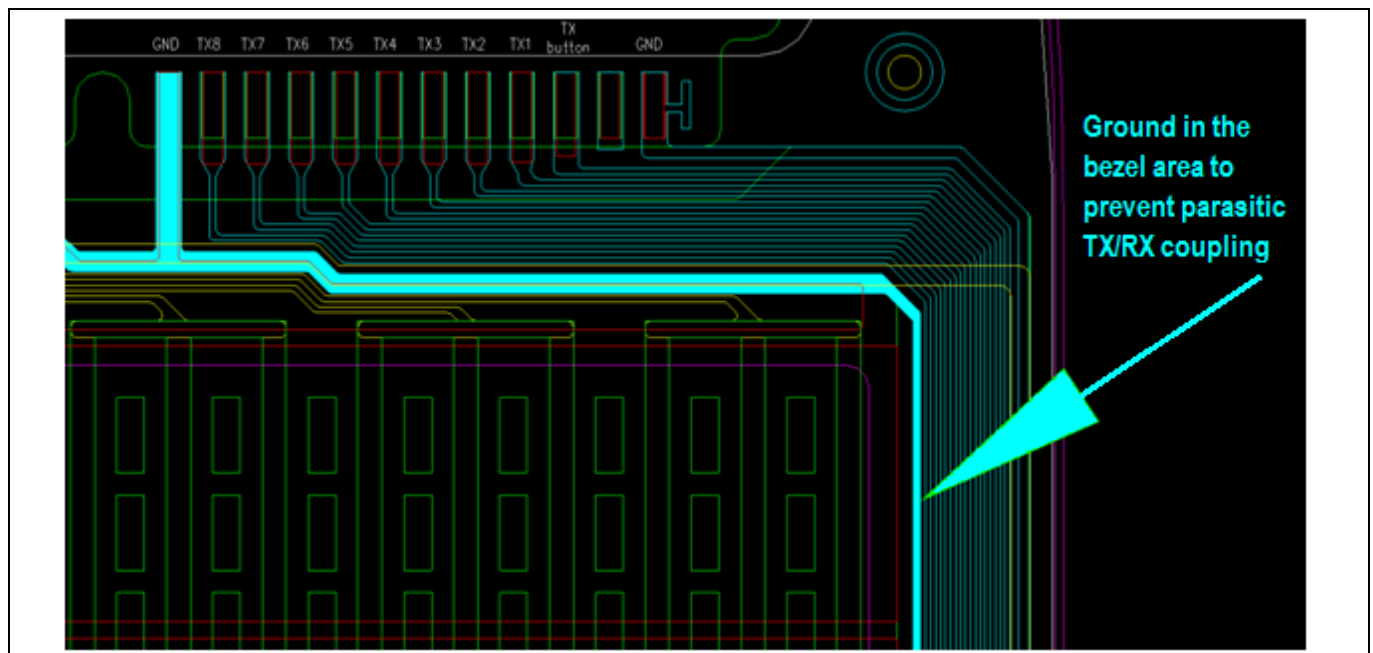


Figure 5 GND Trace protects against TX/RX Coupling during finger touch on edge

2.3.3 Single-layer independent multitouch (SLIM) patterns

A summary of SLIM pattern recommendations is given in [Table 11](#).

Single-Layer Independent Multitouch (SLIM) patterns are a Infineon proprietary technology. SLIM offers mutual-capacitance sensed multi-touch on a single ITO layer without metal bridges, which significantly reduces manufacturing cost. Infineon currently supports the SLIM-E pattern for SLIM designs. The “E” in the pattern name does not indicate a revision, but refers to the shapes of the electrodes in the sensor: they look similar to the letter “E”. See [Figure 6](#) for an example of what a SLIM-E sensor pattern looks like.

With SLIM patterns, some of the traces that connect the touchscreen controller to the sensor electrodes are made with ITO because that part of the routing is within the active area of the panel (see the vertical traces in [Figure 6](#) that are labeled “Routing Channel”). The traces to the electrodes are routed within the active area of the sensor from either one or two ends of the panel.

In this specification, a SLIM panel with routing channel traces that exit only one end of the panel is called “Single End” routing. Other names you may hear in the industry are “Zero Bezel” and “Bottom Bezel” routing. A SLIM panel with routing channel traces that exit two ends of the panel is called “Dual End” routing. Other names used in the industry are “Double” and “Top & Bottom” routing. [Figure 6](#) illustrates the difference between single-end and dual-end routing for the SLIM-E panels. Dual-end routing is normally a higher-cost option because it requires either metal traces routed around the panel through the bezel area, or a second FPC tail. Dual-end routing typically provides better touch accuracy due to a smaller routing channel width.

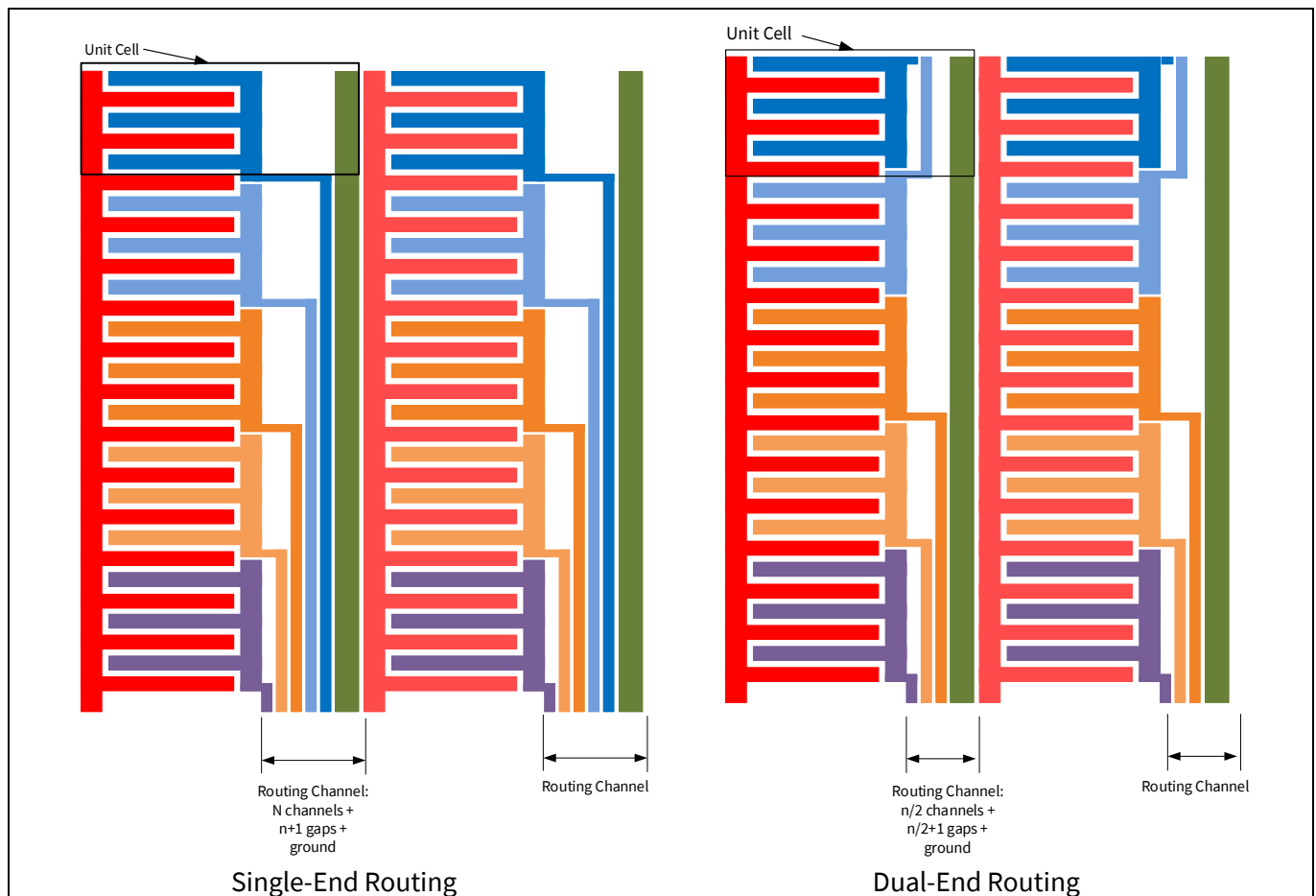


Figure 6 SLIM-E single - (bottom bezel) and dual-end (top/bottom bezel) routing, routing channel width

Touch panels

There are two electrode types in a SLIM panel:

- **S1** – A continuous electrode that spans an axis of the touch panel. This electrode can be assigned as RX or TX depending on the number of rows and columns of unit cells and the number of RX channels available in the touch controller (see section [Pin assignment](#)).
- **S2** – A discontinuous electrode that mechanically spans a unit cell. This electrode can be assigned as RX or TX depending on the number of rows and columns of unit cells and the number of RX channels available in the touch controller (see section [Pin assignment](#)). S2 electrodes are electrically connected to an edge of the panel through ITO traces in the active area, and connected to each other through metal traces in the border area and FPC tail. This creates electrodes that span the other axis of the panel.

The S1 electrodes in [Figure 6](#) are shown in various shades of red and form column electrodes in this panel. The S2 electrodes in [Figure 6](#) are shown in multiple colors and form the row electrodes in this panel. All S2 electrodes of the same color are electrically connected outside the active area or in the FPC to form a single row electrode.

With SLIM-E panels, one of the routing channels is on an edge of the sensor array instead of between columns. In [Figure 7](#) this edge routing channel is shown on the right side of the figure and is masked by the bezel paint. The routing channel area still responds to capacitance changes when a finger is present, but at a lower sensitivity. By excluding touch reports in this edge-area, the accuracy within the viewing area is improved.

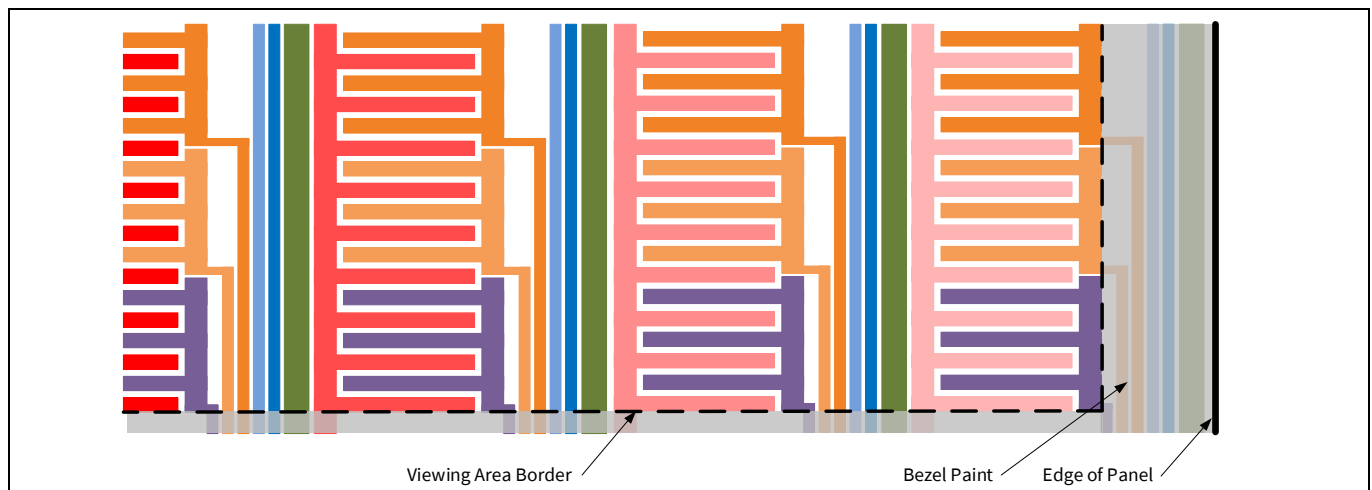


Figure 7 SLIM-E edge routing channels

SLIM designs make use of segmented dummies (see [Figure 8](#)). These dummies are floating ITO material between S1 and S2 electrodes to ensure the SLIM pattern is not visible. Dummies are cut up into pairs to reduce parasitic coupling between S1 and S2. These dummies are typically only used with glass substrates.

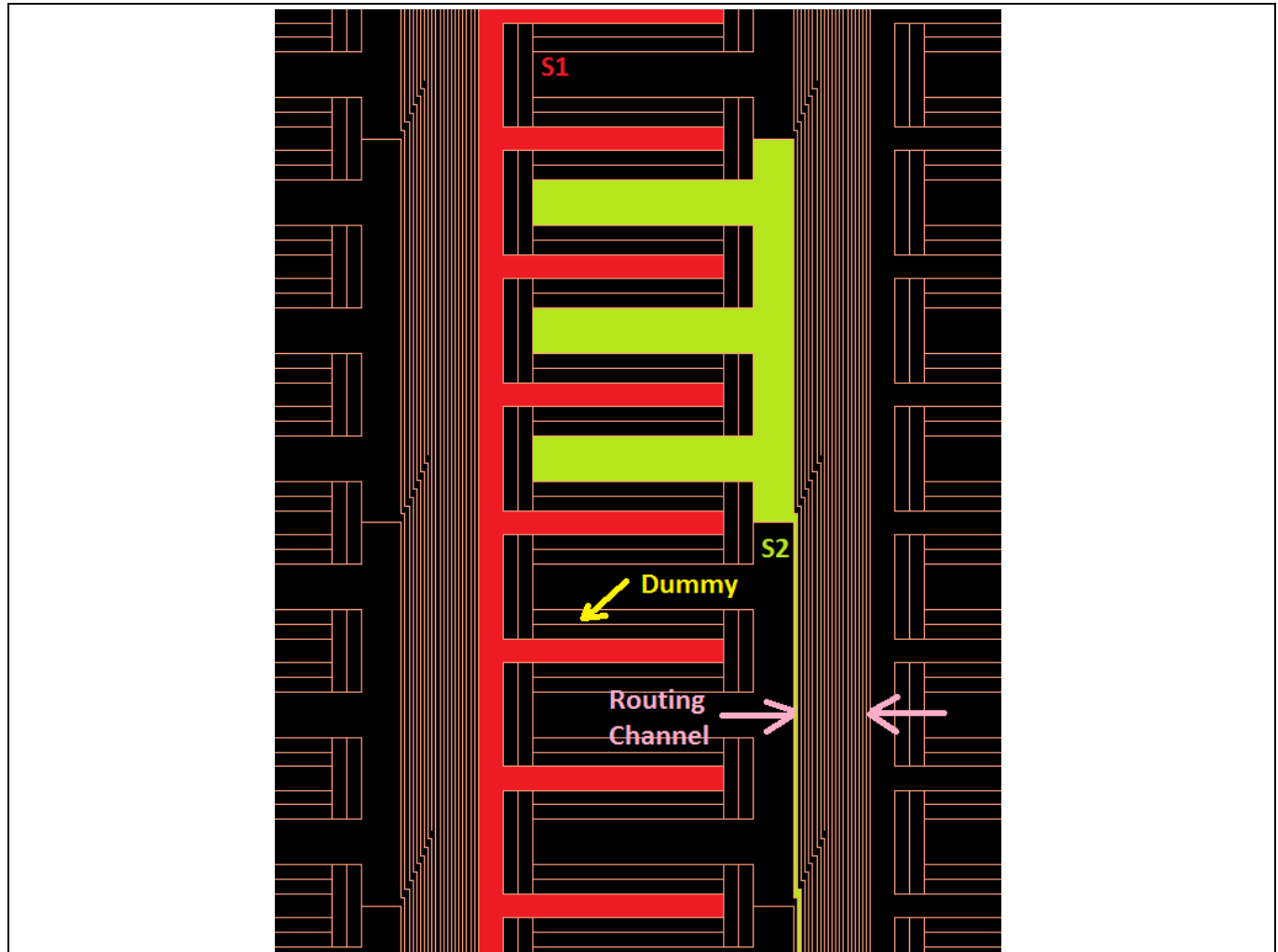


Figure 8 SLIM-E pattern with dummies

The critical design points for SLIM designs are:

- **Resistance** – The longest and narrowest ITO trace in the routing channel determines the maximum resistance to the S2 electrodes. SLIM sensors are typically slower (~150 kHz) than other sensors because of the narrow resistive ITO traces in the routing channel. See [Table 11](#) for the maximum recommended routing channel resistance.
- **Bonding** – See section [FPC/panel bonding](#) for a definition of this term. Depending on the ITO vendor's process technology and mechanical design constraints, the number of bonding pads impacts the overall cost in different ways. [Table 3](#) provides options of routing/bonding that can be evaluated to see what the best option is for a particular panel vendor. In [Table 3](#), “#S1” is the number of S1-type electrodes in the panel design, “#S2” is the number of S2-type electrodes in the panel design, “2 GND” and “GND” describes the number of ground pads need for the type of options, and “2 ESD” is for two bonding pads required to connect a guard ring around the panel. This guard ring is required to prevent system-level ESD events from reaching the RX and TX electrodes, and to provide a discharge path for ESD events entering the product from alternate paths.

Touch panels

- **Tail effect** – Because the traces connecting to the S2 electrodes are routed through the panel (in the routing channel), finger touches interact with these traces and create small signals that look like the finger signal has a “tail”. This effect can be seen in [Figure 9](#) (Note: The tail extends parallel to the S1 electrodes). On the left in the figure is the finger touch with a large signal (red, orange, and yellow unit cells). To the right of the touch are multiple unit cells with higher signal levels (a few hundred counts). These elevated counts are from the finger coupling to the ITO traces in the routing channel.

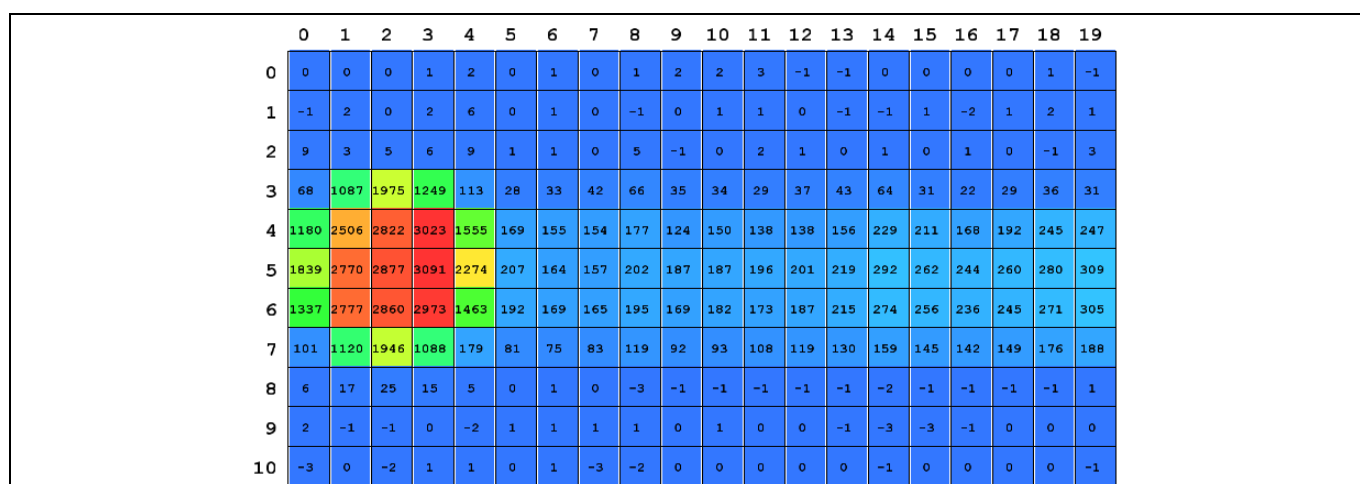


Figure 9 Example of Tail effect with SLIM

Table 3 SLIM options and required bonding pads

SLIM option			FPC		Number of bonding pads	
			Single layer	Multi-layer	Equation	#S1 = 11, #S2 = 20
SLIM-E	Single-End Routing	Single Layer Metal		✓	$\#S1 * (\#S2 + 1 S1 + 2 GND) + 2 ESD$	255
		Multi-Layer Metal	✓		$\#S1 + \#S2 + GND + 2 ESD$	34
	Dual-End Routing	Single Layer Metal		✓	$\#S1 * (\#S2 + 1 S1 + 2 GND) + 2 ESD$	255
		Multi-Layer Metal	✓		$\#S1 + \#S2 + GND + 2 ESD$	34

- **Single layer metal** – Refers to a single layer of metal traces on the panel (typically silver ink). This option allows the vendor to reduce the panel space/cost at the expense of a higher FPC cost.
- **Multi-layer metal** – Refers to multiple layers (typically two conductive layers and one insulation layer) of metal traces on the panel (typically silver ink). This option allows the vendor to reduce the cost/space of the FPC at the expense of a higher panel cost. [Figure 10](#) shows the recommended anti-VIA method for connecting metal traces between the two layers on the panel.

PSoC™ Automotive Multitouch touchscreen controller

Module design best practices

Touch panels

- #S1 and #S2 are the number of S1 and S2 electrodes respectively.
- GND refers to a bonding pad that is required for a V_{SS} ground connection. This is used for a signal ground.
- ESD refers to a bonding pad that is required for a system ground connection. This is used for ESD protection.

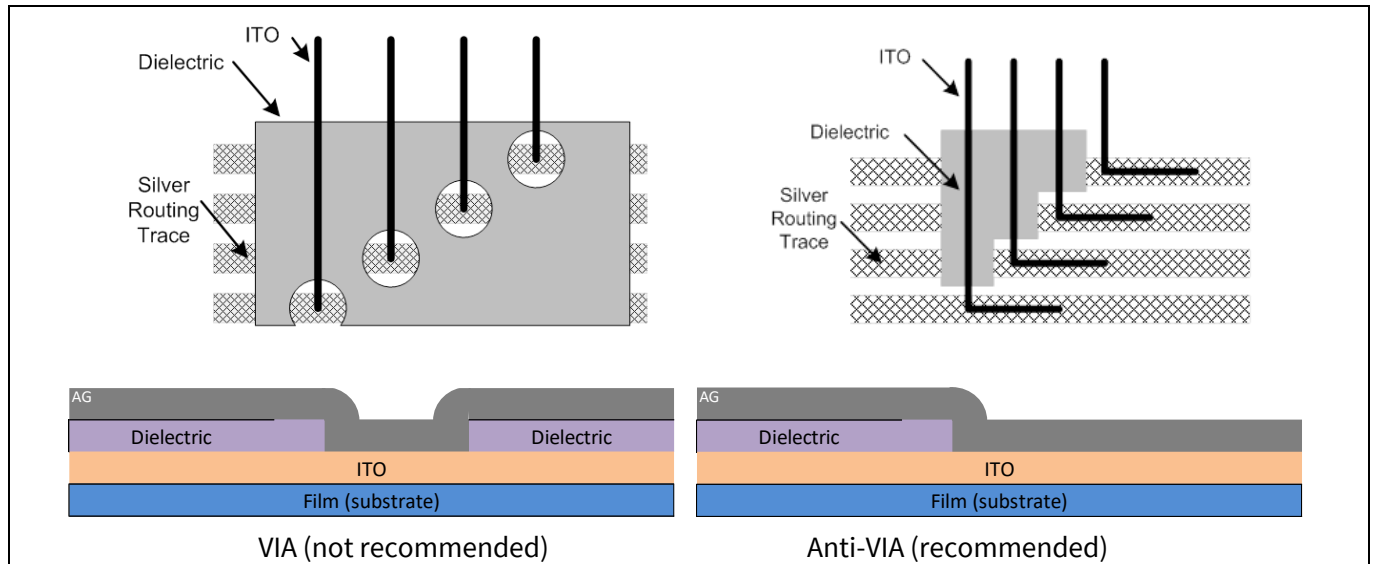


Figure 10 Via and anti-VIA approach for multi-layer metal routing

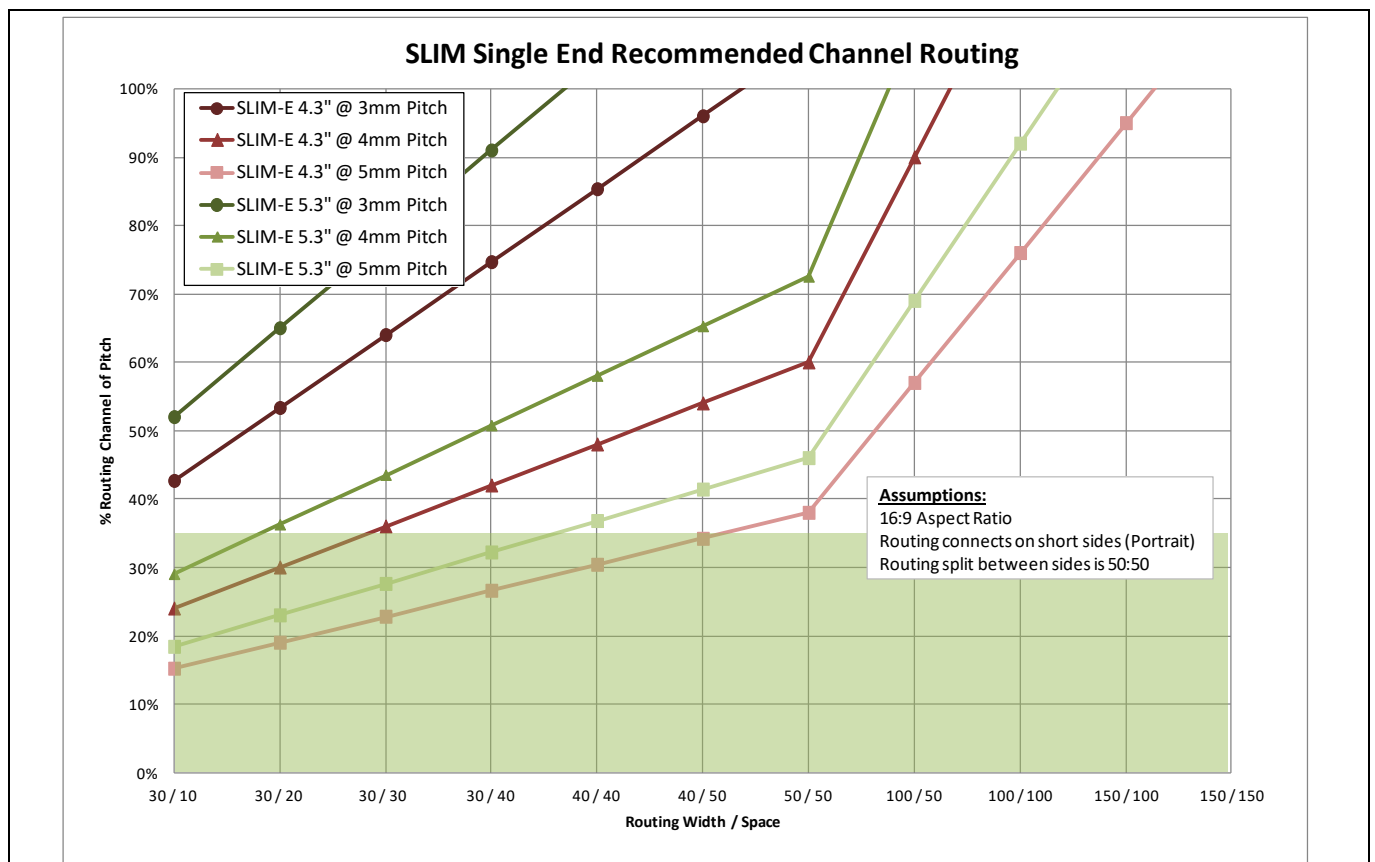


Figure 11 SLIM-E single end channel routing guidelines

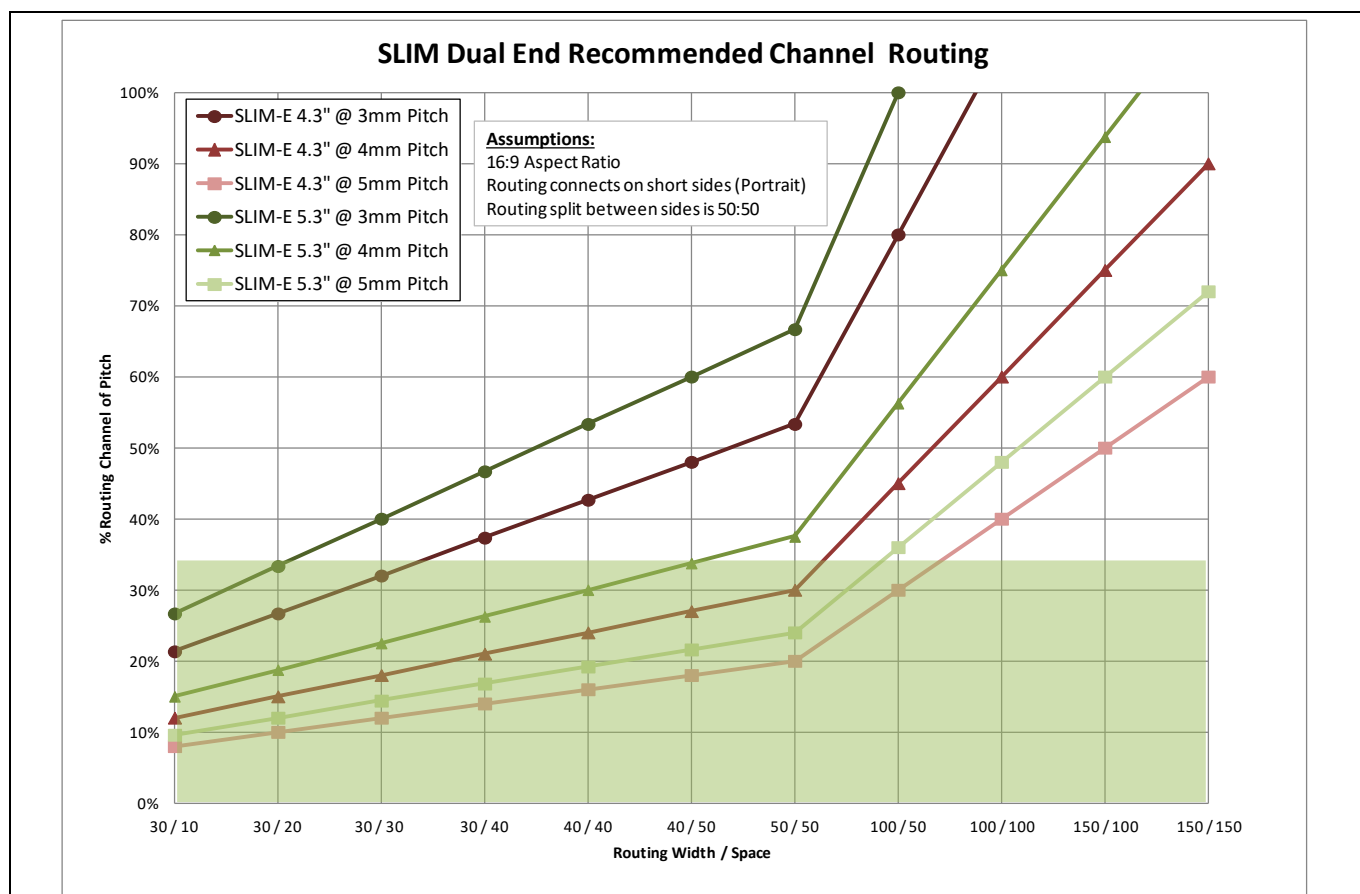


Figure 12 SLIM-E dual-end channel routing guidelines

2.3.3.1 SLIM-E guidelines

Table 11 summarizes the design guidelines for SLIM-E patterns. The definition of the features is shown in Figure 13.

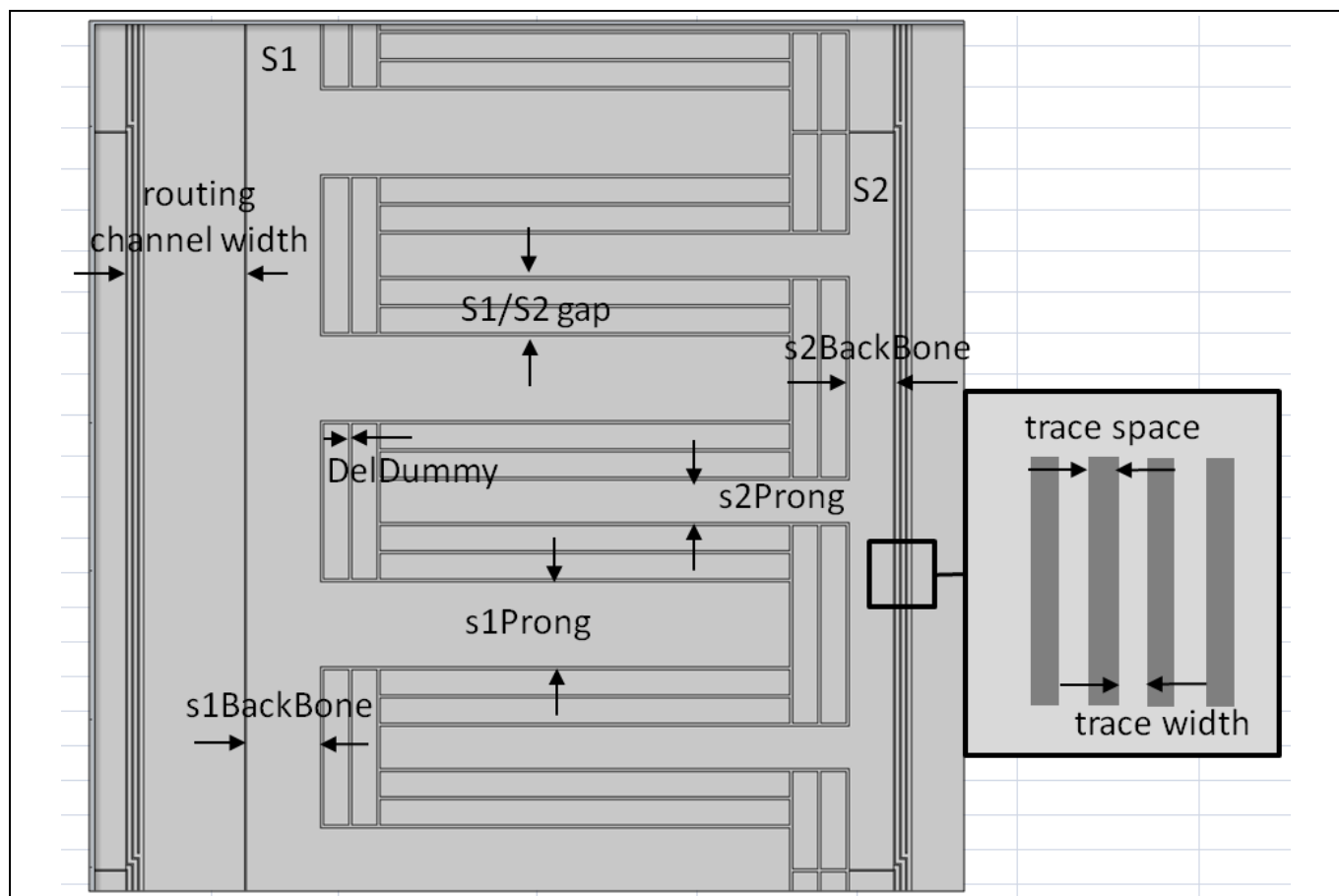


Figure 13 SLIM-E pattern dimension definitions

2.4 Routing

All touchscreen systems require conductive traces to connect the touch panel electrodes to the FPC and to the touch controller. At least one trace is required for each electrode, and (with the exception of SLIM panels) the traces are usually routed along the edges of a panel to the location where the FPC or PCB is attached. The most common routing configurations are single, double, and split, which are discussed in this section. In addition, this section provides guidance on ITO-metal contacts and where to overlap the active and viewing areas with respect to them.

The figures in this section that show example trace routings represent the touch panel sensors using the single solid diamond (SSD) pattern. However, the routing concepts discussed apply to all sensor patterns.

2.4.1 Sensor trace parameters

Two critical parameters related to the sensor routing traces affect the touchscreen design and performance: trace pitch and resistance.

2.4.1.1 Trace pitch

Trace pitch is the sum of trace width and space between the traces, or the distance from centerline-to-centerline of the traces (see [Figure 14](#)).

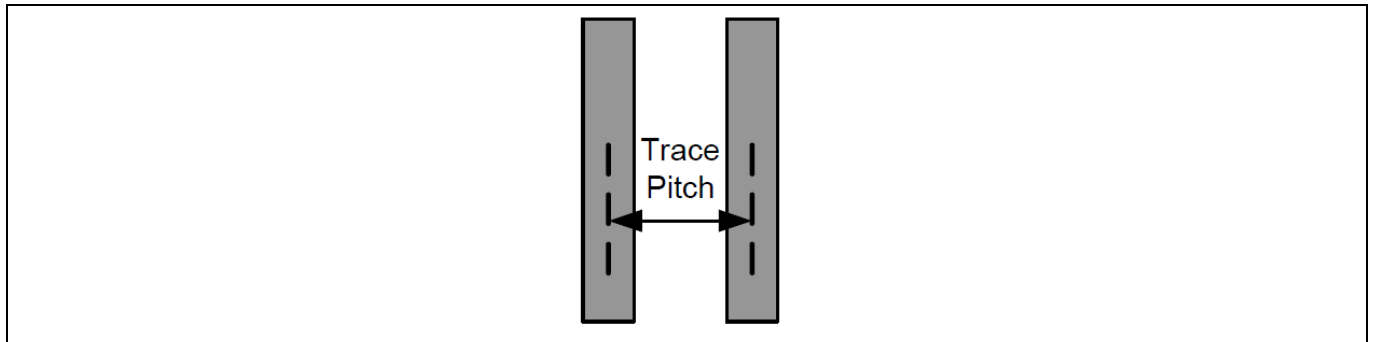


Figure 14 Trace pitch

Trace pitch determines whether it is possible to place all the required traces in the panel border area. Because these traces are often made of opaque materials, they must fit under the bezel of the product to remain invisible. The allowed trace pitch varies from one manufacturing process to another, in the range of 10 µm to 300 µm. This means that not all touchscreen vendors can meet the requirement for designs with a narrow border area.

[Table 12](#) shows the recommended minimum value for trace width and gap.

2.4.1.2 Trace resistance and capacitance

The formula for trace resistance is provided in [Equation 2-1](#).

$$\text{Resistance} = \text{Trace sheet resistance} \times \text{Trace length} / \text{Trace width} \quad \text{Equation 2-1}$$

Trace resistance is evaluated relative to sensor resistance. High trace resistance compared to the sensor resistance degrades the touchscreen performance.

[Table 12](#) shows Infineon's recommendation for maximum trace resistance. The sheet resistivity and width of the traces should be adjusted to meet the recommended value.

See section [Capacitive buttons](#) for how to calculate mutual/self-capacitance of routing traces.

2.4.2 Routing configurations

2.4.2.1 Single routing

In single routing, shown in [Figure 15](#), traces from each row connect along only one edge of the panel.

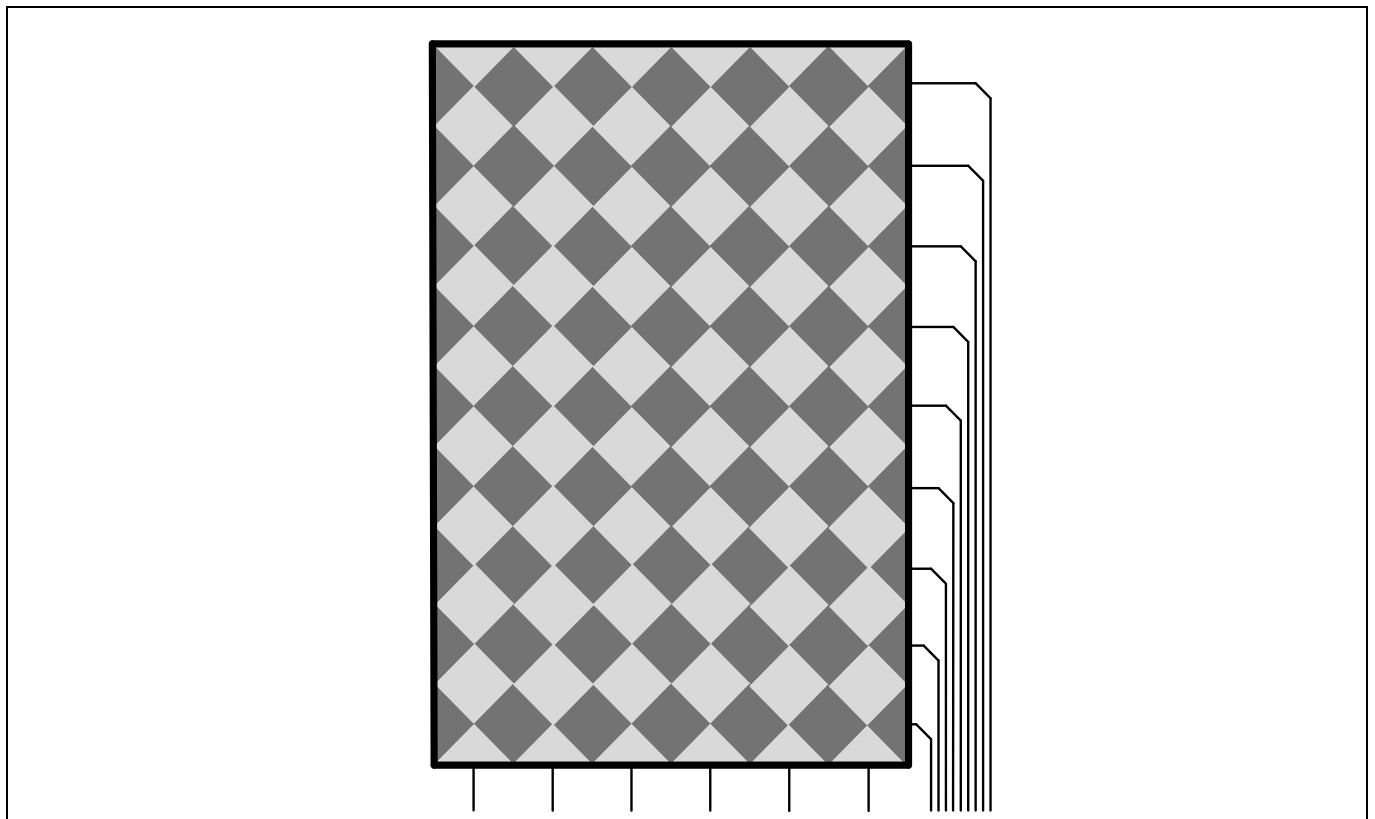


Figure 15 Single routing

2.4.2.2 Split routing

Split routing as shown in [Figure 16](#), is an implementation of single routing in which half of the routes are on each side of the panel. This keeps the bezel width to a minimum, and equal on both sides. The disadvantage is that there is at least one point in the panel where connections to row electrodes switch from one side to another: one row is connected along the right edge and the adjacent row is connected on the left. Touches occurring along these electrodes are attenuated to a different degree due to the different resistive paths. This can result in an additional finger location error.

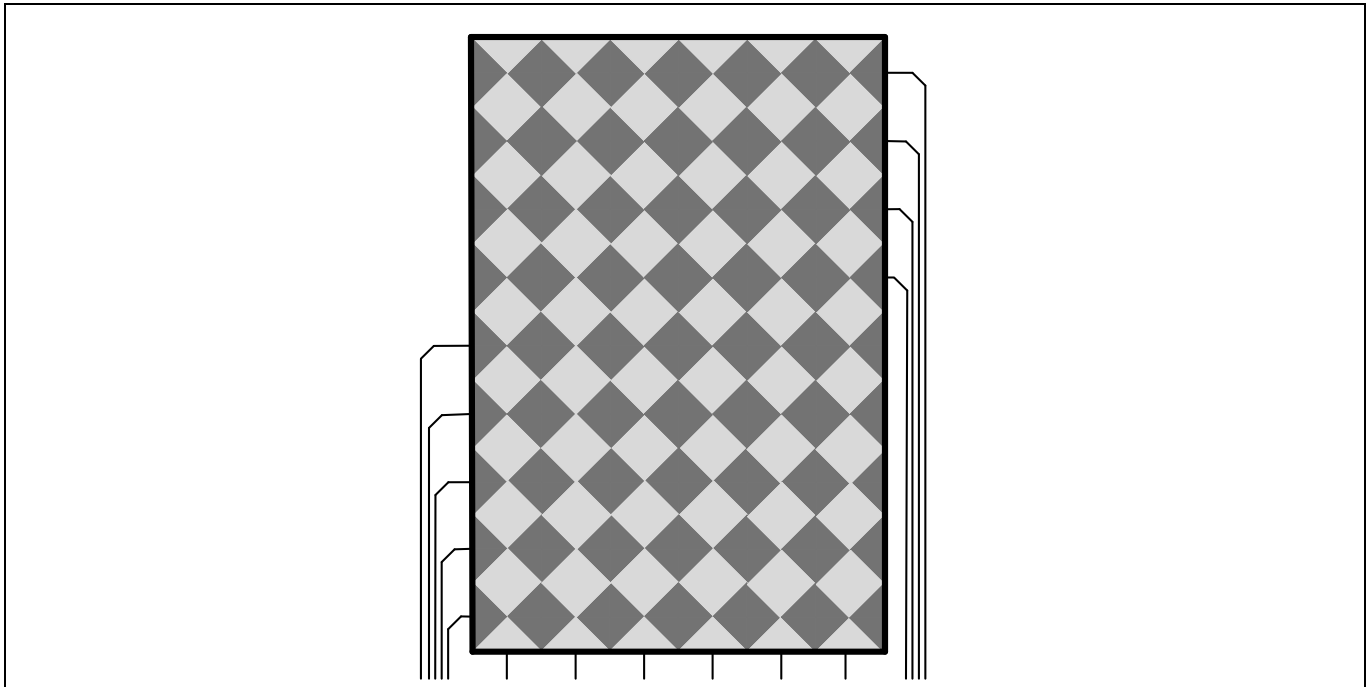


Figure 16 Split routing

2.4.2.3 Interlaced routing

Interlaced routing (a variation of split routing) can be useful in circumstances where other routing options do not work from a layout standpoint, or where there is an overall noise benefit.

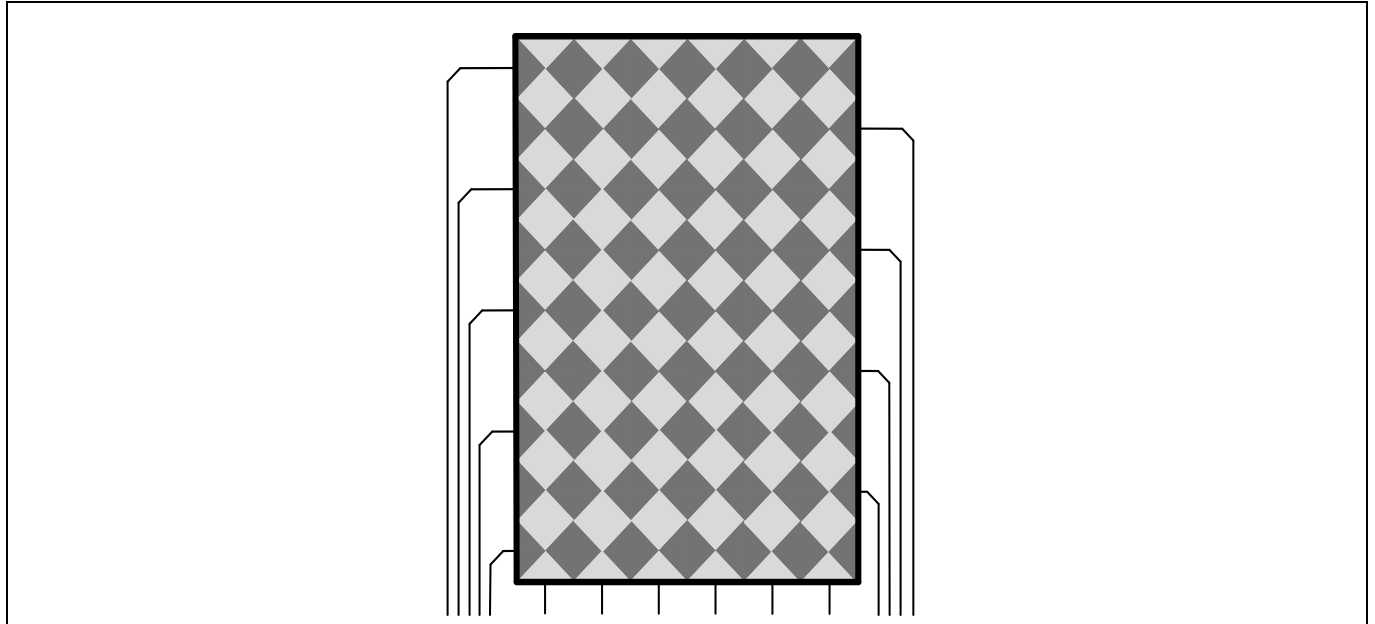


Figure 17 Interlaced routing

2.4.2.4 Double routing

Double routing is the preferred routing method for large screens. In double routing, a routing trace is connected to both ends of each row or column. [Figure 18](#) shows, from left to right, X-double routing, Y-double routing, and XY-double routing. The pairs of traces corresponding to a particular row or column are typically shorted together on the FPC and routed to a single touchscreen-controller sense pin.

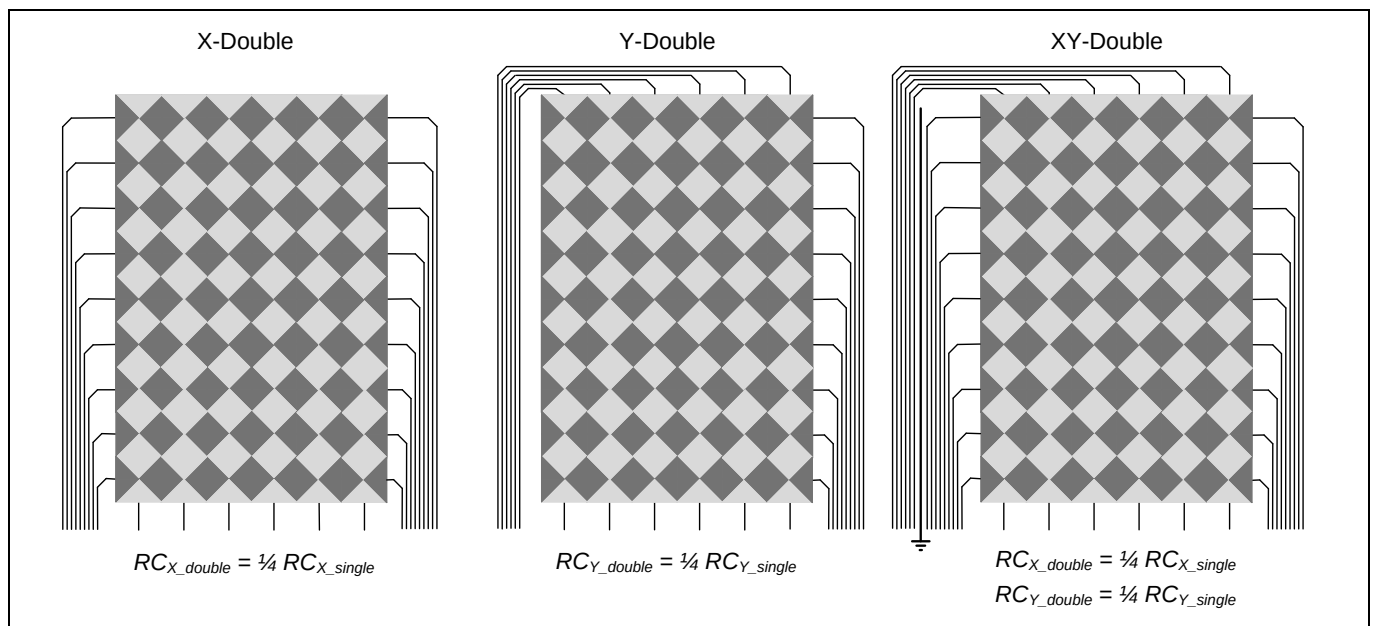


Figure 18 Double routing

Touch panels

Double routing reduces the RC delay of a row or column by a factor of four. This means that the maximum switching frequency can be increased, up to a factor of four, which can significantly improve the maximum refresh rate, noise rejection, and SNR.

Double routing often makes more sense for SSD/DSD sensors because the bridges incur a longer RC delay. Y-double routing should be used for portrait panels, while X-double routing is more appropriate for a landscape panel.

2.4.2.5 Double split TX routing

Double split TX routing is similar to double routing, but with the electrodes cut in the middle of the touchscreen ([Figure 19](#) illustrates double split routing on the X-axis). This type of routing is currently supported only for large screen multichip solutions, where each side of the TX electrode is driven by a different touch controller.

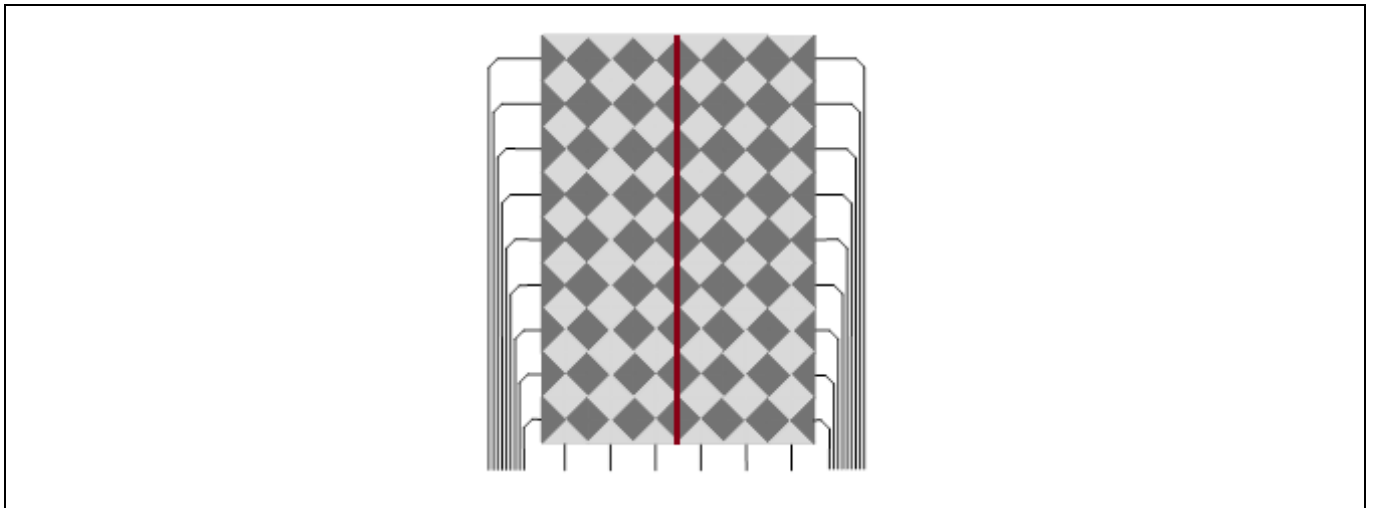


Figure 19 Double split TX routing

2.4.3 Sensor routing guidelines

- For metal traces connecting to TX and RX electrodes, 90-degree corners should be avoided. The corner causes an increase in capacitance, concentrates the associated electric field, and create areas of high mechanical stress on the traces. Corners should be 45 degrees or radiused.

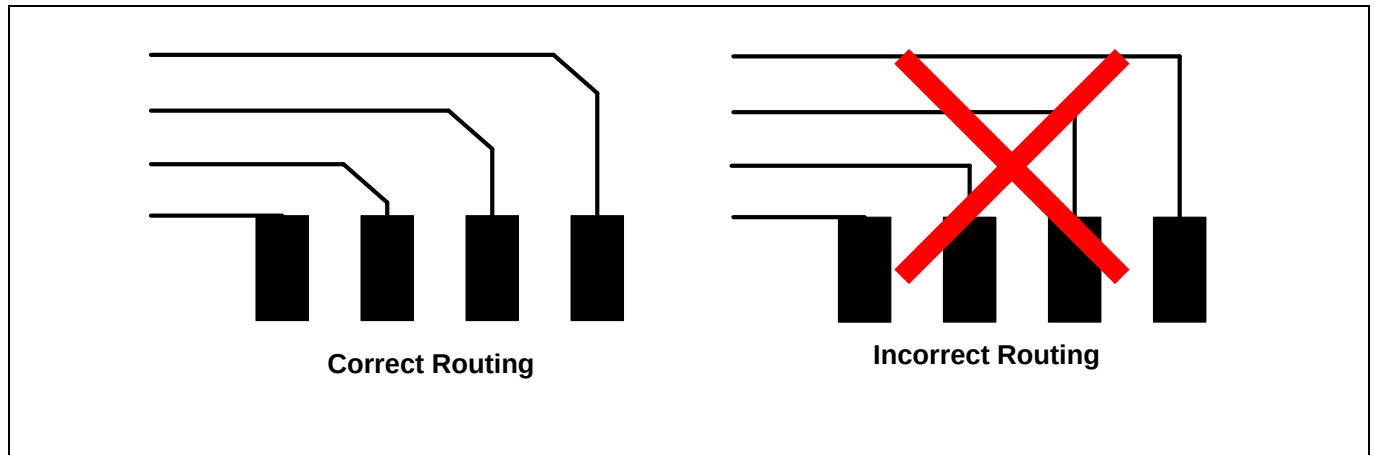


Figure 20 Correct and incorrect routing of corners in metal trace

- Figure 21 shows both the correct and incorrect ways to route traces in the vicinity of the bonding pads. The left figure shows correct bonding pads. Each pad is properly sized to meet the FPC bonding requirements. Connection between the horizontal traces and bonding pads is done by extending the trace and not by extending the bonding pad. The right image shows the incorrect extension of the pads. The inconsistent pad sizes introduce differences in the total parasitic capacitance between the electrodes.

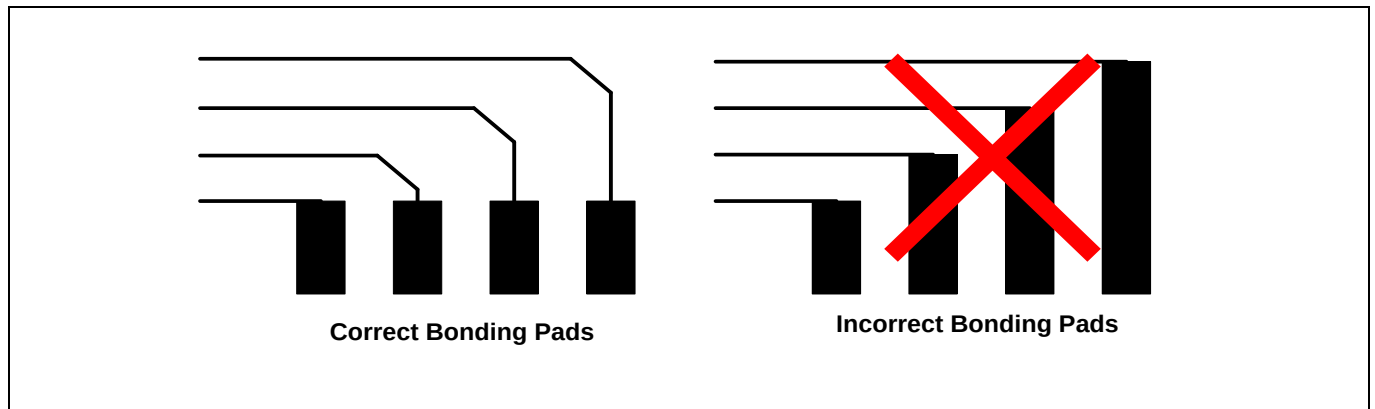


Figure 21 Correct and incorrect routing in the vicinity of bonding pads

Touch panels

- Do not let the TX and RX traces overlap for more than 1 mm. If they must overlap, insert a shield-layer between the TX and RX traces. [Figure 22](#) shows an example of overlapping TX and RX traces.

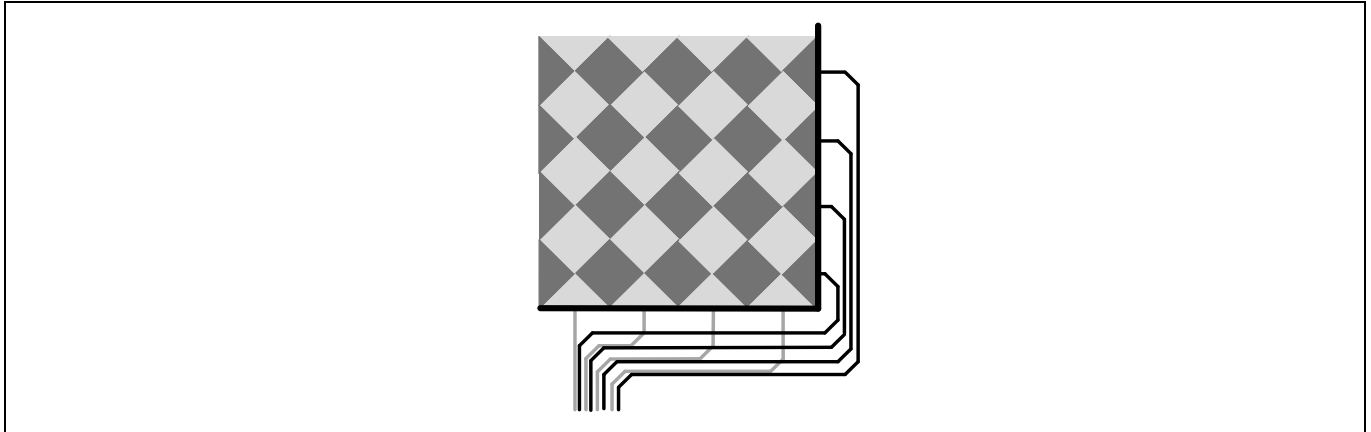


Figure 22 Overlapped routing traces

- For designs where TX and RX traces are routed in parallel (but on the same plane), insert a grounded isolation trace between adjacent TX and RX traces. [Figure 23](#) shows an example of such an isolation trace. This may be either a system ground or the V_{SS} reference of the touch controller. Other than this, ground traces/planes should be kept away from the sensor traces.

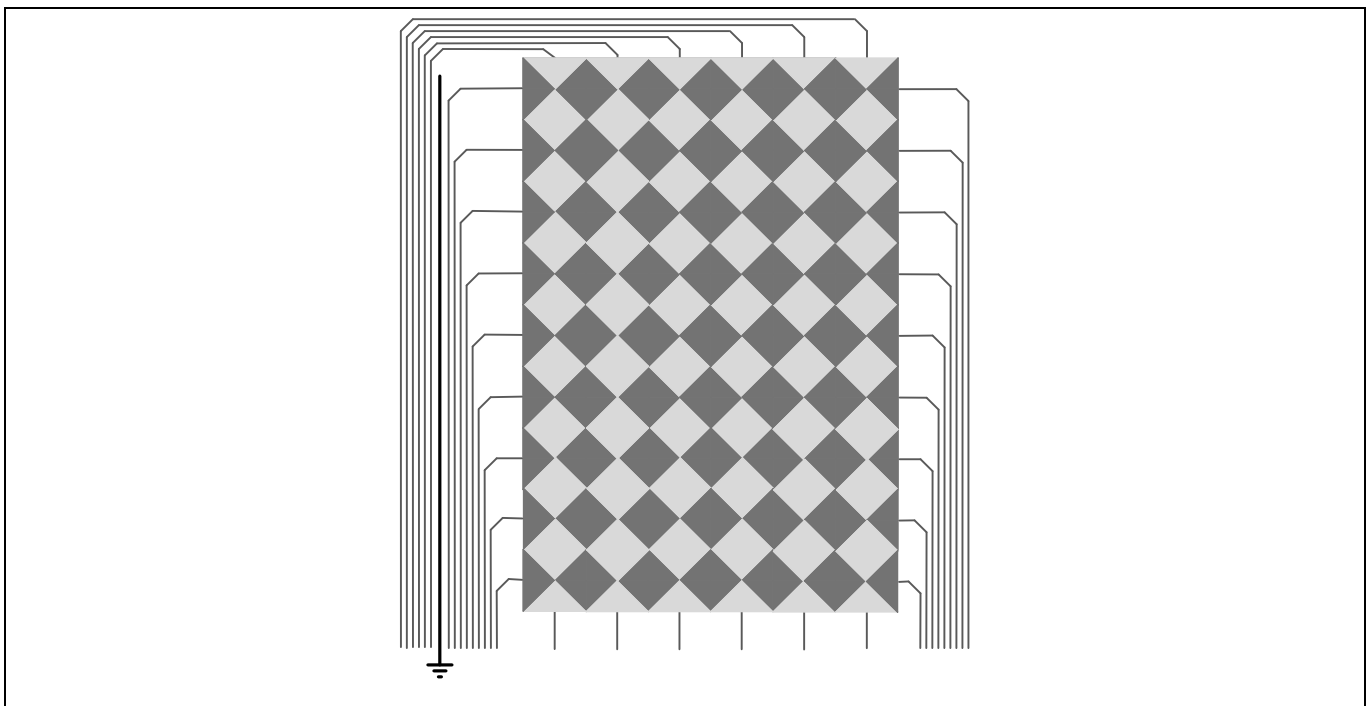


Figure 23 TX/RX grounded isolation trace insertion

Ground planes (on alternate layers) should not overlap with sensor traces. The exception is for FGND hover designs, where traces should be routed on inner layers a hatched ground is recommended on outside layers.

Touch panels

It is important to approximately match the mutual capacitance between all the TX and RX electrode pairs. Matching the mutual capacitance minimizes the required dynamic range of the sensing channels and saves design margin for the unknowns and noise in the system. This is a particularly important consideration when a grounded isolation trace is needed. The width of the isolation trace may be determined using [Equation 2-2](#).

$$W = \left[\frac{\varepsilon \times L}{2 \times PERIOD} \right]^{1.23} \quad \text{Equation 2-2}$$

Where:

- L is the total length of the adjacent TX and RX traces routed in parallel.
- ε is the dielectric constant of the material upon which the traces are printed. If a panel is glass with PET cover lens, or vice-versa, use the average dielectric constants of the two for ε .
- PERIOD is a parameter of the expected tuning. If the parameter is not known, use the default value from the tuning.
 - For TMAx devices, use the larger of the RAMP_INT_UP and RAMP_INT_DOWN parameters.
 - For CYATx devices, use the TX_PERIOD_MC parameter.
- w is a scale factor that is multiplied by the standard RX/TX trace width for a given design.

For instance, consider a film-based panel with a glass cover lens having a dielectric constant of 3 and 7, respectively. If the parallel trace length between TX and RX traces is 33.5 mm and the PERIOD value is 75, then the ground trace width is as follows:

$$W = [(5 \times 33.5) / (2 \times 75)]^{1.23} = 1.15$$

Therefore, for a trace width of 0.1 mm, the grounded isolation trace width is $1.15 \times 0.1 \text{ mm} = 0.115 \text{ mm}$.

- Where border area allows, double-route all panels along the highest-resistance axis.
- If there is sufficient border (bezel) area for single routing, then do not use split routing.
- If the mechanical design cuts into the corners of the active area, cut at angles, not in unit cells (see [Figure 24](#)).

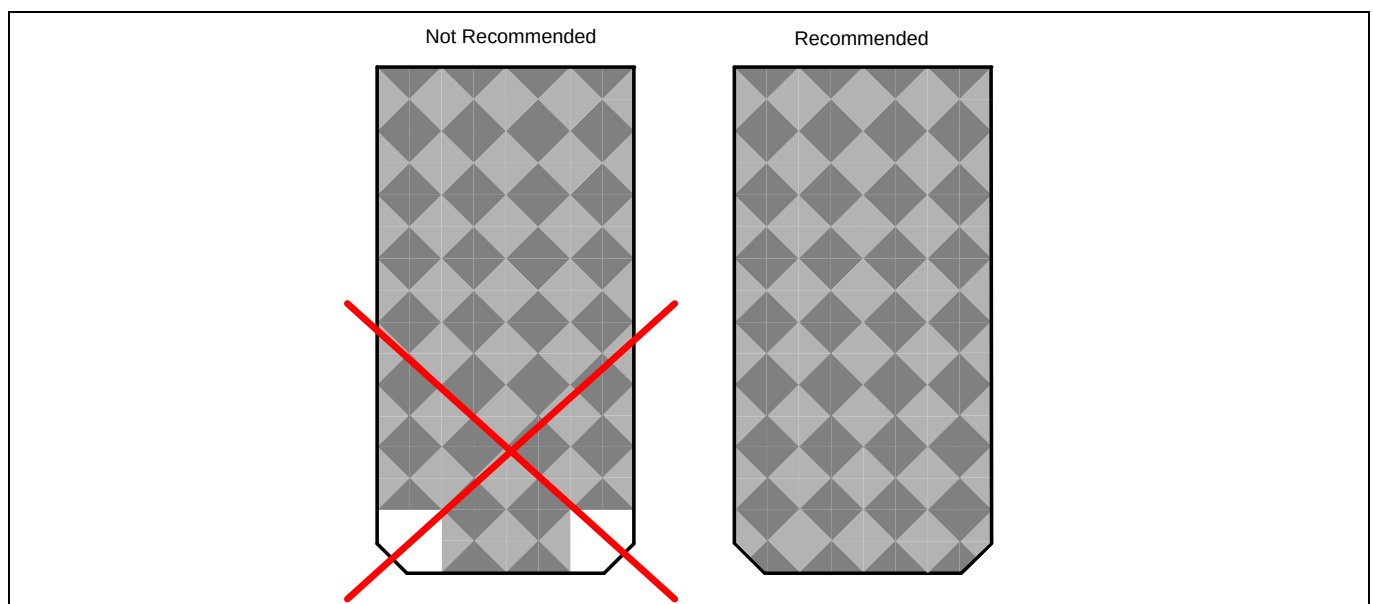


Figure 24 Pattern corner cutting

2.4.4 Metal to ITO contacts

To achieve good yield and electrical performance, a solid contact between the end of each electrode and the metal routing trace is necessary. The position of the contact-area relative to the active and viewing areas can also affect accuracy along the edge of the screen.

The active area and viewing areas are:

- Active area – The area that capacitance is sensed, or in other words, the area that is covered by ITO, excluding the ITO/metal contact area.
- Viewing area – The area that is transparent to the user to view the display.

2.4.4.1 Diamond sensors

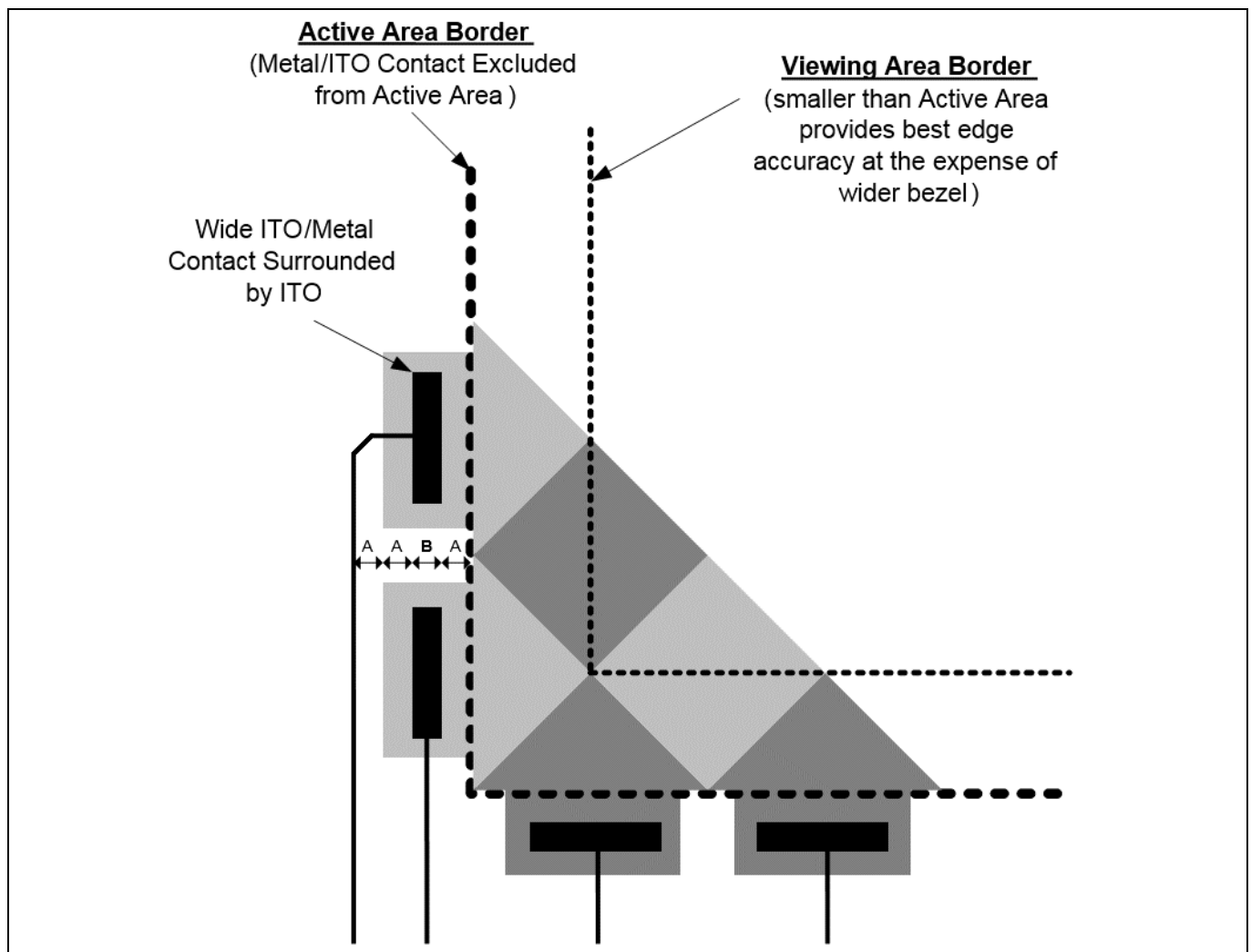


Figure 25 Diamond sensor ITO-metal contact

2.4.4.2 Manhattan sensors

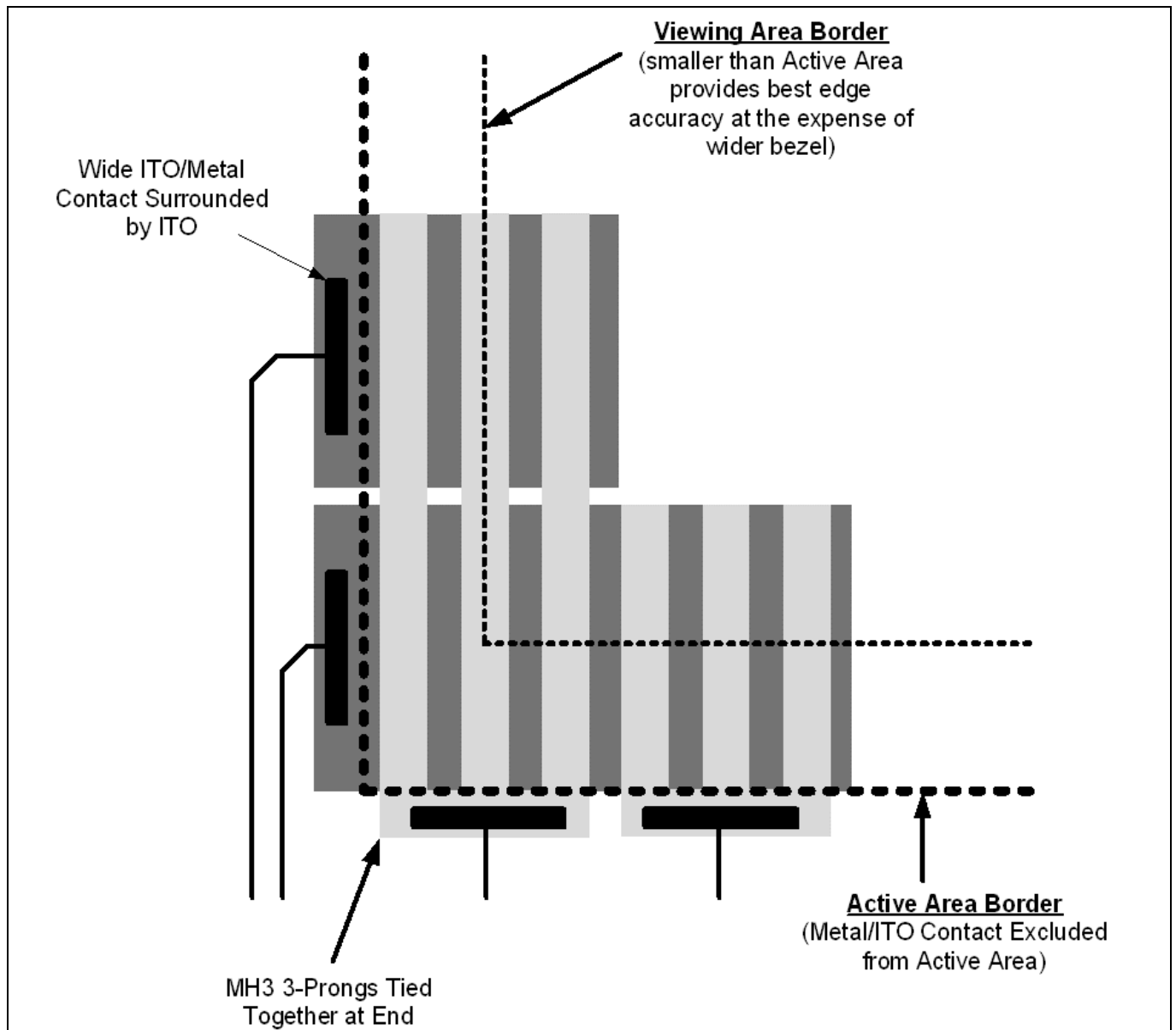


Figure 26 Manhattan sensor ITO-metal contact

2.4.4.3 Extending sensor area to match ITO-metal contact area

The metal-ITO contact area can produce extra signal when touches are present over the contact area. This causes different performance than touches over an edge with no metal-ITO contact area. To balance this extra signal, extend the ITO active area at the far end of the sensor (non-contact edge). This produces consistent touch behavior for all edges. See [Figure 27](#). The metal-ITO contact area adds more ITO area to the half-diamonds on one axis. Adding the same area to the half-diamonds on the other axis helps to match the areas of the two electrodes. If the sensor pattern already extends beyond the viewing area by 0.3 mm–0.5 mm, then this may not be needed.

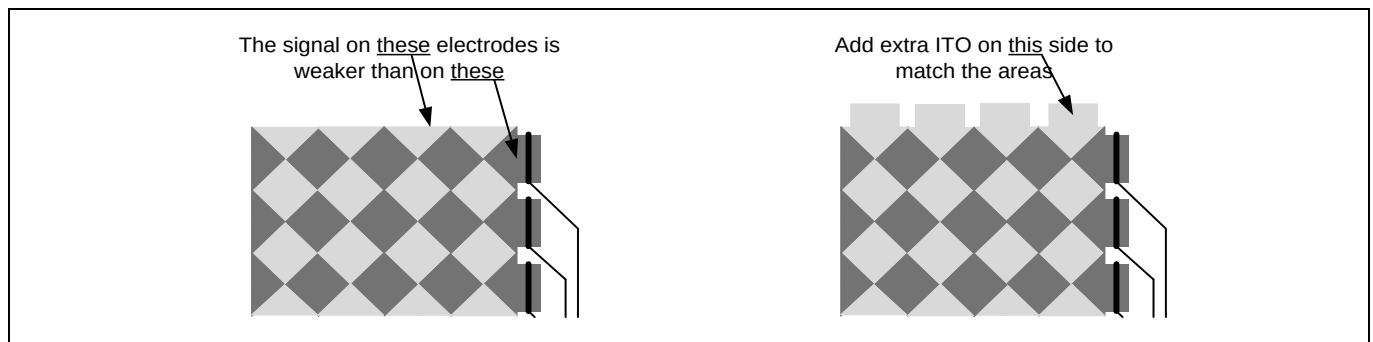


Figure 27 Extending ITO sensors to match ITO-metal contact area

2.4.4.4 Metal to ITO contact guidelines

- Make the metal traces over the ITO landing pads large enough to guarantee low contact resistance, as specified in [Table 12](#) ($< 100 \Omega$). This requires a certain minimal width of the trace over a landing pad; B. The value of B, for a given manufacturing process, is established by the ITO partner. See [Figure 25](#).
- In manufacturing, the metal and the ITO layers do not always align perfectly. This also requires a certain minimal width of the landing pad; A. For a given amount of misalignment (specified by the ITO partner), allow extra space in the following places (see [Figure 25](#)):
 - Between metal trace and visible (sensor) area; otherwise, there is a risk of metal being visible.
 - Between metal trace and the edge of the ITO landing pad; otherwise, the contact area might fall below B.
 - Between a landing pad and an unrelated metal trace; otherwise, shorts may occur.
- A combination of the rules in the previous bullets dictate that the minimum width of the ITO landing pad must be $2A + B$, as illustrated in [Figure 25](#), where the values of A and B are established by the ITO partner.
- For PET substrates with silver ink routing traces, Infineon recommends a minimum of 0.3 mm contact width.
- For glass substrate with metal routing traces, Infineon recommends a minimum of 0.1 mm contact width.
- Infineon recommends a minimum contact height of 90% of the electrode pitch, independent of substrate materials.

2.5 Capacitive buttons

This section contains the design requirements for implementing capacitive buttons with Infineon touch controllers. Capacitive buttons can be implemented in either copper or ITO. All Infineon touch controllers support self-capacitance and mutual-capacitance buttons. For applications where the button need not be transparent (i.e., it is not positioned over the display), using copper electrodes on either an FPC or PCB is strongly encouraged. The capacitive buttons should always be placed on the conductive layer closest to the cover lens. Copper electrodes provide higher yield, lower cost, and better performance than comparable ITO sensors.

IMPORTANT – In many cases, the button shape will be driven by the end product's industrial design. Therefore, the example button shapes below may not be feasible to implement. Additionally, the additional self and mutual capacitance from button routing must not exceed the total C_M and C_P specified in [Table 7](#).

In all cases, it is recommended that the button design, including routing traces, be shared with Infineon for simulation. The simulation will verify that the button will have appropriate performance.

2.5.1 Self- vs mutual-capacitance buttons

Self-cap designs have the advantage of only requiring one XY pin per sensor. Mutual-cap designs require more XY pins, but can provide superior water rejection capability over self-capacitance buttons when scanned in DualSense mode. Mutual-capacitance buttons also are less susceptible to performance degradation due to high parasitic capacitance ($C_p > 15$ pF) than self-capacitance buttons. Capacitive button matrices implemented with mutual-capacitance buttons can provide higher SNR with a similar number of XY pins compared to self-capacitance buttons.

2.5.2 Mutual-capacitance button designs

Mutual-cap designs require TX and RX lines. To minimize XY pin use when multiple mutual-capacitance buttons occupy the same panel, use one or more common TX lines with one RX line per button.

For automotive parts, it is recommended to use dedicated TX lines for buttons; that is, the TX pin should not be shared with a TX in the main panel.

[Figure 28](#) and [Figure 29](#) show examples of two recommended button patterns (one single-layer and one two-layer) for combined self/mutual scanning, or DualSense. This allows for the most reliable CAPSENSE™ performance, including water rejection. Note that the button dimensions in the figures are for typical panel stack-ups and materials, and will not be ideal for every design. Therefore, as noted above, Infineon recommends that the button design be simulated to ensure proper self-capacitance and mutual-capacitance.

If simulation is not possible, [Equation 2-3](#) gives an alternate check for the self-capacitance of the RX electrode.

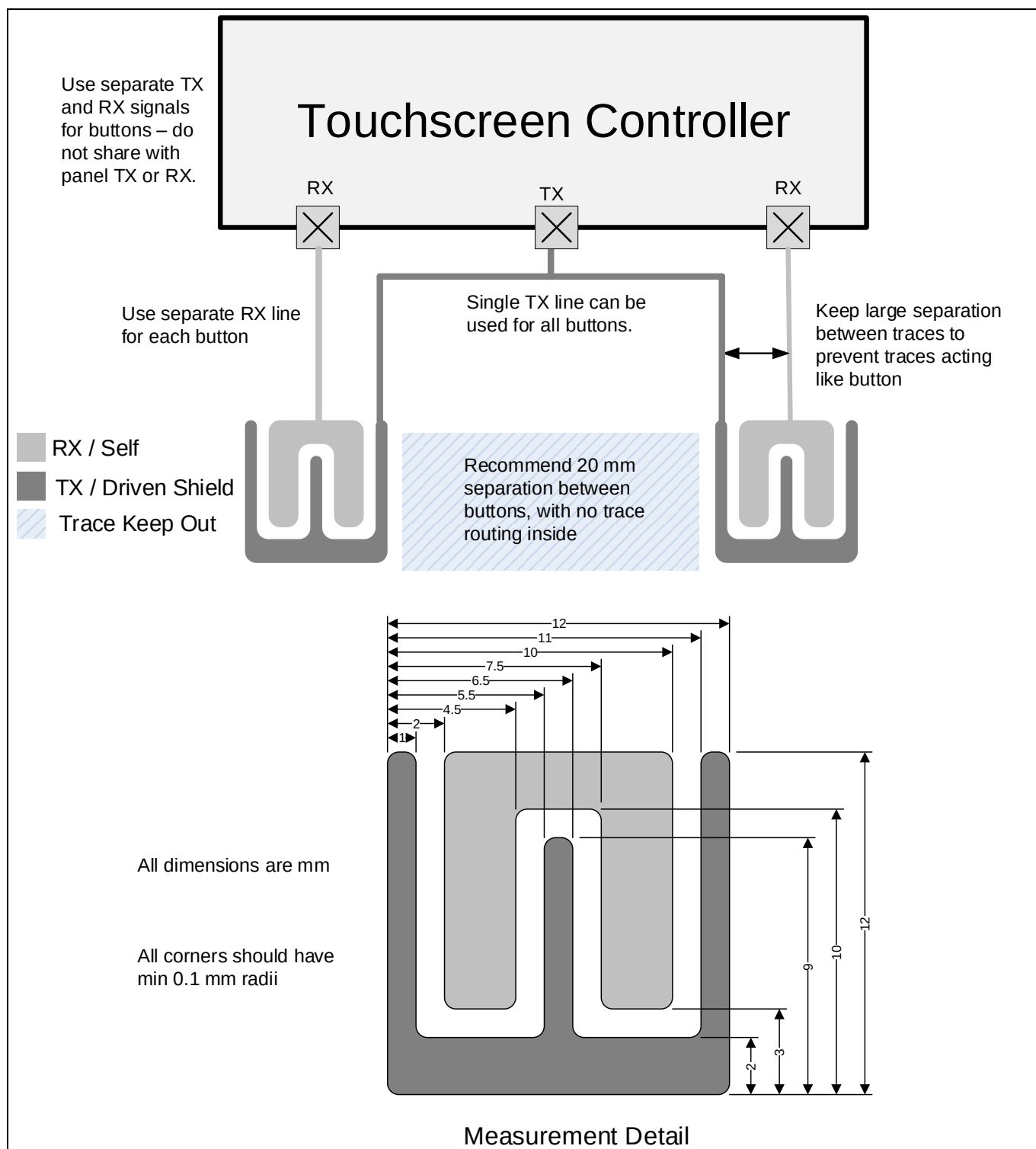


Figure 28 Single layer, mutual capacitance buttons with 12 mm pattern

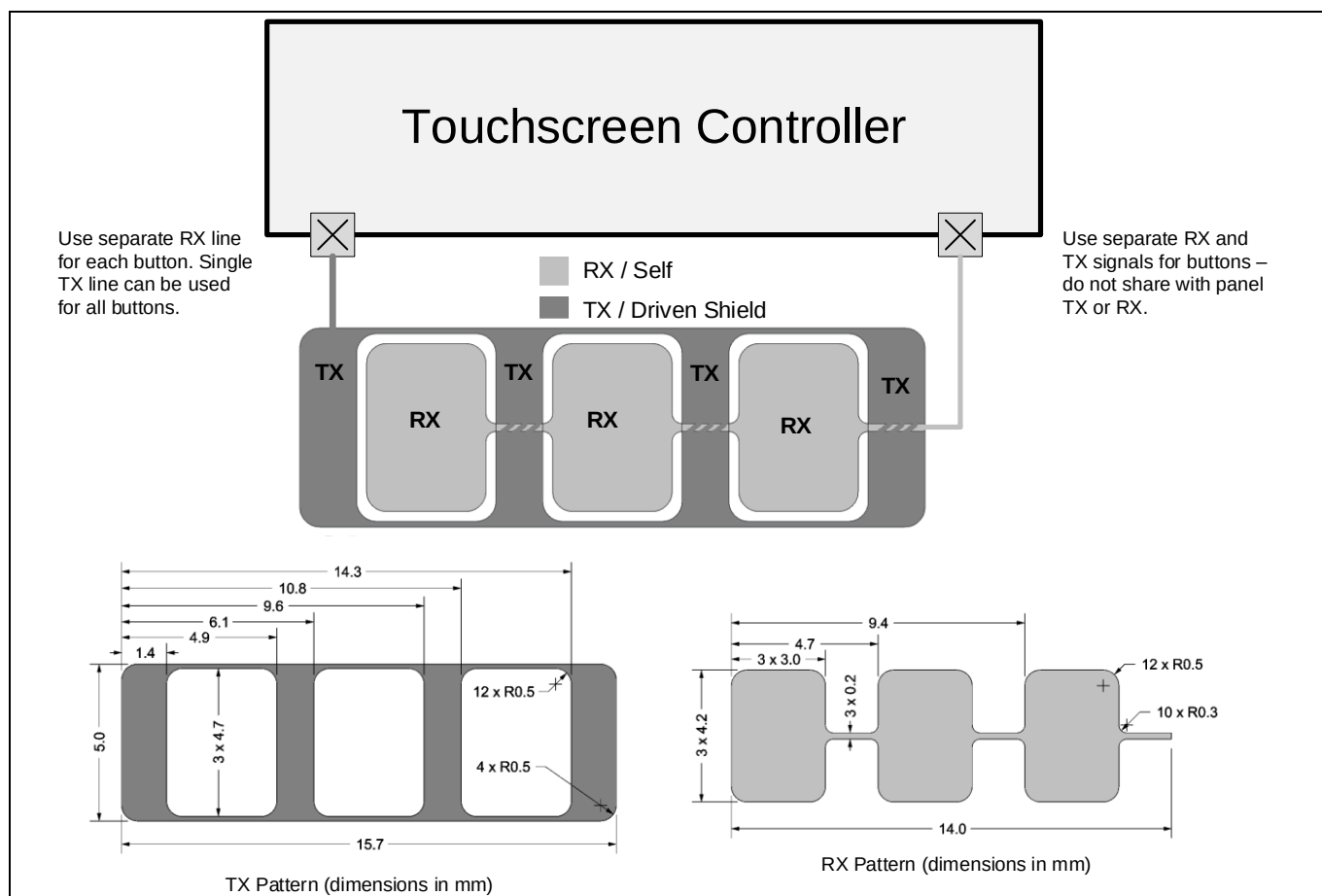


Figure 29 Two-layer button design with rectangular aspect ratio

2.5.3 Self-capacitance button design

The advantage of self-capacitance buttons over mutual-capacitance buttons is that they require only one XY pin per sensor. These buttons can be any shape as long as their length to width (where length is the always the longest dimension) aspect ratio does not exceed 5/3 and their geometry does not contain any corners with an included angle less than 90 degrees. The ideal shape for self-capacitance sensors is a circle; a square is also acceptable, but it is important that the corners are reduced (at least 0.5 mm radius) to avoid electric field concentrations, which can degrade the sensor performance. [Figure 30](#) shows a variety of sensor shapes, from ideal to bad, based on these guidelines.

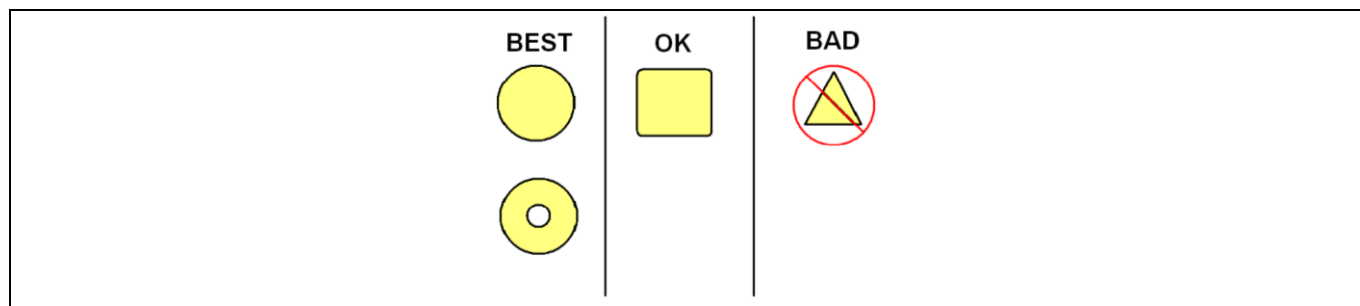


Figure 30 Self-capacitance sensor shapes

Touch panels

Sensor area is a critical parameter for self-capacitance sensors. For a given cover lens thickness (t), the minimum sensor area (a) is given by [Equation 2-3](#).

$$a = \frac{0.25 \text{ pF} \times t}{\epsilon_0 \times \epsilon_r} \quad \text{Equation 2-3}$$

Where ϵ_0 is the permittivity of free space (8.85×10^{-15} F/mm) and ϵ_r is the dielectric constant (also known as relative permittivity) of the cover lens. [Table 4](#) provides ϵ_r of several materials commonly used for a touchscreen device cover lens.

Table 4 Relatively permittivity of common cover lens materials

Material	Dielectric constant (ϵ_r)
Glass	6.5–8
Acrylic	2.7
PET (Mylar)	3.2

2.5.4 ESD compliance for buttons

To reduce the impact of ESD on buttons, a ground ring is recommended around each button or group of buttons. The goal is to divert the ESD energy to the system ground instead of through the button. A side effect of such a ground ring is that it reduces the effectiveness of a driven shield for water rejection. To maximize the effectiveness of the driven shield, the clearance between a metal ground ring and a button RX should be at least 1.5 mm. To limit coupling of the energy in the ESD pulse into the touch controller, this ground ring should be connected to the system ground, and isolated (where possible) from the device V_{SS} .

2.5.5 Button routing

Avoid routing button traces between buttons. Touch coupling to traces between buttons can trigger a touch on adjacent buttons. See [Figure 28](#) for details on the keep-out area for button routing.

2.6 Shielding

This section provides guidelines for implementing a shield layer for touchscreen designs. It does not apply to designs supporting hover detection with the CYAT817x device family. For hover shielding, please refer to section [Hover guidelines](#).

Capacitive touchscreens operate by detecting changes in electric field coupling caused by a finger or other conductive object. This makes them inherently sensitive to electromagnetic interference (EMI). Touchscreen devices typically contain multiple EMI sources, with the display often being the worst offender because of its close proximity to the touchscreen. An effective mitigating technique for designs challenged by EMI from the display is to add an ITO shield layer between the touchscreen and the display.

The advantages of a shield layer include:

- Increases C_p uniformity, resulting in more robust performance; if the shield is driven (active shield) by the touch controller, the effect is magnified
- Suppresses display noise, offering better SNR
- Reduces raw-count changes due to panel flexing (for assemblies with an air gap between the touch sensor and display)
- Reduces parasitic capacitance (when driven as an active shield)

In touchscreens with an air gap between the sensor and the display, if the cover material of a panel is not sufficiently rigid (for example, PMMA) and the sensor substrate is made of PET Film, a shield layer (or a self-shielding pattern like MH3) is mandatory to prevent raw count changes due to panel flexing.

2.6.1 Shield design

The ITO shield layer is applied to the bottom-side of a stack up (see section [Stack-up](#)), and must cover all the ITO row and column electrodes. The purpose of the shield is to prevent electronic noise coupling from the display into the touchscreen. If not shielded by the FPC, all the traces routed near the display controller, display tail, or display driver must also be covered. For the shield to be effective, its sheet resistance must be less than or equal to that of the electrodes it is shielding.

For applications that only require the EMI shield function of this layer, the ITO shield is connected to system ground on the FPC. The routing traces should be made with either metal (preferable) or silver ink, and the termination at the ITO shield should be routed around the entire perimeter of the ITO shield layer (under the bezel) to minimize the shield resistance (see [Figure 31](#)). The diamond shapes in [Figure 31](#) are the touchscreen electrodes, and the light green area is the shield. The routing traces are on the same layer as the touchscreen electrodes, while the metal ground ring (wide black line) is on the same layer as the shield.

Because this shield layer is grounded and closer to the row and column sensing electrodes than the display's ground reference, use of a shield increases the parasitic capacitance (C_p) for self-capacitance sensing, and the load on the TX driver for mutual capacitance sensing. This can reduce the overall dynamic range of the measurement system for those touch controllers that do not have hardware baseline compensation. The magnitude of the parasitic capacitance is determined by the thickness and dielectric constant of the substrate and attachment layers of the shield.

Touch panels

For applications that require sensing of objects at significant distance from the top side of the touch sensor, or objects that generate very small signal changes, it is also possible to actively drive this shield to reduce or remove most of the parasitic capacitance (C_p) of the X and Y sensor electrodes. When measuring self-capacitance, a significant amount of the dynamic range of the RX channel is used in measuring the parasitic capacitance of these X and Y electrodes. By driving the shield layer in matched phase and amplitude with the electrodes being measured, that parasitic capacitance is removed. For this to operate the shield layer must be connected to an XY pin on the touch controller, which is then driven with the same shield signal used for X and Y electrodes not being measured for self-capacitance.

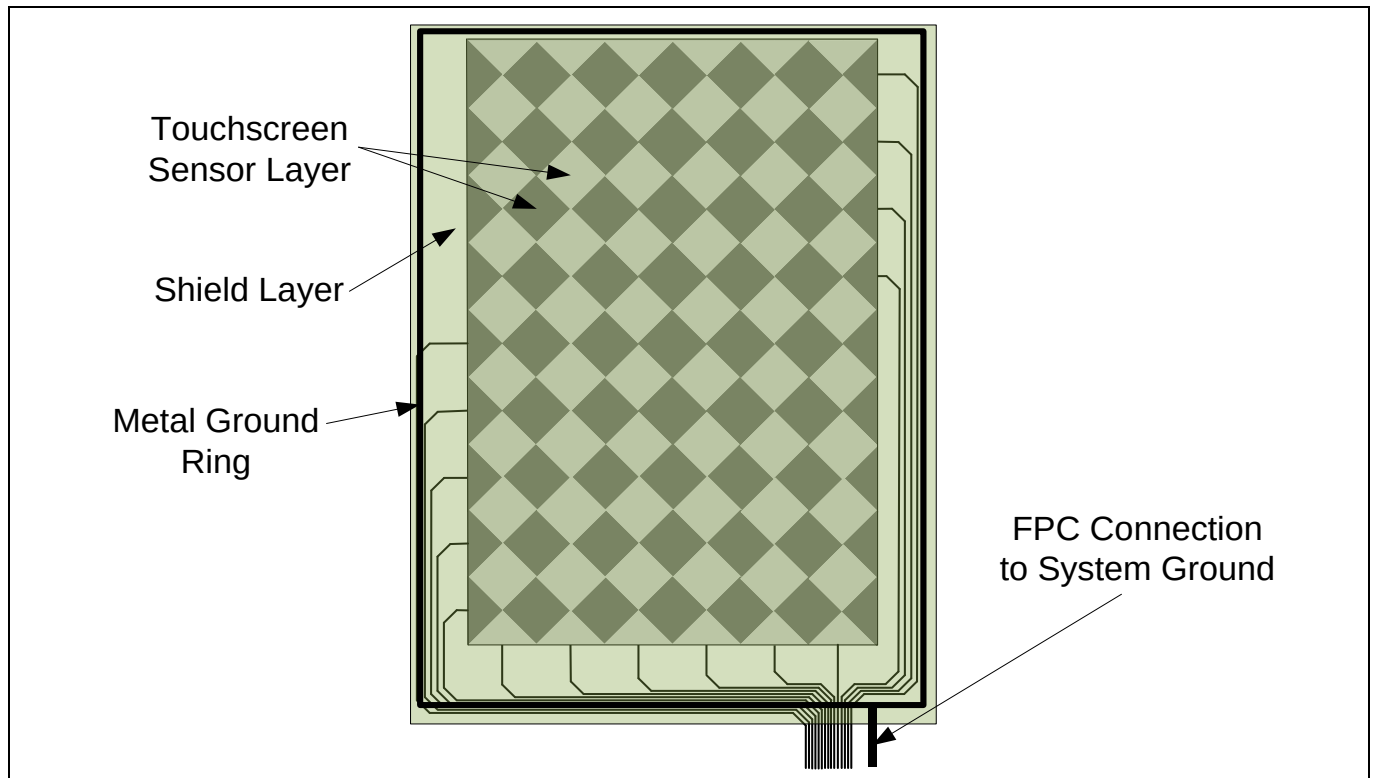


Figure 31 ITO shield and metal ground ring

2.7 ESD and electrical overstress protection

A touchscreen is inherently susceptible to damage from electrostatic discharge (ESD). Protecting against ESD requires design elements on the PCB (Section [PCB design](#)) and on the touch panel.

This section provides guidelines for an ESD protection ring around the periphery of the touch panel.

2.7.1 Panel guard ring guidelines

To help ensure the ESD pulse does not jump directly to the sensor routing, place an ESD guard ring around the perimeter of the touch sensor. The guard ring is a grounded trace around the periphery of the touch sensor panel.

As shown in [Figure 32](#), when a guard ring is present and there is a discharge near the edge of the display, the ESD discharge path follows the guard ring to chassis ground, bypassing the touch controller.

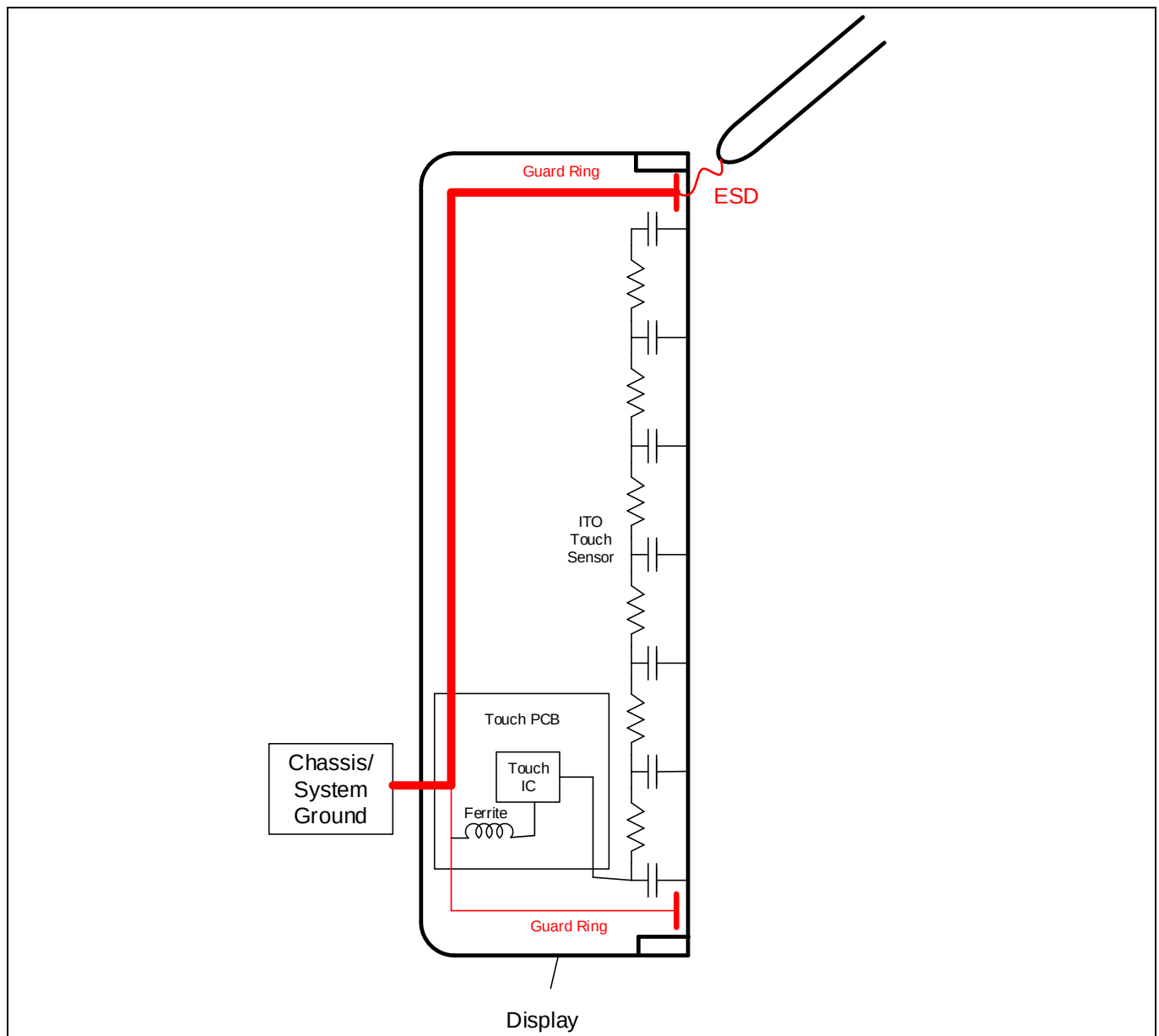


Figure 32 ESD discharge to guard ring during test near bezel

Touch panels

The guard ring must be optimized for the specific sensor. Multi-layer sensors have different requirements from single-layer sensors, but of the following guidelines apply to both types:

- Configure each guard ring with a gap, shown at the top of the panel in [Figure 33](#), to avoid creating a loop antenna.
- The guard ring should be as wide as possible and extend as close to the outer edge of the cover lens as allowed by the selected manufacturing process.
- The resistance of the guard ring should be minimized by use of metallic plating, silver-loaded ink, or other low-resistance material. ITO material is not recommended for the guard ring because of its high resistance.
- The guard ring must not be tied to the touch controller ground (V_{SS} pin of the package), but must instead be tied to the system ground. This directs the energy from any ESD discharge away from the touch controller.
- The guard ring must be connected to the system ground through a low-resistance / low-inductance path. Generally this means that the ring must be metal, not ITO.
- The guard ring should not be routed close to the row or column routing traces to limit parasitic capacitance that would negatively impact sensing.
- Where possible, increase the size of the touch sensor substrate to cover unused surface area of the cover lens, and provide greater surface area for guard ring capacitance.
- When a passive FPC is used to route the TX and RX signals from the touch sensor to another circuit board, the outer edges of that FPC must maintain the guard ring ground, and keep it spaced from the TX/RX traces to prevent ESD arcing into the TX/RX traces. This should especially be followed where a flex circuit is routed through an opening in the chassis.
- In cases where the guard ring must be done in ITO, the resistance of the trace should be less than 5% of the lowest resistance row or column electrode in the panel.

2.7.1.1 Guard ring for single-layer sensors (such as SSD, DSD, SLIM)

For touch sensors where row/column traces are on the same layer as the guard ring, ensure that the parasitic capacitance between the outer panel trace and the guard ring does not add excessive parasitic capacitance (C_p) to the overall design. See [Table 7](#) for maximum parasitic capacitance guidelines.

2.7.1.2 Guard ring for multi-layer sensors (such as MH3)

- In any multi-layer touch sensor, a guard ring is required on both layers.
- The guard ring on each layer should extend as close to the panel edge as allowed by the manufacturing process.
- The guard ring on the conductive layer closest to the cover lens should be maximized.
- The second guard ring (for example, TX layer with MH3) should fill-in any gaps not covered by the guard ring on the upper (RX) layer. This maximizes the capacitance that provides an alternate discharge path.
- The guard ring should be as wide as possible without interfering with the active display area.
- Routing a wide guard ring on the conductive layer closest to the cover lens may extend over the row and column routing traces on a different layer, and increase their parasitic capacitance.
- The best implementation for ESD maintains a solid guard ring over these traces. If they are used to connect to TX electrodes using mutual capacitance sensing, there is no impact on sensing.
- If the electrodes connected by these traces are also sensed for self-capacitance, the impact of additional parasitic capacitance must be evaluated. This capacitance can be reduced by changing from a solid to a hatched fill in areas over these signal traces.

Touch panels

- If the additional parasitic capacitance is still too high with a hatched fill, remove the additional fill over these traces.

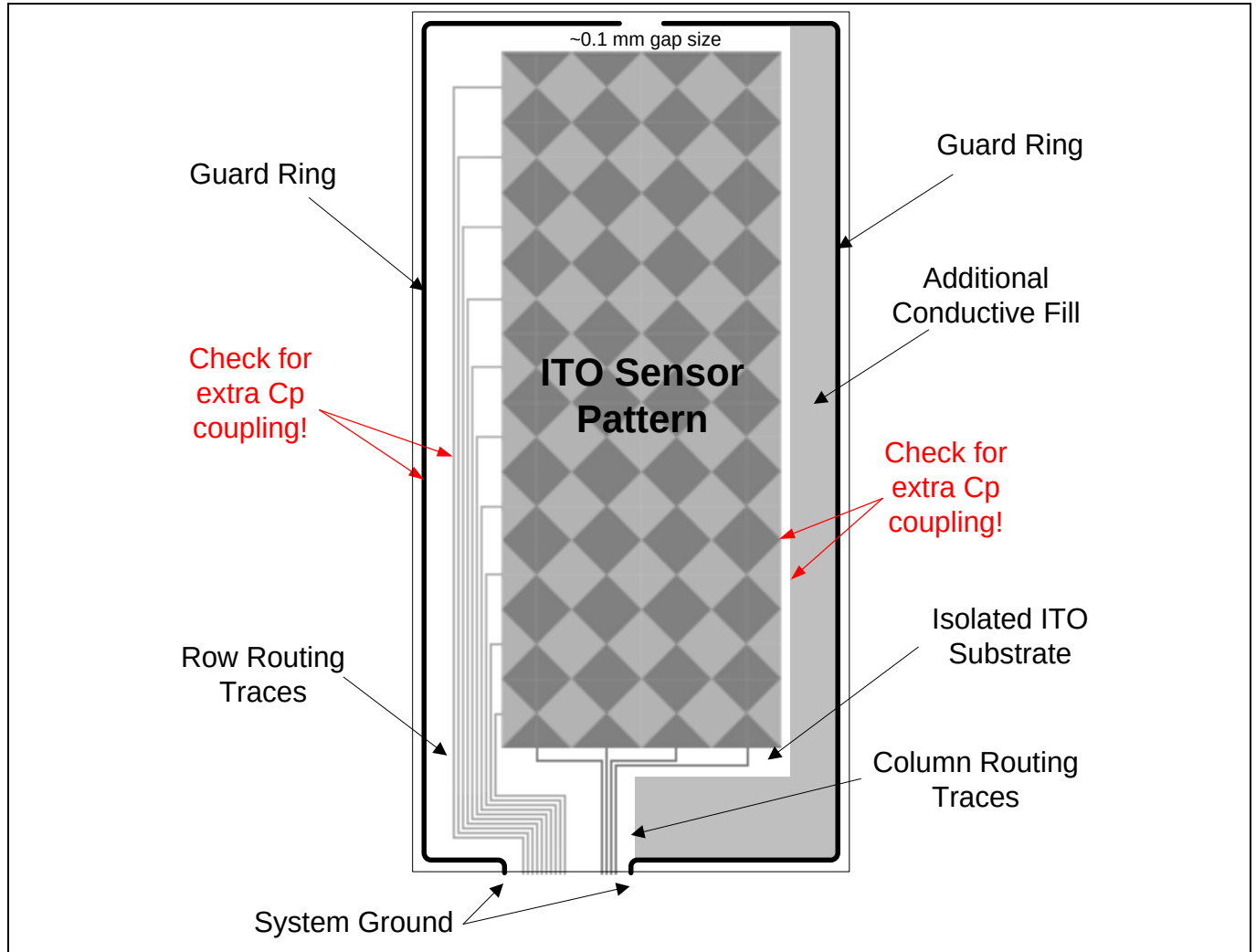


Figure 33 Example of panel with ESD dissipation guard ring traces

3 PCB design

This section covers all aspects of the PCB that holds the touch controller. The sections are:

- Pin assignment - optimize the mapping between the touch controller and the rows/columns of the touch panel.
- Schematic guidelines - guidelines for the schematic design.
- Layout guidelines - list of items for the PCB layout.
- PCB schematic and layout parameters - table showing recommended parameters for components and layout.
- Common PCB layout errors - a list of common errors made in the design of the PCB.
- PCB manufacturing - guidelines for PCB manufacturing.

The datasheet for each touch controller provides information on package footprint, dedicated pin assignments, and required external passive components.

Note: the recommendations in this section represent best practices to be followed to ensure a successful design. In some case, for example protection components for ESD, it may be found in testing that some protection components can be removed to reduce cost. It is therefore recommended that these components be included in the initial design, but removed or not installed in the final design. The intent is to provide guidance as to where to put pads on the circuit board that can be used if passing testing becomes a problem. Testing is often done late in the development process where a change to the circuit board is difficult.

3.1 Pin assignment

Infineon touch devices allow fully customized sensor pin selection, giving hardware designers the flexibility needed to conserve valuable routing areas. However, touchscreen performance can be negatively affected by poor pinout choice. This section discusses methods and tradeoffs for optimizing the touchscreen controller pinout.

3.1.1 Unrestricted pinout parts

All but one of the current touch controllers support unrestricted pinout. The exception is the 103-sensor CYAT817x part, which has 5 TX-only pins (XY98–XY102). Otherwise, any of the XY-pins can connect to either TX or RX electrodes.

3.1.1.1 Select number of pins based on required electrode pitch

The number of electrodes required to achieve a particular pitch (D) with a given panel size is given in [Equation 3-1](#) and [Equation 3-2](#). See section [Electrode pitch \(DX, DY\)](#) for information on tradeoffs involved with pitch selection.

$$n_X = \frac{X_{dimension}}{D} \quad \text{Equation 3-1}$$

$$n_Y = \frac{Y_{dimension}}{D} \quad \text{Equation 3-2}$$

3.1.1.2 Assign TX and RX electrodes to X or Y

CYATx devices almost always have enough RX channels to support one axis. Assign that axis to the RX sensors. If both axes are less than the number of RX channels, assign RXs to the longest axis.

Either with legacy TMAx devices, or in an extreme case, both axes may have more sensors than the RX channels in the target device. In this case, the panel axis assigned to TX/RX electrodes must be determined. All touch controllers have multiple RX channels. While each RX channel can have multiple XY pins that can connect to it, each XY pin can normally connect to only one RX channel. The configuration of pins available for each part can be viewed by selecting the 'Channel View' tab in the 'Project Configuration Wizard' of the Touch Tuning Host Emulator (TTHE).

Figure 34 shows an example for TMA46x where RX channel 0 can be configured to use pins XY07, XY15, XY23, and XY31. Each of those pins can only connect to RX channel 0.

	XY07	XY15	XY23	XY31
Channel 0				
Channel 1	XY08	XY08	XY16	XY24
Channel 2	XY01	XY09	XY17	XY25
Channel 3	XY02	XY10	XY18	XY26
Channel 4	XY03	XY11	XY19	XY27
Channel 5	XY04	XY12	XY20	XY28
Channel 6	XY05	XY13	XY21	XY29
Channel 7	XY06	XY14	XY22	XY30

Figure 34 Example TMA46x TTHE channel view

During a scan of the panel, the touch controller uses a construct called an *RX scan slot* to configure the available RX channels for sensing. The number of RX scan slots needed to address the full panel often determines whether X or Y electrodes are configured as TX or RX. Equation 3-3 and Equation 3-4 give the minimum number of scans required to sense a full panel, depending on the axis assigned to RX (#TX * RX scan slots). To minimize panel scan time, calculate both values and select the RX axis that results in the fewest number of scans. If the number of scans is equal, use either.

$$n_{scans} = n_X * \text{ceiling}\left(\frac{n_Y}{N_{ch}}\right) \text{ (RX on Y)} \quad \text{Equation 3-3}$$

$$n_{scans} = n_Y * \text{ceiling}\left(\frac{n_X}{N_{ch}}\right) \text{ (RX on X)} \quad \text{Equation 3-4}$$

Example TX / RX selection

The Infineon CY3290 EVK TMA400 EVK has 12 sensor electrodes on the X-axis and 21 on the Y-axis. The device had 8 RX channels. The results of [Equation 3-3](#) and [Equation 3-4](#) are:

$$12 * \text{ceiling}\left(\frac{21}{8}\right) = 36 \text{ (RX on Y-axis)}$$

$$21 * \text{ceiling}\left(\frac{12}{8}\right) = 42 \text{ (RX on X-axis)}$$

The optimal solution uses TX on the X-axis (12 sensors) and RX on the Y-axis (21 sensors), which requires 36 measurements instead of 42.

ceiling(x): function that returns the smallest integer not less than x (i.e. round up to the nearest integer).

Figure 35 Example TX / RX selection

3.1.1.3 Select type of routing

The performance tradeoffs of the routing choices are discussed in section [Routing](#). Since all XY pins are capable of being assigned to RX or TX electrodes, the choice of routing does not directly affect the pinout. However, it does affect the ease of achieving the optimal scan performance. See section [Allocate pins](#) for more information on how routing affects the optimal pinout.

When using split routing, splitting the XY pins assigned to TX electrodes usually results in an easier pin allocation than splitting the routing to the RX electrodes.

3.1.1.4 Allocate pins

[Equation 3-3](#) and [Equation 3-4](#) give the minimum number of TX-scans required for a full panel scan. [Equation 3-5](#) gives the total number of TX-scans for a particular configuration. In order to ensure these two equations are equal (and scan time is minimized) each RX channel can have no more than $\text{ceiling}\left(\frac{n}{n_{ch}}\right)$ pins assigned to it (the definition of ceiling can be found in the [Figure 35](#)). [Figure 36](#) shows an example of a non-optimal pin assignment for TMA46x, which requires an extra (unnecessary) RX scan slot. In this example: $\text{ceiling}\left(\frac{16}{8}\right) = 2$. RX channel-7 exceeds the recommended number of pins with three assigned to it. The RX scan slot view shows that Y8 is assigned a third RX scan slot with just one pin, causing a longer panel scan. RX channel-0 has only one RX pin assigned. If electrode Y8 was moved from RX channel-7 to channel-0 then the number of RX scan slots would be reduced from three to two, and total scan time for the panel would be minimized.

$$n_{scans} = n_{TX} * n_{Scan Slots}$$

Equation 3-5

The RX electrodes should be evenly distributed across the RX scan slots. For example, if there are ten RX electrodes but the device has only eight RX channels, the panel scan must be broken into two RX scan slots, with each RX scan slot having five electrodes (evenly distributed). This ensures each RX scan slot has sufficient data to run the common-mode filter. RX scan slots can be observed on the 'Slot' view of the TTHE pin configuration wizard. The RX scan slots for the TX pins are for self-capacitance scans only. However, the RX scan slots for the RX pins are shared between the self-capacitance and mutual-capacitance scans. Self-capacitance and button scans have the same pinout constraints as mutual. For example, three buttons measured for self-capacitance require that each button is assigned to a different RX channel to remain in one RX scan slot.

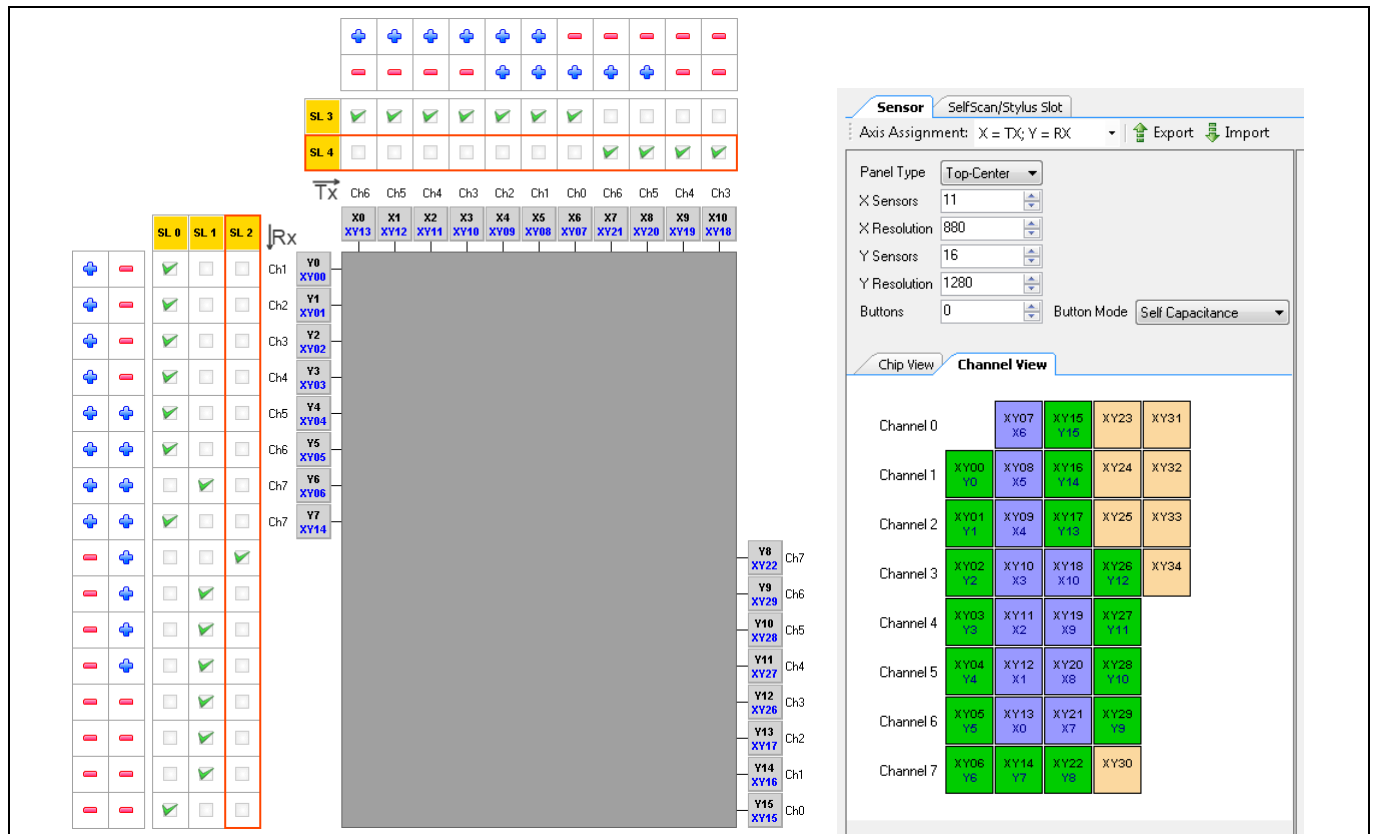


Figure 36 Extra scan slot from sub-optimal number of pins per channel

3.1.1.5 Single routing

For TMAx devices, this routing method starts with the lowest numbered RX or TX electrode. Assign the XY00 pin of the touch controller to the selected electrode and assign the pins incrementally across the RX/TX space.

For CYATx devices, consult the device TRM, as the chip pinout is optimized for split-routed and double-routed designs, as these are the most common. Ensure that the number of RX scan slots is minimized; usually only 1 slot is used. Using the CYAT817x as an example. These devices have up to 103 TX/RX pins and 64 RX-channels. The 64 RX-channels are assigned in order in the middle of the XY-array (for example, XY16–XY79 in the 103-sensor part), so this is an ideal region to assign as RXs. The TXs can then be assigned to the remaining pins at the start and end of the XY-array.

3.1.1.6 Split routing on TX

This is the easiest split-routing method to achieve optimal pin configuration. It is similar to the single routing case.

For TMAx devices, start by assigning the lowest numbered TX electrode to XY00. Continue assigning the first half of the TX electrodes sequentially. When assigning the RX electrodes, use the channel view in TTHE (an example is shown in [Figure 34](#)) to distribute the electrodes evenly across scan slots using the next available XY pins. After filling in the RX electrodes, continue filling the TX electrodes sequentially until all electrodes are assigned.

[Figure 37](#) shows an example of pin assignment with split TX routing.

For CYATx devices, follow the same rules as for single routing (section [Single routing](#)). This is the most common routing method; therefore, these devices have pinouts designed to make routing easier.

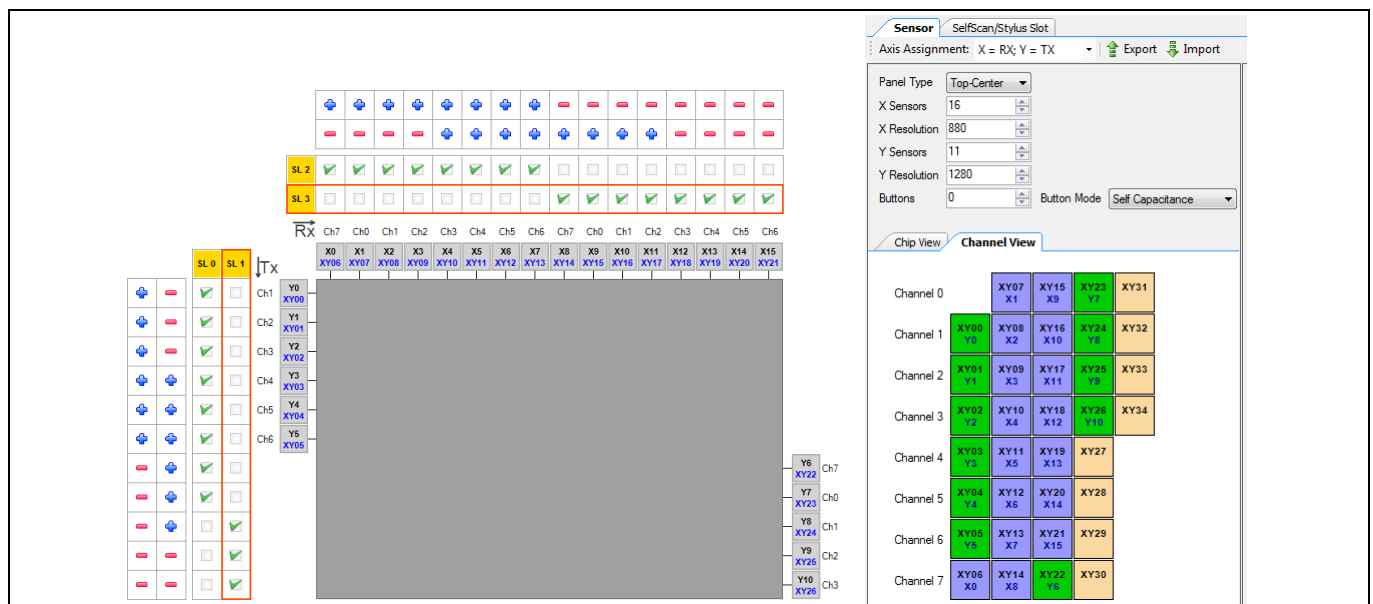


Figure 37 Split TX routing

3.1.1.7 Split routing on RX

When splitting the routing for RX electrodes, obtaining an optimal pin assignment may require skipping XY pins and may be difficult to achieve in a pin constrained system.

For TMAx devices, begin at one end of the touch controller and assign the first group of RX electrodes sequentially starting with pin XY00. Make note of the channel number of the last RX electrode assigned. Now assign the TX electrodes in order immediately following the first set of RX electrodes. Make note of the RX channel number that corresponds to the last pin assigned to a TX electrode. Before assigning the RX electrodes in group two, skip enough pins that the channel number of the first RX electrode in group 2 is the next channel number after the last RX electrode in group 1 (refer to [Figure 34](#)).

For CYATx devices, follow the same rules as for single routing (section [Single routing](#)).

An example of split RX routing is shown in [Figure 38](#). The first set of RX electrodes begins assignment with Channel 1 and ends assignment on Channel 0. In this example, pins XY19–XY23 have been skipped so that the second slot of RX electrodes begins on Channel 1.

There may be cases where the pin count of the project does not allow assignment with skipped pins. In this case, the routing of the second set of RX electrodes should begin immediately after the last TX electrode assigned. This is shown in [Figure 39](#). If pins had been skipped, the second set of RX electrodes would have started pin assignment on XY33, leaving no space for the required pins. Note that the RX electrodes have been evenly distributed across the scan slots by manually moving the checks to the desired box in the scan slot view on the left.

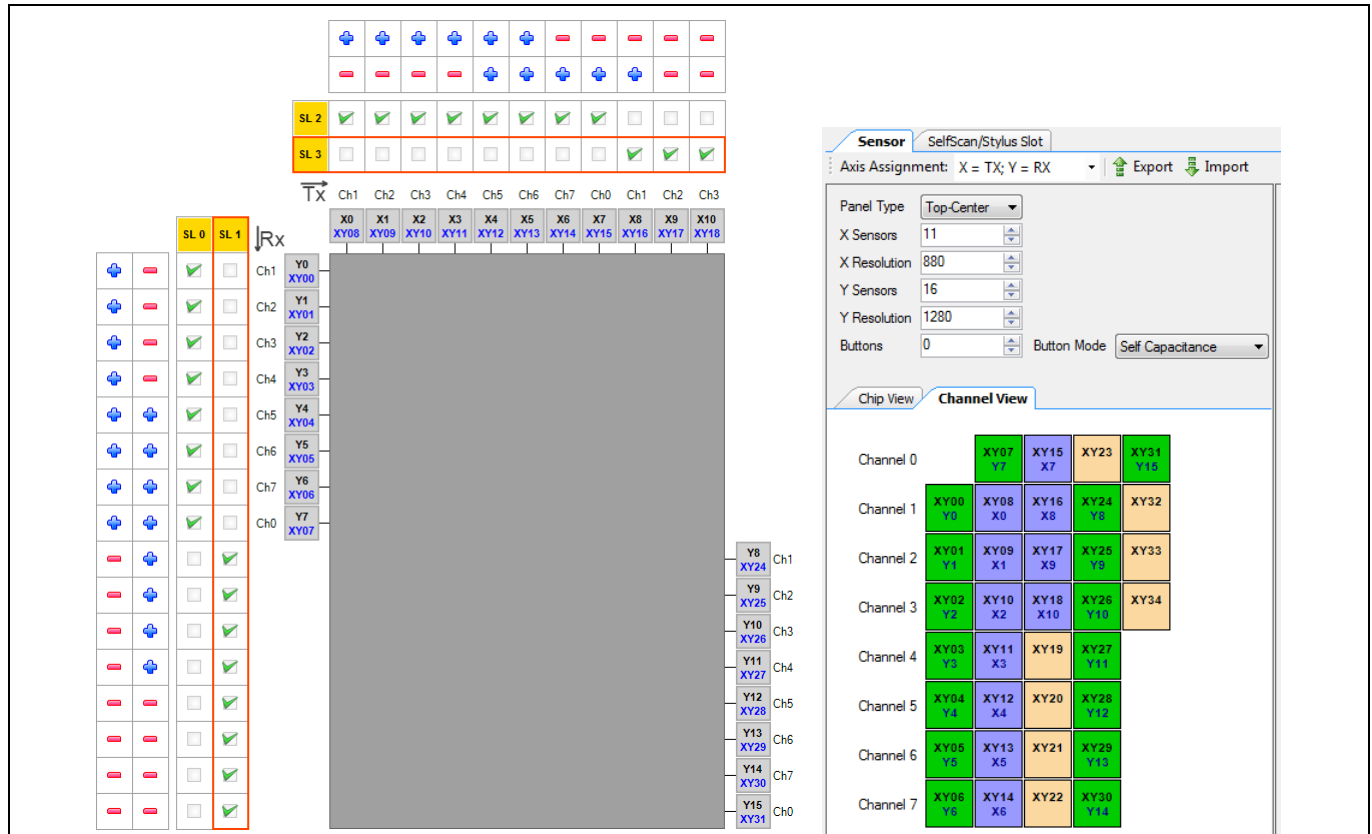


Figure 38 Split RX routing with skipped pins

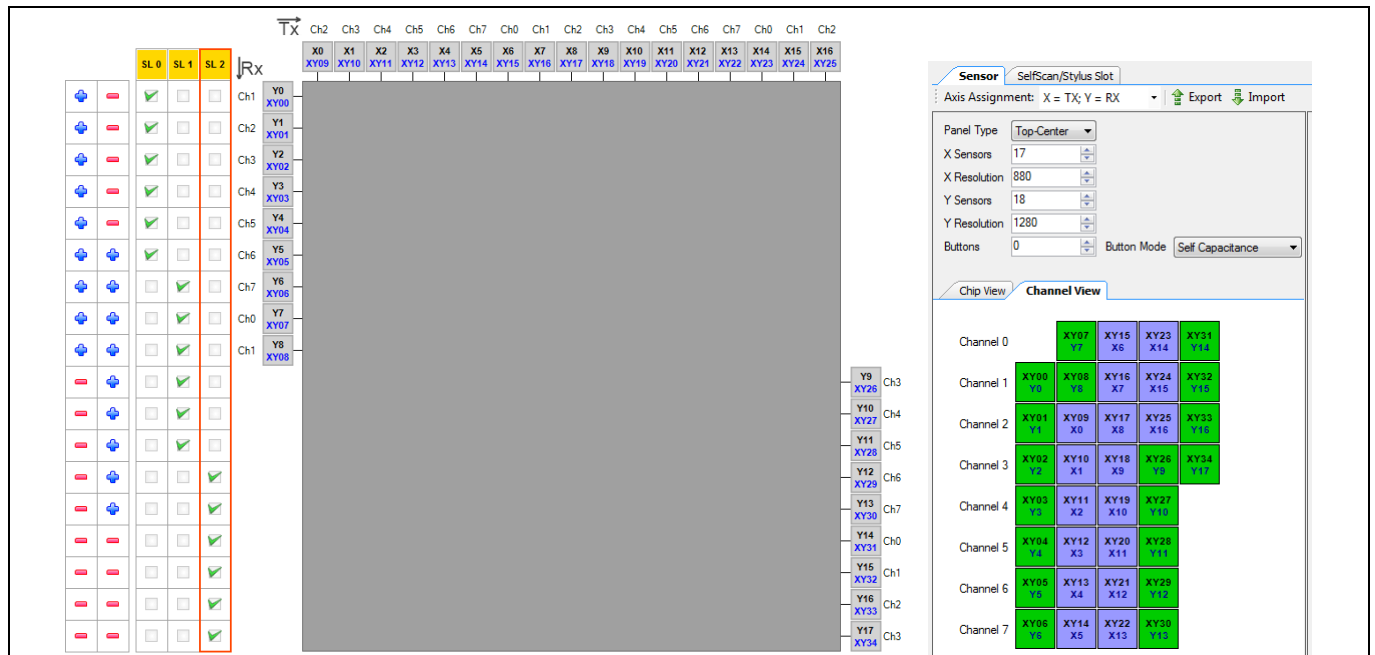


Figure 39 Pin constrained split RX with no pin skipping

3.1.1.8 Double routing

Double routing a panel results in nearly identical pin assignment to single routing. Allocate the RX and TX into two separate groups. Start at one end of the panel and assign the X-axis electrodes first and the Y-axis electrodes second. Vias are required if traces must cross, which is generally the case when the X-axis is double routed. If only the Y-axis is double routed, then the traces for the top can be routed along the side opposite to the X-axis traces.

3.1.1.9 Multichip routing

Multichip designs are recommended to double route the longest signal path, which would be the TX electrode. Depending on the resistance of the TX electrode, Infineon may recommend splitting the TX electrode in the middle (only applicable to single slave designs), see section [Double split TX routing](#).

Split TX routing is similar to TX double routing, but with the TX electrodes cut in the middle of the touchscreen. Each side of the TX electrode is driven by a different touch controller.

The multichip solution has one master device and one or two slave devices. Pin assignment should be done for each device separately by choosing the device from the dropdown menu and mapping panel electrodes (on the right) to the appropriate device pins (on the left). If the multichip solution consists of three devices (not applicable to split TX routed designs), the device in the middle (master) can have TX electrodes assigned to both the left and right columns. However, it cannot be assigned to both TX electrodes in the same row to avoid a TX driving conflict. THE pin configuration with “Split TX” set to “Disabled” is shown in [Figure 40](#).

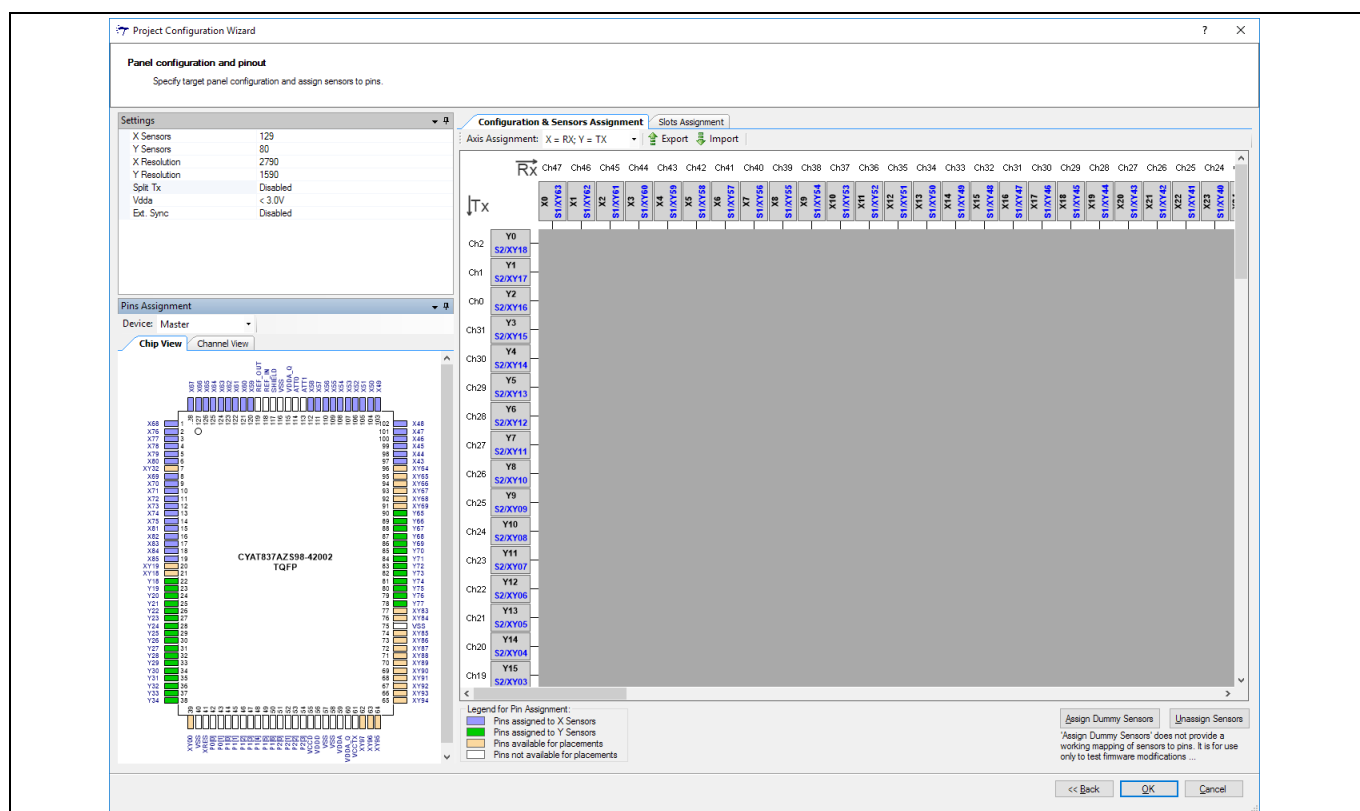


Figure 40 Multichip pin assignment without split TX routing

PSoC™ Automotive Multitouch touchscreen controller

Module design best practices

PCB design

TTHE pin configuration with “Split TX” set to “Enabled” is shown in Figure 41. With the split TX feature enabled in the TTHE configuration, a second TX column is present in the “Configuration & Sensors Assignment” tab. The two columns correspond to the left and right side of the panel. The device on the left must have all of its TX electrodes assigned to the left TX column. Equally, the right-side device must map to the right TX column.

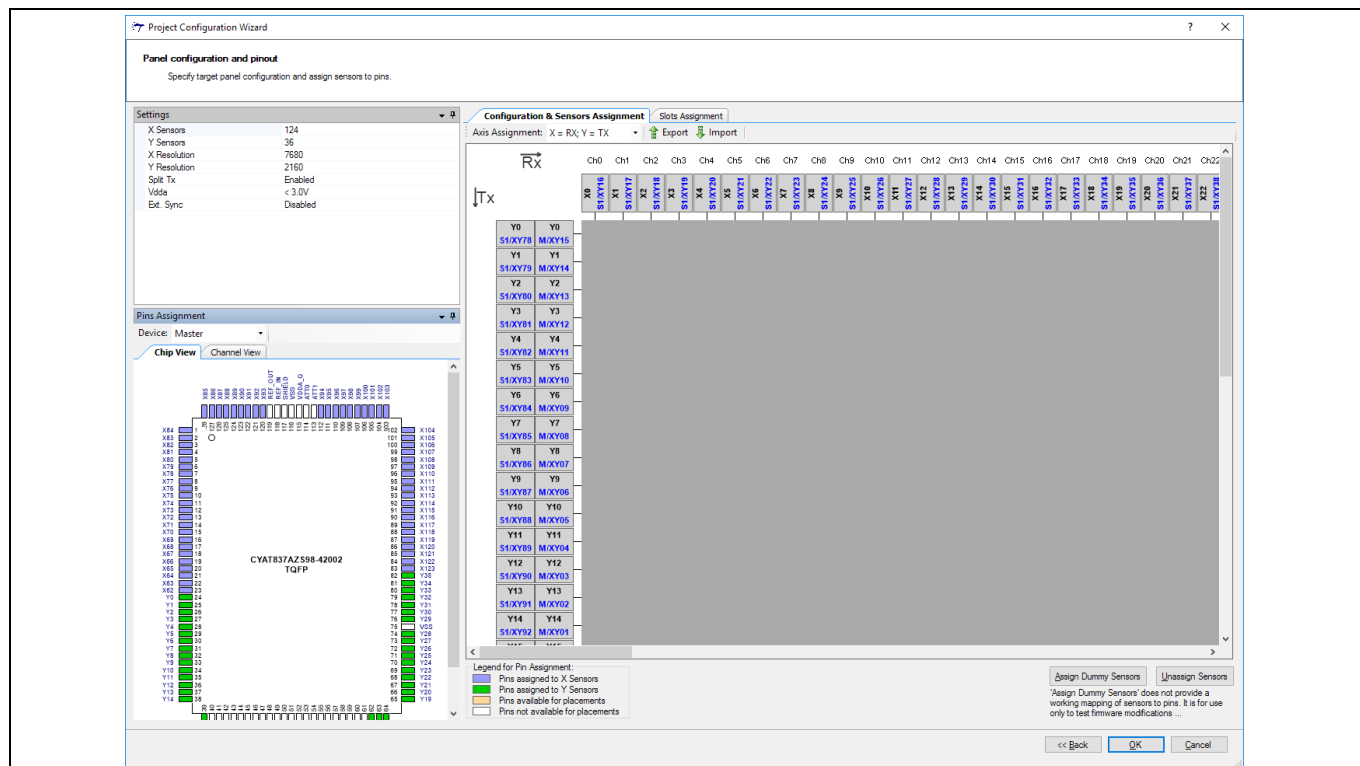


Figure 41 Multichip pin assignment with split TX routing

For either multichip routing method, the maximum number of RX electrodes per device is 64, and they must be configured into a single scanning slot.

3.1.2 Restricted pinout parts

Only 2 sets of devices have a restricted pinout. For all other devices, any sensor pin can be assigned as a TX or RX.

The non-automotive TMAx family of parts has a restricted pinout. These devices are no longer recommended for new designs.

The 103-sensor CYAT817x part has 5 TX-only pins (XY98–XY102). The other 98 sensor pins can connect to either TX or RX electrodes. The 5 TX-only pins can only be used for mutual-cap TX, they cannot be used for mutual-cap RX or self-cap.

3.2 External oscillator

If an external oscillator is required by the design, then the emissions should be considered. External oscillators can cause significant emissions in the 30 MHz to 1 GHz range. It is therefore recommended to use an external oscillator with a differential signal output to help reduce the emissions.

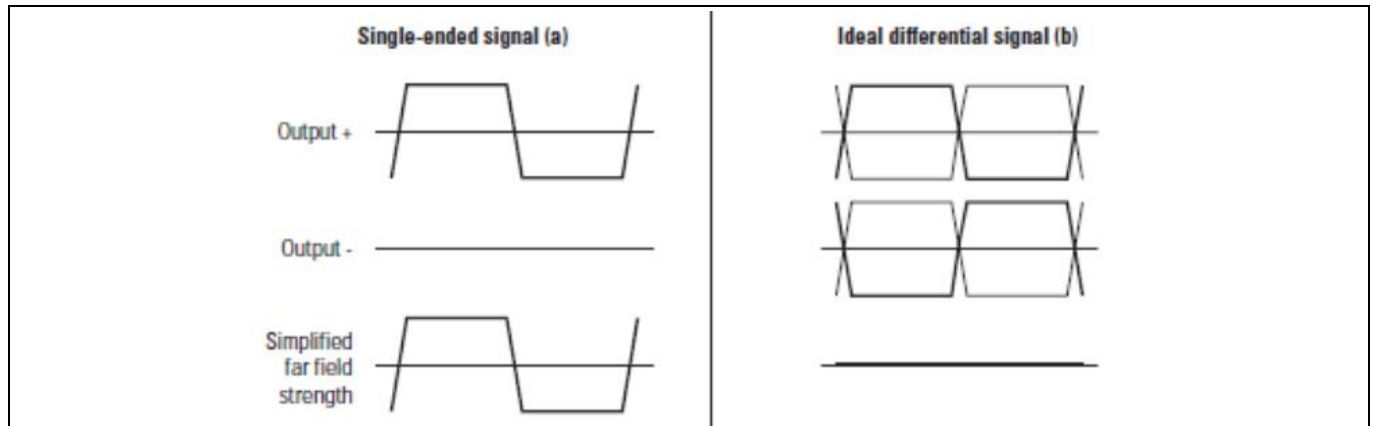


Figure 42 External oscillator signal output

Follow these design considerations:

- Minimize output impedance mismatch in signals of every pair.
- Tightly couple and symmetrically route signals in every pair.
- Closely match the length of each signal trace in every pair to minimize output skew.
- Route the clock signals on internal layers and increase separation between adjacent clock traces that are not pairs to minimize coupling.
- Follow proper termination guidelines to eliminate reflections:
 - AC termination resistors should be close to the source output pins.
 - DC termination resistors should be close to the receiver input pins.

The physical layout of the differential clock signals should also be well controlled. Following are the recommendations for the trace/space distances. With reference to Figure 43, maintain the following relationships:

- $S < h$ – Space between differential pairs should be less than the distance between the neighboring layers.
- $W > S$ – Clock trace width should be greater than the space between the pair.
- $X > 2W$ – Space between a clock trace and any other signal should be greater than 2x the clock trace width.

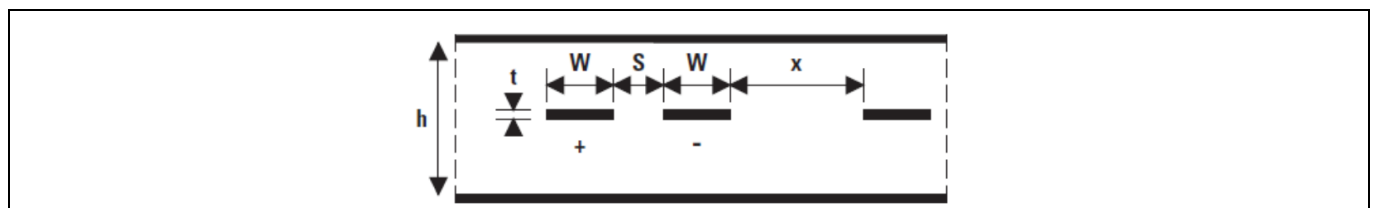


Figure 43 PCB cross-section for external oscillator differential pair layout

For more information see a useful application report from TI at:
<https://www.ti.com/lit/an/snaa217/snaa217.pdf>.

3.3 Schematic guidelines

This section gives guidelines for the schematic design, including component and circuit selection.

3.3.1 General guidelines

- The device datasheet will give basic circuit examples, including bypass capacitors and filters, for the touch controller device. One implementation is shown in [Figure 45](#).
- All QFN packages with an EPAD require the EPAD to be grounded. Connect the grounded EPAD to the V_{SS} pins on the touch controller.

3.3.2 Power bus guidelines

3.3.2.1 ESD considerations

To protect the power buses from voltage swings caused by ESD transients, the power pins need to be clamped below the absolute maximum voltages allowed for the IC (specified in the device datasheet). As noted in section [ESD simulator](#), power buses with reasonable amounts of bulk capacitance present are already protected by the ability of the capacitor to absorb the charge of the ESD pulses. In most cases, ensuring sufficiently low dynamic impedance of a power bus is sufficient to protect the device from any overstress.

Many touch controllers contain two additional power domains (V_{CCTX} and V_{CCD}) that (depending on configuration) do not have sufficient capacitance to absorb these transients directly. For those domains, additional components may be needed to protect the device.

The V_{CCTX} supply pin can be configured as a direct input, or as a filter and reservoir node of the on-die TX charge pump. When the V_{CCTX} supply is provided from an external regulated source, the presence of sufficient bulk capacitance can prevent this node from moving when ESD events occur. But when the charge pump is used, a much smaller capacitor (typically 220 nF) is present. While making this capacitor larger can improve tolerance to ESD, it can also degrade overall device performance by increasing the time needed to charge this capacitor following deep sleep intervals. A larger value is recommended for projects that do not require deep sleep.

For devices having a separate V_{CCD} domain, when the V_{DD} supply to the device is ≥ 2.0 V, the V_{CCD} pin is an output of the core regulator and provides a 1.8 V nominal supply to the device's core logic and memories. In this configuration, the recommended capacitance on the V_{CCD} regulator supply is only 100 nF, which is insufficient to absorb external transients injected into that supply.

V_{CCTX} and V_{CCD} are not normally accessible to direct outside ESD events, but can be impacted through the touchscreen panel.

In an environment where significant charge passes through the touchscreen electrodes, that charge can move the potential of the device substrate relative to these weak internal supplies. If the coupling through these electrodes is sufficiently high (relative to the alternate paths for charge), additional protection may be required on these power nodes.

When developing a design, if sufficient board area is available, it is recommended that board locations be provided for a transient voltage suppressor (TVS) on the V_{CCTX} supply (D1 in [Figure 45](#)). To select the TVS diode, the following guidelines should be followed:

- The TVS should be unidirectional (not bi-directional). A unidirectional TVS protects against both positive and negative voltage excursions. As an example, a unidirectional 5 V TVS protects against voltages above 5 V and below zero, while a bi-directional version protects against voltages above 5 V and below -5 V.
- The TVS response time should be < 1 ns (most Zener and avalanche diodes are not acceptable).

PCB design

- The TVS must support operation at the normal working voltage of the system. Note that touch controllers are available with V_{CCTX} supplies operable at different voltages (10 V or 5 V nominal, and some programmable to other regulation levels). This operating voltage is referred to in most datasheets as the “rated standoff voltage,” or the “reverse working voltage”. This is the voltage at which the TVS remains high impedance and does not draw current. With the touch controllers, this voltage must be greater than the TX voltage, but it should be as low as possible. The objective is to select a TVS that minimizes the excess voltage to which the controller is exposed.
- The capacitance of a TVS only matters if the frequency on a signal line is high. In this usage the TVS is protecting a power supply, so the capacitance does not matter.
- The TVS diode should be placed as close as possible to the touch controller.
- The Infineon ESD114-U1-02EL was used in qualification testing by Infineon and provided acceptable protection.

3.3.2.2 V_{DDD} ramp-up and ramp-down

The VDDD ramp-up and ramp-down specifications in the relevant device datasheet must be met. If they cannot, assert XRES during ramp-up and/or ramp-down where the requirements cannot be met. This is especially important for projects using base firmware prior to build 1123934 (TBD for multichip).

3.3.3 Single-point ground guidelines

The ground for the touch controller must be connected to the system ground at a single point. The ground on the touch controller side of the single-point connection must include the EPAD on the bottom of QFN packages.

Note: In some packages, the EPAD is the *only* device V_{SS} connection.

- The layout of the FPC includes a series ferrite in the V_{SS} path (L1 in [Figure 45](#)). The ferrite should be placed such that there is only a single point connection between V_{SS} and the system ground, through the ferrite.
- The mechanical design of the system affects the impedance of the ground path to the touch controller; this determines the magnitude of the spectral components within the ESD signal that reach the controller. It is recommended that testing start with a ferrite with its maximum impedance in the 1 GHz range. The peak impedance should be hundreds of ohms or more. See [Figure 44](#) for an example of impedance vs. frequency curves for a family of ferrite beads. Other ferrites, with different peak impedance and with maximum impedance at different frequencies may be tested to see which is most effective in preventing ESD side effects. If testing shows a ferrite is not needed, then the ferrite can be replaced with a 0 Ω resistor.
- The ferrite should have its peak impedance in the 100 MHz to 1 GHz range.
- The impedance of the ferrite over the 100 MHz to 1 GHz range should be more than 100 Ω .
- The Murata BLM18BD252SN1D was used in qualification testing by Infineon and provided acceptable protection.

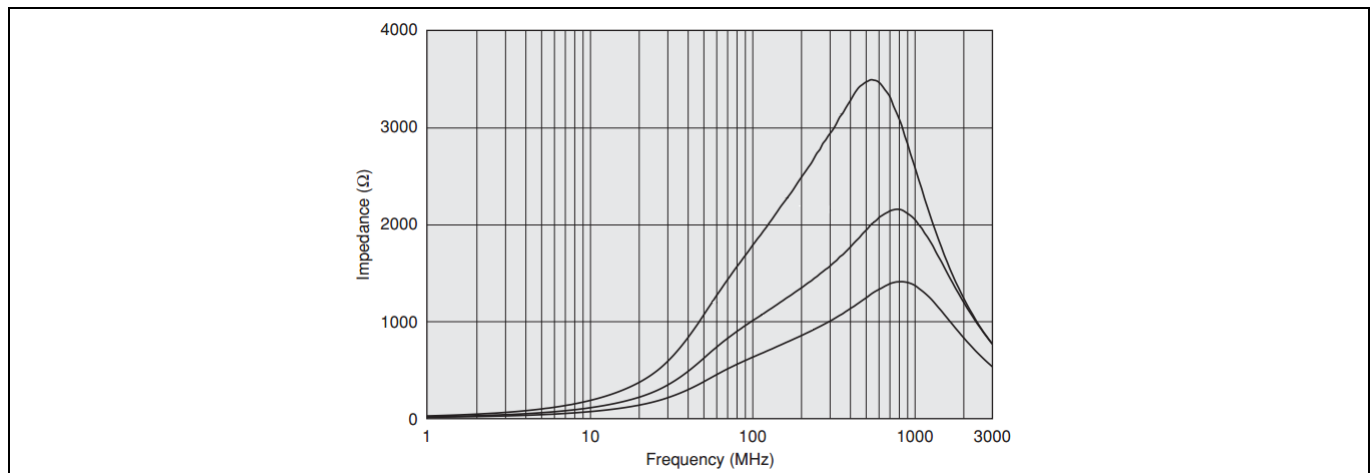


Figure 44 Typical impedance vs. frequency curves for ferrite beads

Although not common, the ferrite bead between system and device ground may in some cases have an adverse effect on I²C and SPI communication. The failure mode that has been observed in some systems is the communication lines lock up such that the device can no longer respond to the host. In this case, the system resets or power cycles the touch controller when it identifies the halted or stuck condition. This reset may result in breaks in finger tracking during electrical fast transient (EFT) testing.

If this is found to be the case, then a lower-impedance ferrite should be selected, and the tradeoff between ESD and EMI will need to be analyzed to find a suitable ferrite.

- Connecting the EPAD to a conductive stiffener that is then connected to the chassis (or system ground) is not recommended. This creates a low-inductance path between the system ground and V_{SS} which violates the single-point ground and bypasses the ferrite bead.

3.3.4 TX/RX line guidelines

Infineon recommends series resistors in all TX and RX lines between the touch panel and controller. The purpose is to reduce the effect of radiated electromagnetic interference (EMI), which could otherwise induce signals on the lines and result in false touches. The recommended resistor value is 2 kΩ for all Infineon controllers. The resistors should be placed as close as practical to the controller. Resistors R1 and R2 in [Figure 45](#) show examples of these resistors.

Adding these resistors increases the panel time constant, and therefore might have a small impact on the time it takes to scan the panel. For most projects, Infineon will have already simulated the panel speed, and can calculate the impact on scan time.

3.3.5 Communication line guidelines

To prevent signal corruption from electrical fast transient (EFT) burst noise, it is recommended to connect a TVS diode from each communication line to system ground. The pads for these TVS diodes can be put on the PCB and when the system is tested it can be determined if the TVS diodes are needed for the design. See D2 and D3 in [Figure 45](#).

Although not common, in some cases the I²C or SPI communication may generate EMI. If this is the case, then series ferrites should be installed in series with the communication signals. The ferrite values should be chosen so that they do not affect the normal communication. L3 and L4 in [Figure 45](#) are examples of these components.

The interrupt line should have an external pull-up/down. If relying on the touch controller's internal pull-up/down, it should be noted that the internal pull-up/down will be disabled (touch controller pin will be in the Hi-Z state) when the touch controller is in reset. This includes an approximate 5ms period, 20ms after the exit bootloader command is received.

The error pin should be connected to the host, and monitored. If a physical connection to the error pin is not possible, the error register can be read. For most devices reading the error register once after device startup is sufficient. Refer to the individual device's TRM.

3.3.6 Typical schematic with protection circuitry

Figure 45 shows a typical circuit for power and ground for a touchscreen controller.

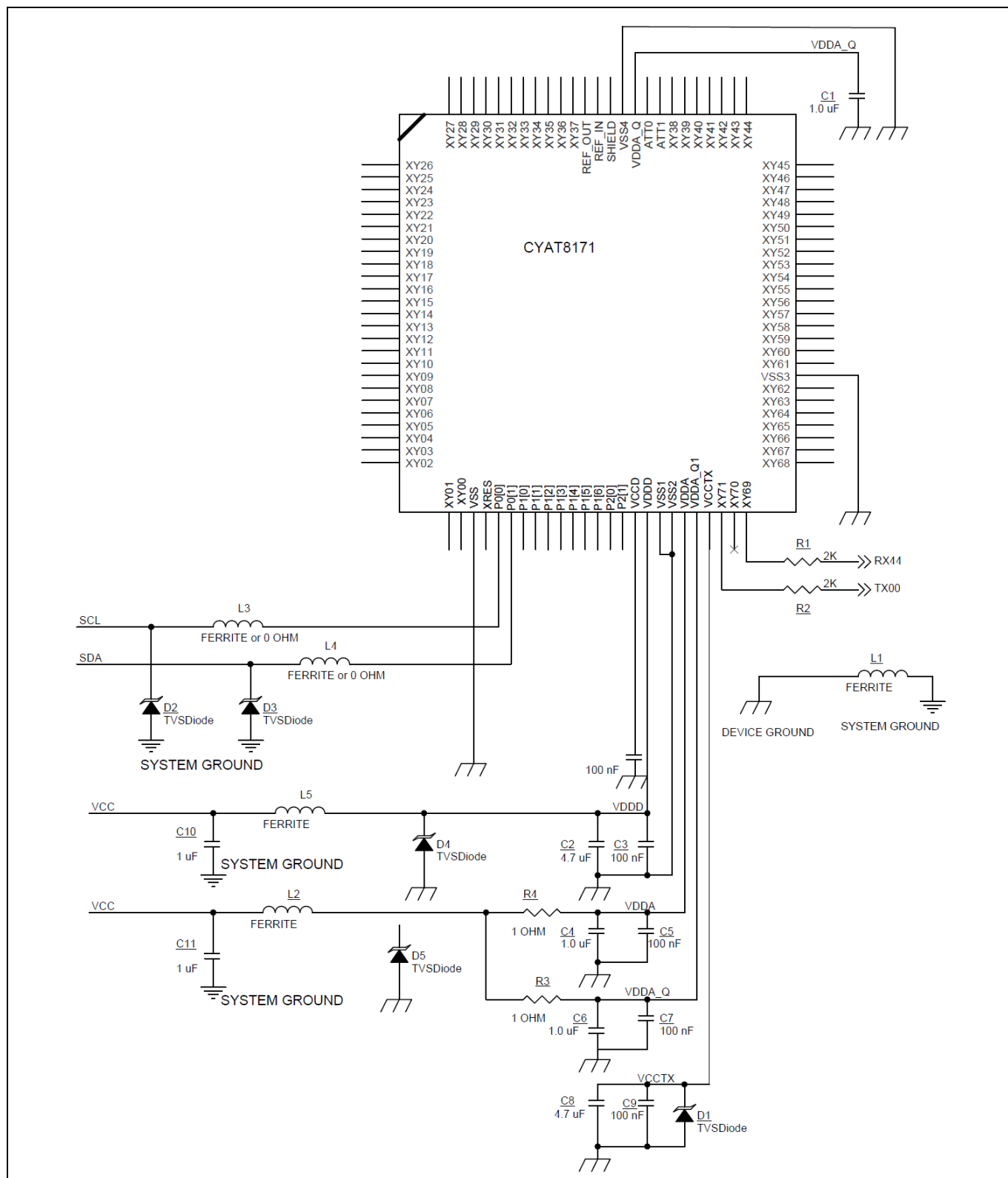


Figure 45 Typical schematic diagram

3.3.7 Additional ESD options

If system testing results in ESD failures, even with the countermeasures above, then additional countermeasures in this section can be applied. Alternatively, the PCB can be initially designed to accept these components, and if not needed then the components can be left off the board.

- Add Pi filters with two decoupling capacitors (around 1 μ F) and a ferrite bead to work as EMI chokes on the V_{DD} and V_{DDA} paths (L2, L5, C10 and C11 in [Figure 45](#)). Note the capacitance of ceramic capacitors can decrease dramatically when a DC voltage is applied; therefore, it's important to ensure that the right capacitance is used with bias voltage present. If testing shows that a specific Pi filter is not needed, the ferrite(s) can be changed to 0 Ω resistors and the capacitors can be changed to "Not Installed".
- Add unidirectional transient voltage suppressors (TVS) between each of the device supply voltages and the V_{SS} pin of the touch controller (D4 and D5 in [Figure 45](#)). The TVS must be unidirectional and at an appropriate low voltage for the system design. If testing shows that a TVS is not required, it can be changed to "Not Installed". To properly handle ESD events, the response time of the TVS must be < 1 ns. The TVS should be placed as close as possible to the touch controller.

3.3.8 Unused pins

Leave all unused pins unconnected.

3.3.9 Ground pins

The touch controllers only have one type of ground (V_{SS}). The power pins are divided between analog (V_{DDA}) digital (V_{DD} , V_{CCD}), and TX driver (V_{CCTX}) supplies. Inside the touch controller the ground nets are split between analog and digital and the two are connected with a single point inside the controller. Because the analog signals are not sourced or referenced outside of the controller, there is no need for an external single point ground (with the exception of ground segmentation for ESD, as documented in section [Single-point ground guidelines](#)).

3.3.10 Provision for power cycling

In extreme cases, the ESD event may be enough to put the touch controller into a latch-up state. For both latch-up and Electrical Overstress (EOS) the best approach is to prevent the energy from going through the device. Latch-up is a recoverable condition, where cycling power (turning power off and then back on) to the device clears a latch-up condition. While all touch controllers have a reset input (for example, `XRES`), assertion of reset does not clear a latch-up condition. This is because the thyristor is made of parasitic transistors there is no power-on way to clear this latch.

To enable system-level recovery of latch-up events, designs using touch controllers should allow the power to the touch controller to be switched on and off independent of the host processor. This can be done with a dedicated channel of a PMIC (Power Management IC) or by inserting a power PFET in the V_{DD} and V_{DDA} supplies.

The release of the latch-up requires a complete power cycle of the device. If the system hardware allows power to be cycled to the touch controller, then this can be automatically triggered by the host. For example, The Touch Driver for Android (TTDA) periodically checks for the absence of activity from the touch controller due to an unexpected halt of communication and performs a power cycle.

If the `XRES` pin is left HIGH when V_{DD} is removed, then the voltage on the `XRES` pin can back-power the device through the protection diodes. This is true for any digital pin (other than I²C pins) that is externally driven high. This back-powered voltage could keep the latch-up from being cleared. To prevent this from happening `XRES` (and any other digital inputs controlled by the host processor) should be driven LOW before removing power, or be biased through a high enough impedance to not sustain latch-up.

This is not an issue for the I²C pins because they are designed to allow the I²C bus to be operating while the device is not powered.

3.3.11 Debug connector/adaptor PCB

It is strongly recommended to provide a debug connector. In production, it is acceptable for this connector to be marked as “no stuff”. The connector should include the following touch controller signals:

- Power (V_{DDD} , V_{DDA} , and Gnd)
- Programming pins (ISSP or SWD)
- Communication pins (I2C or SPI)
- Reset (`XRES`)
- Debug-enable signal to the host (optional - see below)

There must be some method to allow the debug connector to be the I2C/SPI bus master. This can be either a software command to the host, or a physical signal on the debug connector. A simple example is to route the debug connector's debug-enable signal to the host's reset pin.

If space is available, the debug connector should be a 8x2 100 mil connector, compatible with the Infineon Touch Tuning Bridge (TTBridge). Usually space is not available, so an adapter PCB should be used to convert the debug connector to the TTBridge connector. The programming pins require external pull-ups. These are provided on the TTBridge, or they can also be mounted on the adapter PCB.

3.4 Layout guidelines

- Route digital communication traces perpendicular to sensor traces.
- Ground considerations:
 - The routing of the signal lines (programming pins, communication pins, etc.) should be done so that their coupling to the system ground is minimized (although ground is preferable to other traces). Long sections with traces parallel to the system ground should be avoided, and the minimum trace to ground spacing (see item 19 in [Table 14](#)) must be observed.
 - All QFN packages with an EPAD require the EPAD to be grounded. Connect the EPAD to the VSS pins on the same PCB/FPC layer rather than connecting through vias on different layers.
 - Insert a grounded shield trace between the communication and sensor traces when routed in parallel. Connect this trace to system ground, not the chip's VSS pin.
- To avoid interference from digital communication signals, coupling capacitances between the digital communication traces and the ITO RX sensor traces should be minimized. In addition, the speed of the digital clock or data edge should be kept as slow as possible. These requirements are intended to minimize the coupling capacitance between communication and sensor traces.
- The RX electrodes and corresponding traces are most susceptible to noise. Follow these rules to avoid coupling noise into these signals:
 - Shield the RX traces from other traces if routed in parallel.
 - Route the RX traces 90 degrees to other noisy traces.
 - Route the RX traces farthest from the display.
- TX traces are relatively low-impedance and do not need to be shielded. TX traces may be routed in parallel with other TX traces, but if TX traces must cross RX traces the intersection angle should be as close to 90 degrees as possible.
- Place external signal conditioning components as close to the touch controller as possible. This requirement is critical to reducing common-mode noise coupling.
- The placement of the voltage reference bypass capacitor is important to achieving good noise performance. Follow these guidelines:
 - The distance of the capacitor to the designated pin should be no more than 3 mm.
 - Minimize the length of the return path to the V_{SS} pin or to the grounded EPAD.
- In QFN and MLF packages:
 - Connect the EPAD to a solid-filled pad on the same layer that the QFN package is mounted.

3.5 PCB schematic and layout parameters

Table 5 provides numerical specifications for the most important layout parameters. The distance specifications assume ± 0.1 mm tolerance unless otherwise noted.

Table 5 Layout parameters

Item	Description	Specification
Passive components	Distance from the passive component edge to the pin	Max. 0.5 mm
	Resistor tolerance	Max. 10%
	Capacitor working voltage rating for non-10 V chips	Min. 6.3 V
	Capacitor working voltage rating for 10 V chips	Min. 16 V
	Capacitor voltage coefficient	Capacitance value must consider DC voltage; check capacitor datasheet
	Capacitor dielectric temperature coefficient	X5R, X7R or equivalent
TX/RX electrode separation	Distance from TX trace to RX trace	Min. 0.4 mm
	Distance from I ² C/SPI/INT to TX or RX	Min. 0.4 mm
V _{DD}	Width of V _{DD} trace	Min. 0.4 mm
	Distance from V _{DD} to COMM/INT	Min. 0.4 mm
V _{SS}	Width of V _{SS} trace	Min. 0.4 mm
FPC tail length	Minimize tail length to preserve signal integrity and minimize coupling to external noise sources	Max. 50 mm
FPC/PCB trace width	Width of trace unless otherwise specified above; ± 10 μ m tolerance	Typical 50 μ m
FPC/PCB trace separation	Distance between traces unless otherwise specified above; ± 10 μ m tolerance	Typical 50 μ m

3.6 Common PCB layout errors

3.6.1 Parallel TX and RX trace error

TX and RX traces cannot be parallel to each other on the same or different layers, unless separated by a system ground isolation trace or plane. This prevents signal cross-talk from TX to RX. Two examples of such errors are shown in Figure 46 and Figure 47.

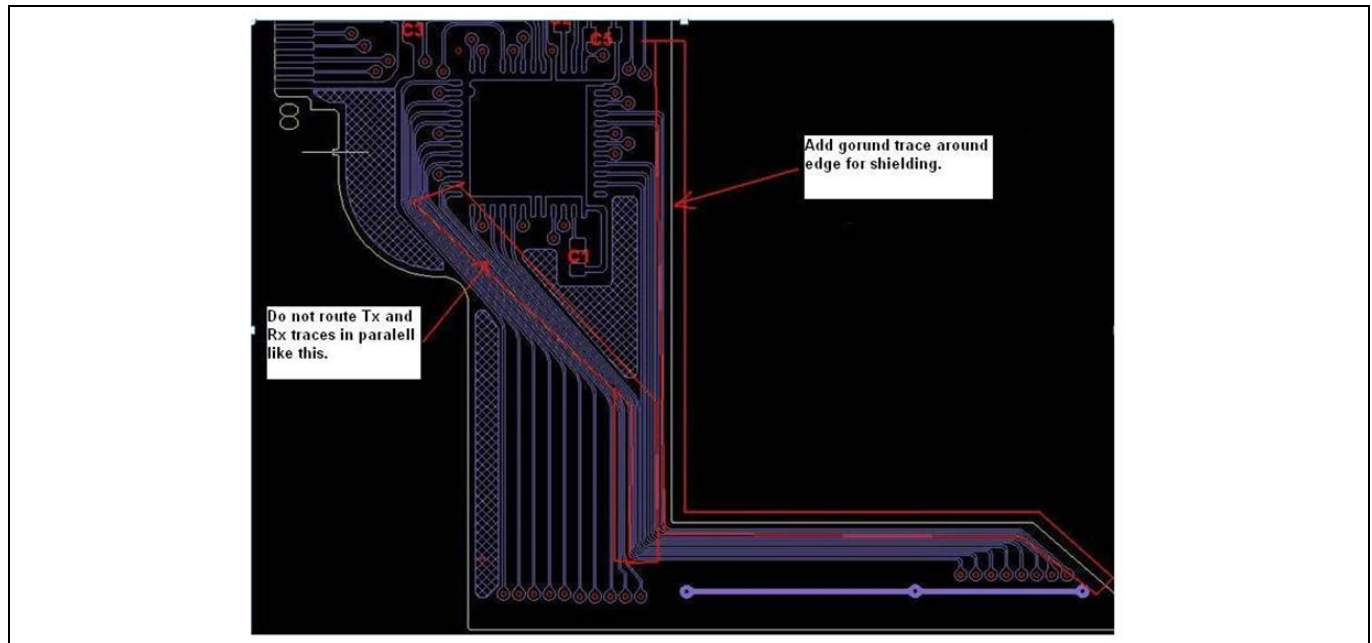


Figure 46 Parallel TX and RX trace error - same layer

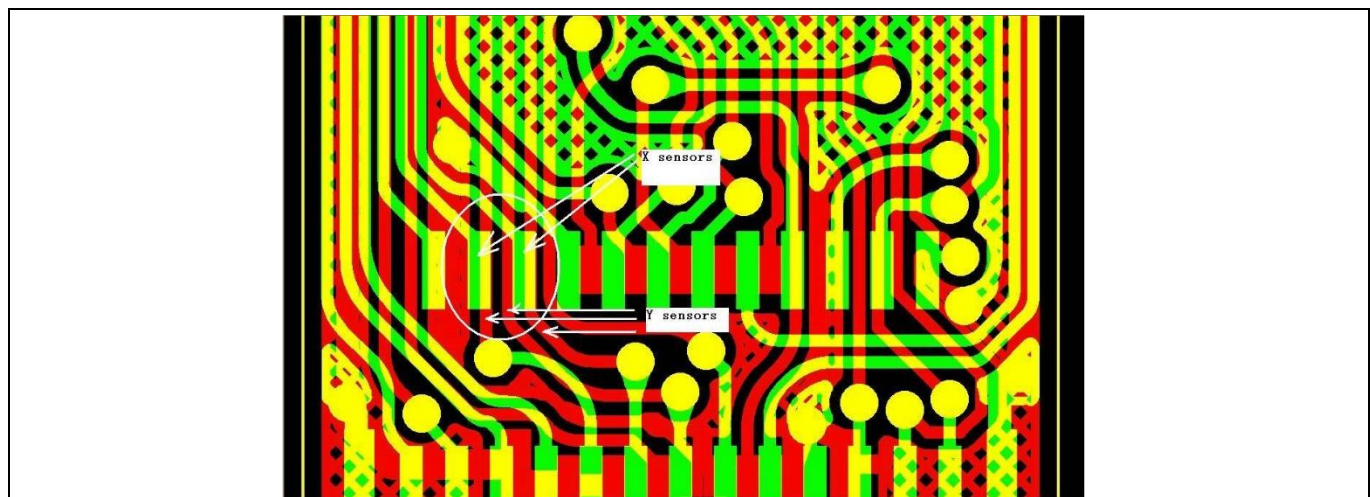


Figure 47 Parallel TX and RX traces error - different layers

To correct these errors, do one of the following:

- Insert system ground trace between the parallel TX and RX traces.
- Rearrange the TX and RX traces on different layers so they are perpendicular to each other.

3.6.2 Parallel electrode (TX or RX) and communication signal trace error

Digital communication traces running parallel to sensor traces can inject noise in both the sensor and communication signals. Two examples of this error are shown in [Figure 48](#) and [Figure 49](#).

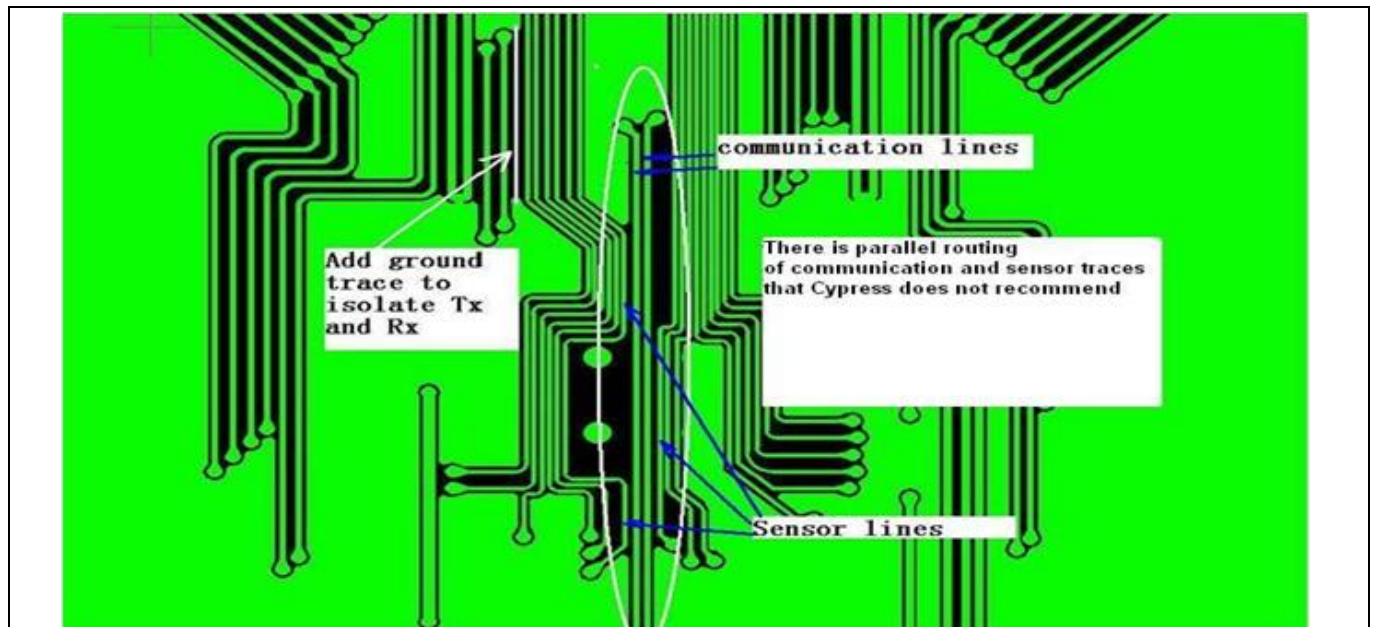


Figure 48 Parallel sensor trace and communication trace error - same layer

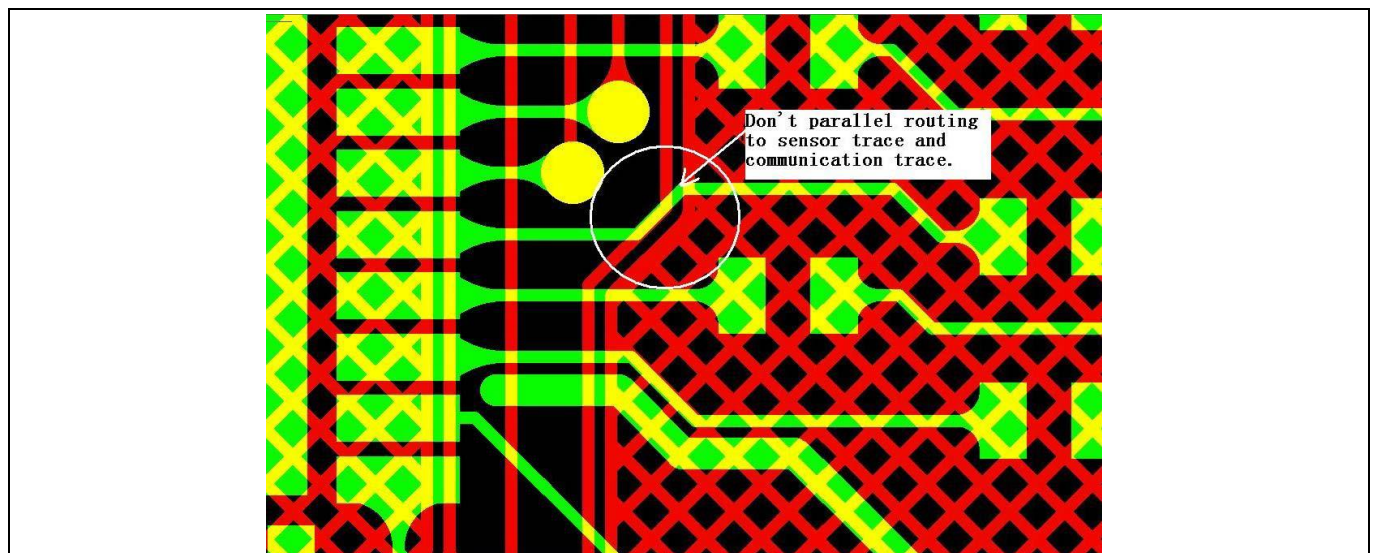


Figure 49 Parallel sensor trace and communication trace error - different layer

To correct the error, do one of the following:

- Insert system ground trace between the parallel TX and RX traces.
- Rearrange the TX and RX traces on different layers so they are perpendicular to each other.

3.6.3 Narrow power supply and ground traces

Excessively narrow power supply and ground traces, as shown in [Figure 50](#), increase the susceptibility to ground bounce and increase inductance. To avoid this, the power supply and ground traces should be at least three times as wide as the signal routing traces. If possible, use a whole layer as ground plane.

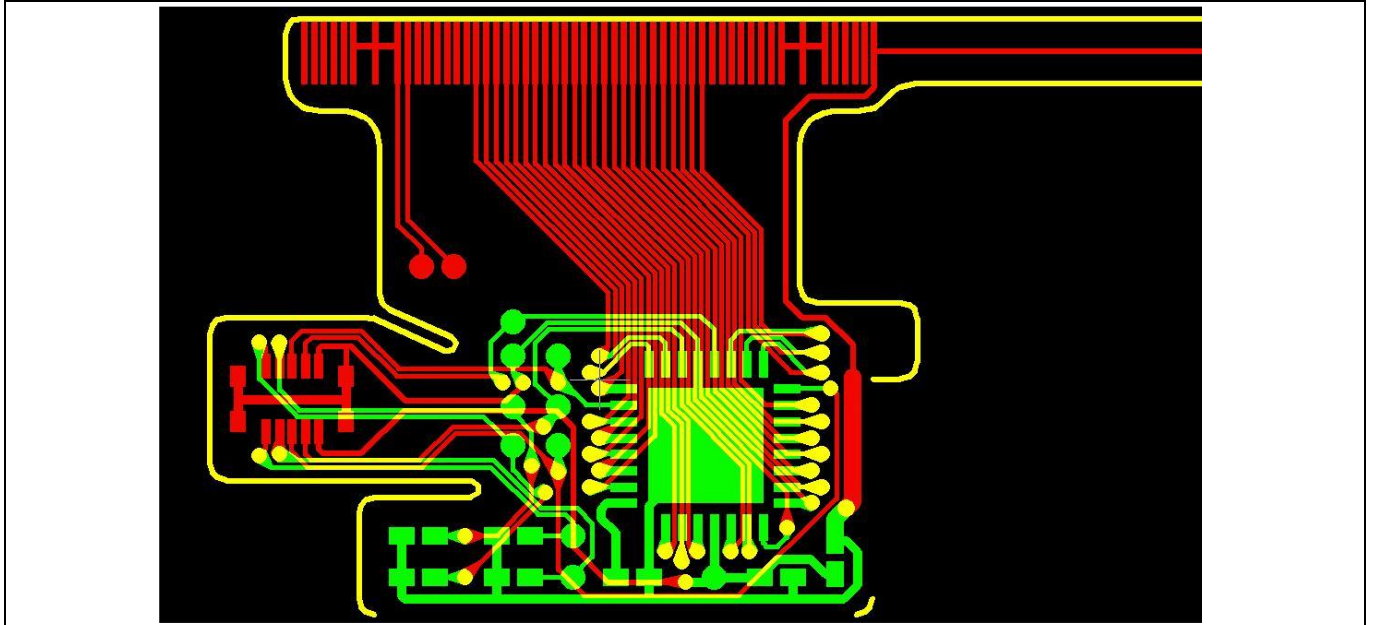


Figure 50 Example of narrow power supply routing traces without ground plane

3.6.4 Sensor trace under voltage-reference bypass capacitor

No sensor signals should be routed under the voltage-reference bypass capacitor (for those designs that include same). [Figure 51](#) shows an example of this error.

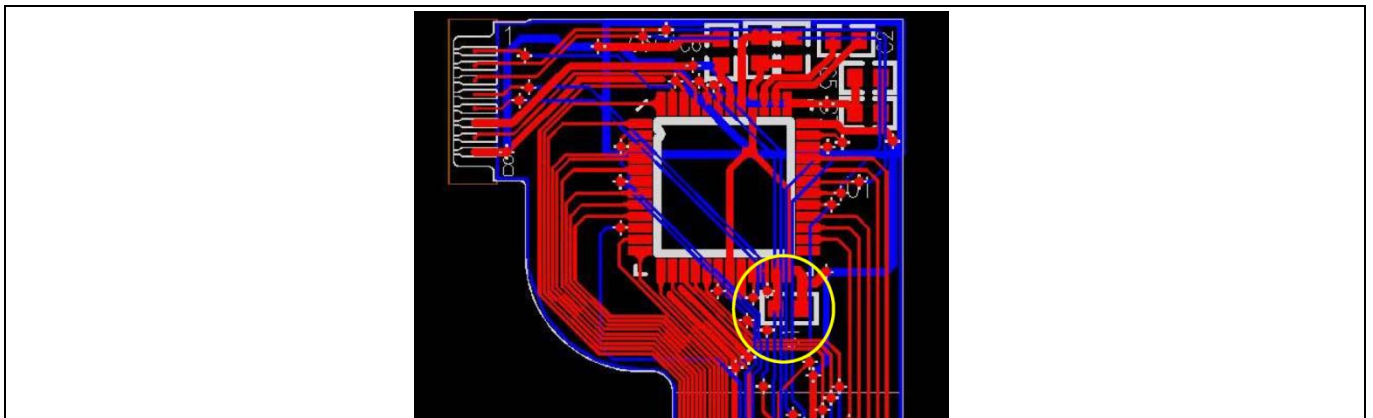


Figure 51 Example of sensor trace under the voltage-reference bypass capacitor

4 FPC guidelines

This section focuses on guidelines for the flexible printed circuit (FPC) that is typically used to connect the PCB to the touch panel. The sections are:

- Stack-ups
- Location guidelines
- Layout guidelines
- FPC/panel bonding
- Common FPC layout errors

4.1 Stack-ups

There are two prevailing types of FPC stack-ups: two-layer ([Figure 53](#)) and four-layer ([Figure 54](#)).

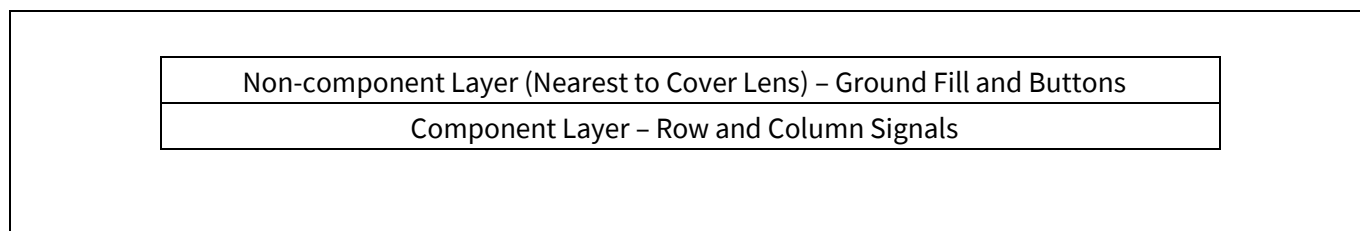


Figure 53 Two-layer stack-up

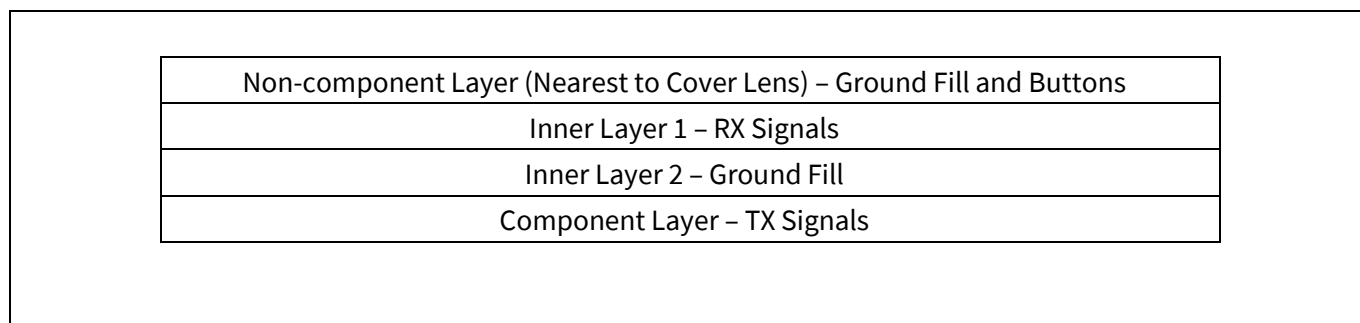


Figure 54 Four-layer stack-up

4.2 Passive FPC location guidelines

This section is for passive tail designs where the chip is on a main PCB and TX/RX signals are routed on a passive FPC to the touch panel. This includes production designs and manufacturing test boards. [Figure 55](#) provides guidelines on multi-layer routing, FPC tail length, and FPC tail shielding.

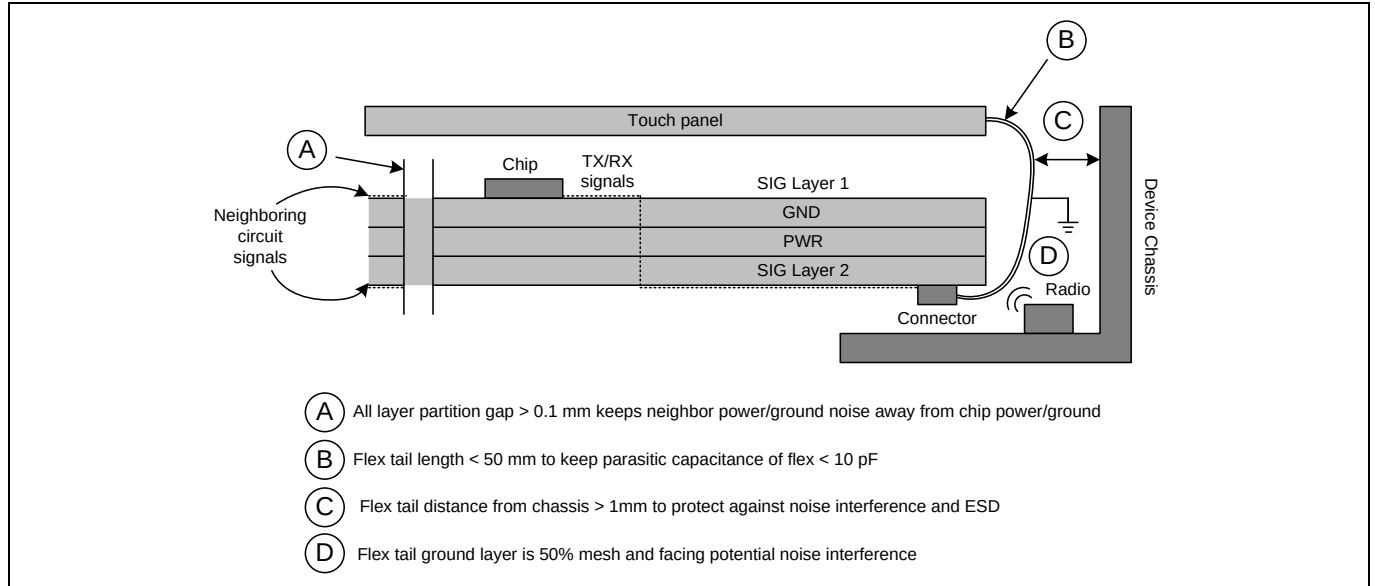


Figure 55 PCB multi-layer routing and passive flex tail guidelines

4.3 Layout guidelines

- The FPC should always have a shield layer with ground fill. In addition, any unused space on the FPC should be filled by ground plane. The ground fill should be tied to the system ground, and not the V_{SS} ground of the touch device. This helps to isolate noise and redirect ESD energy away from the touch device. Another benefit of adding ground fill wherever possible is that it reduces the impedance of the ground net, improving its high frequency response. The disadvantage of doing so is the increase of self-capacitance between sensor traces and the ground plane. To reduce this side effect, use 30% to 60% copper hatch fill for ground areas that are directly above or below row/column signals. For designs that use only mutual-capacitance scanning, use solid fill.

For designs that use self-capacitance scanning, if the traces are less than 5 cm (assuming 0.1 mm trace width, with the flex layer thickness greater than 0.05 mm) use solid fill; otherwise, use 10% copper hatch fill ground.

- Make the shield layer (ground fill) large enough to cover the FPC tail and display-driver pads. Connect the ground to system ground, not the controller's V_{SS} pin. The design of the FPC “tail”, where the sensor traces are routed off the panel to connect with the touch controller, is critical. [Figure 56](#) is an example of a poor layout. The display driver is located near the bottom of the display, and the row/column pixel drivers are routed to pads at the bottom of the display, outside the viewing area. These pads are driven with high-speed digital signals whose logic-level transitions can couple noise to sensor traces routed in close proximity.

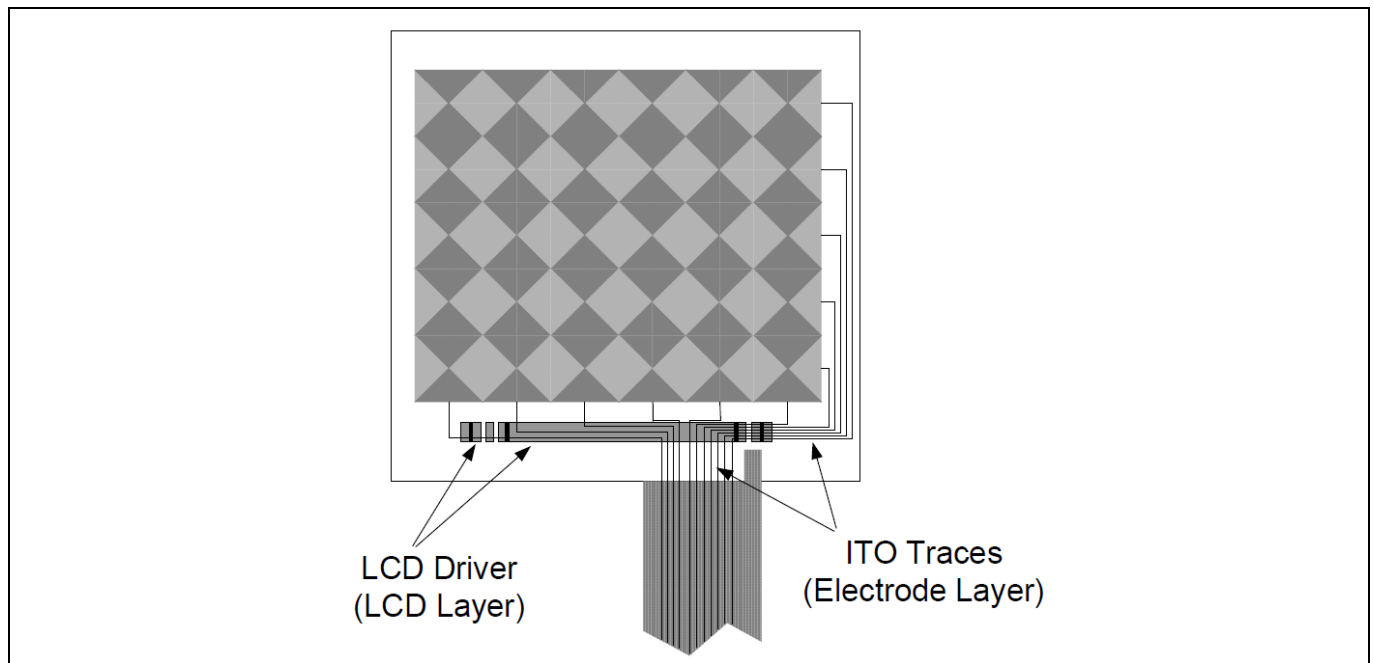


Figure 56 ITO traces not shielded from display driver

- To solve this problem, increase the size of the FPC and corresponding shield, as shown in [Figure 57](#), so that it covers the display driver pads. This design consideration applies to both two-layer and four-layer stacks.

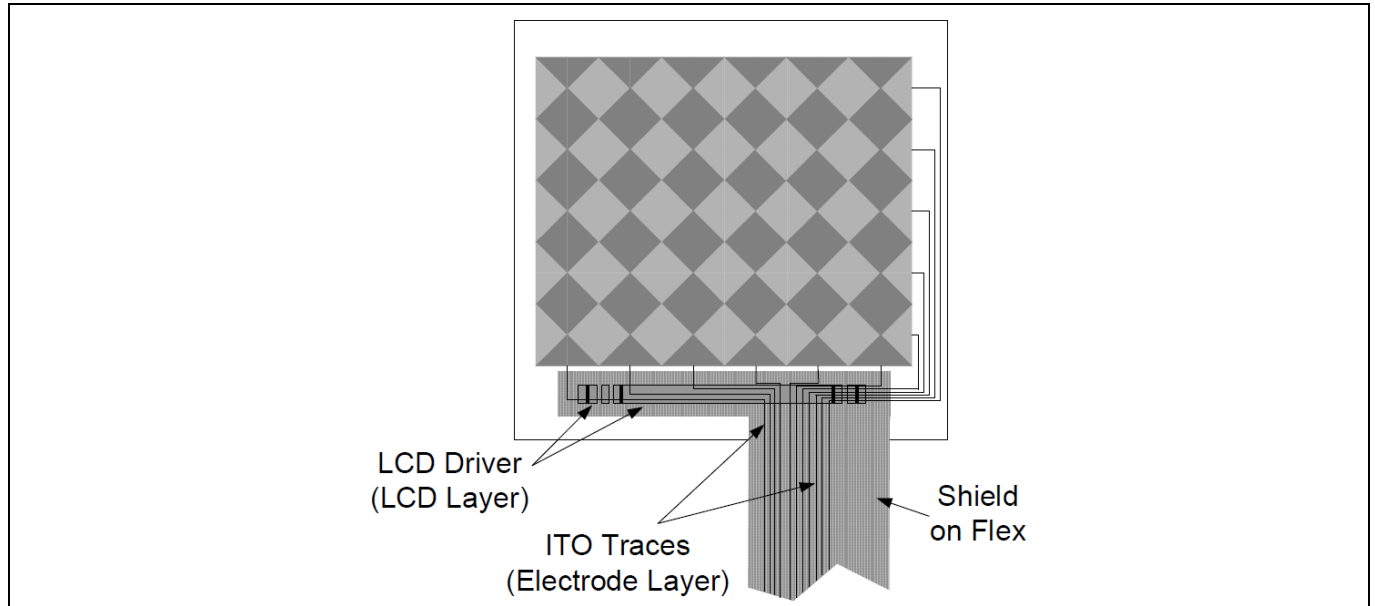


Figure 57 ITO traces not shielded from display driver

- Be careful of display driver Chip-on-Board (COB) interfering with ITO panel FPC routing.

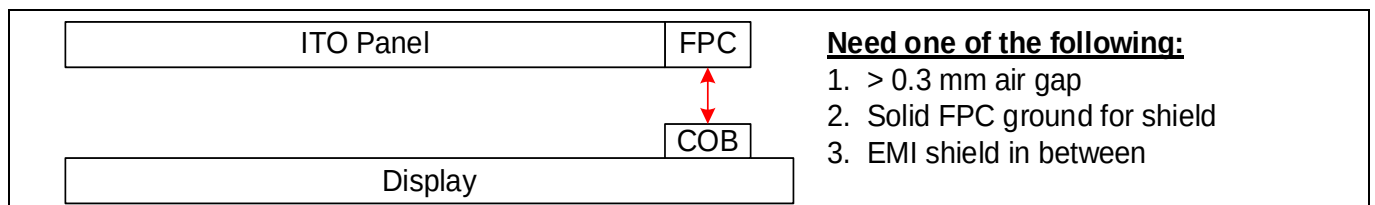


Figure 58 ITO FPC interference with display driver COB

- The traces between the touch controller and electrodes should be kept as short as possible.

FPC guidelines

- RX sensor traces should be shielded for the entire length of their run on the FPC. The area in blue in [Figure 59](#) is the bottom layer of the FPC and is the suggested shield area because it is closest to the display driver signals. It is important to shield the sensor traces in this case because the display driver is directly underneath the touchscreen column traces. Another technique for reducing the coupling between the sensor traces and the display driver is to connect the FPC tail on the opposite end of the module from the display driver.

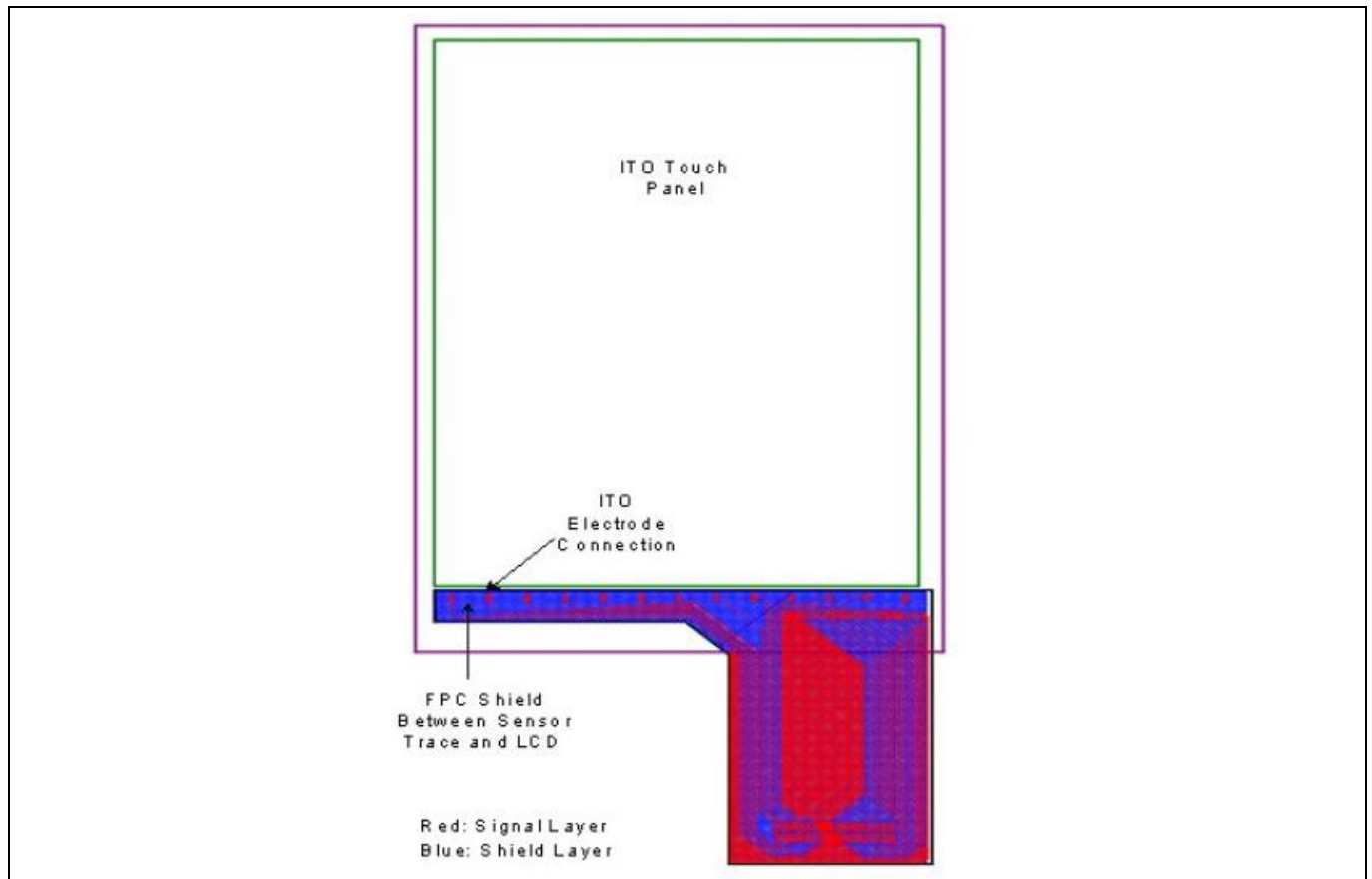


Figure 59 FPC layout with copper shield coverage

- The RX electrodes and corresponding traces are most susceptible to noise. Follow these rules to avoid coupling noise into these signals:
 - Shield the RX traces from other traces if routed in parallel.
 - Route the RX traces 90 degrees to other noisy traces.
 - Route the RX traces farthest from the display.
- TX traces are relatively low-impedance and do not need to be shielded. TX traces may be routed in parallel with other TX traces, but if TX traces must cross RX traces the intersection angle should be as close to 90 degrees as possible.

4.4 FPC/panel bonding

Touchscreen sensors are generally bonded to the FPC with anisotropic conductive film (ACF). To have reliable contact, Infineon recommends contact density for the ACF material between 1:5 and 1:10 for the ratio of sphere size to gap between the conductive spheres; see [Figure 60](#). Furthermore, the contact pads interfacing the FPC and touch panel should be drawn based on the geometries (width, spacing, and height) recommended by the panel and ACF manufacturers, whichever offers the highest contact or yield rate. For instance, typical width and space of the contact pads on PET Film is 0.5 mm and 0.5 mm, respectively.

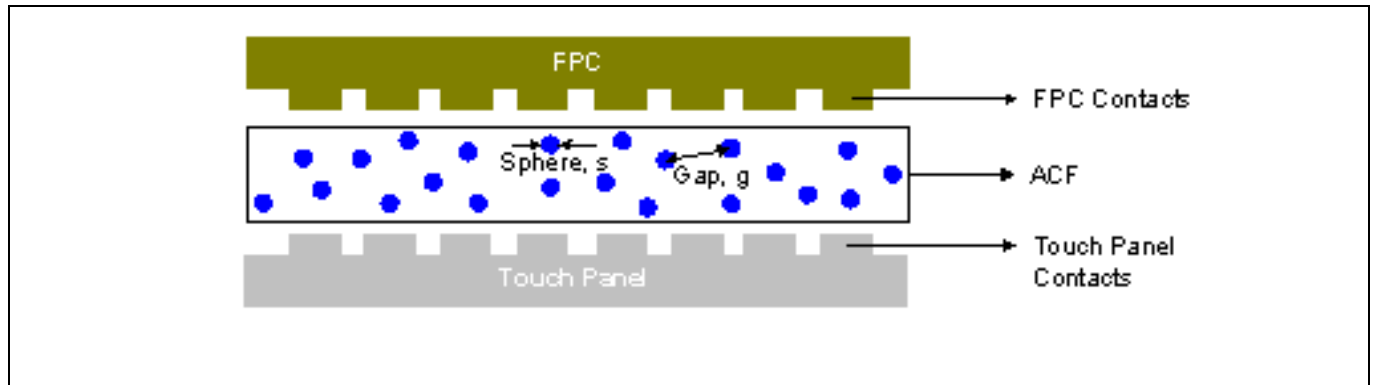


Figure 60 ACF bonding

4.5 Common FPC layout errors

4.5.1 Errors similar to PCB errors

The following typical errors are described in section [Common PCB layout errors](#), and apply equally to the FPC design:

- Parallel TX and RX trace error.
- Narrow ground traces.
- Ground fill around outer periphery.

4.5.2 Sensor routing underneath CAPSENSE™ button error

Some designs may include CAPSENSE™ buttons on the FPC. In these cases, do not route sensor traces underneath the buttons.

5 Hover guidelines

This section covers the design guidelines for detecting and tracking hovering fingers.

5.1 Introduction

The CYAT817x family includes devices that support hover sensing. Hover sensing allows the touchscreen sensor to detect and track the movements of a finger that is not in physical contact with the touchscreen. The capacitance change from a hovering finger is very small, therefore it is important to implement a system design that maximizes this signal and minimizes noise.

This section gives design guidelines for the touch panel and electronic circuitry to enable successful hover functionality.

5.2 Panel design guidelines

A single-solid diamond (SSD) pattern is recommended, because it has good symmetry of capacitance between TX and RX electrodes.

The panel must include a low-resistance (100 Ω) ITO shield layer covering the entire sensor and routing area. The shield layer can be on a separate substrate or on the display's color filter for on-cell systems. The shield should be connected to the touch controller's shield signal with a low-resistance connection. Ideally the shield connection should be made around the entire periphery of the panel as shown in [Figure 62](#), but connecting only along the one long edge of the shield (as shown in [Figure 63](#)) is usually sufficient.

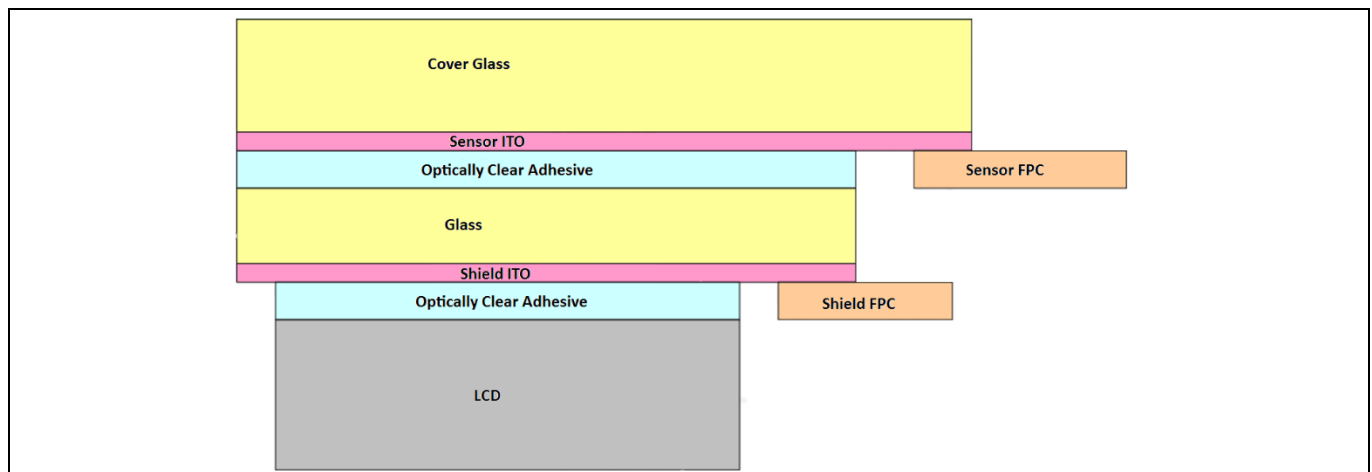


Figure 61 Typical stack-up showing shield

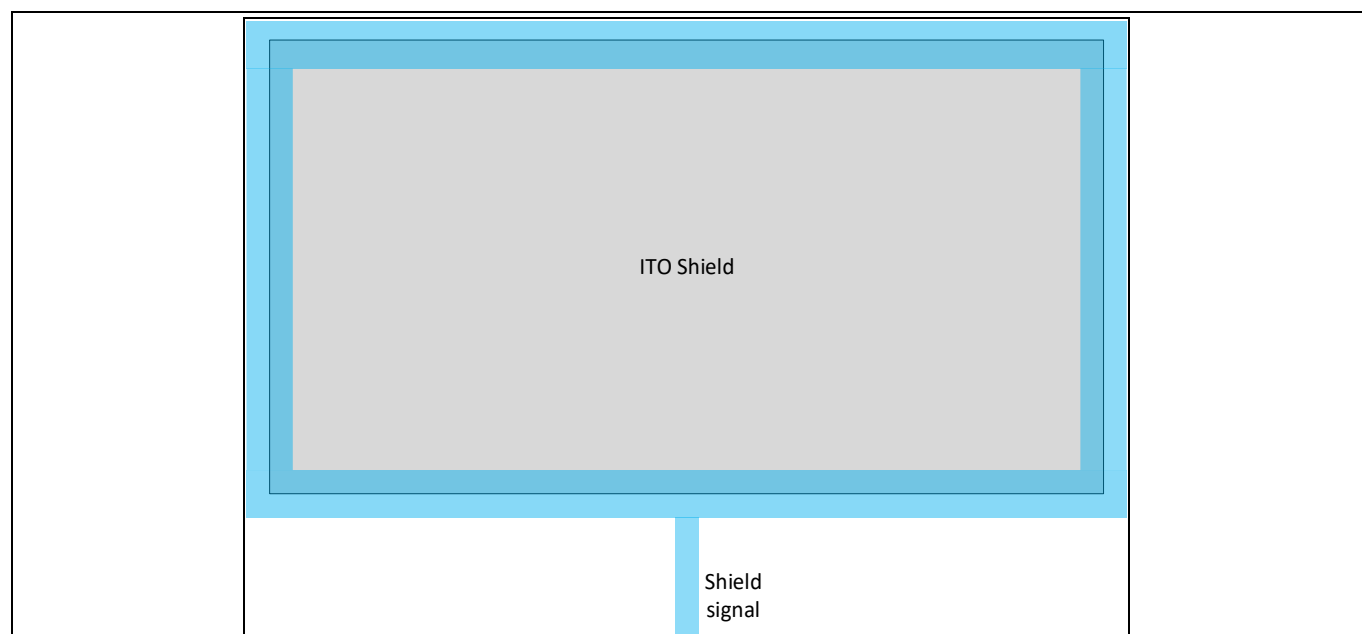


Figure 62 Connection to signal around periphery of shield

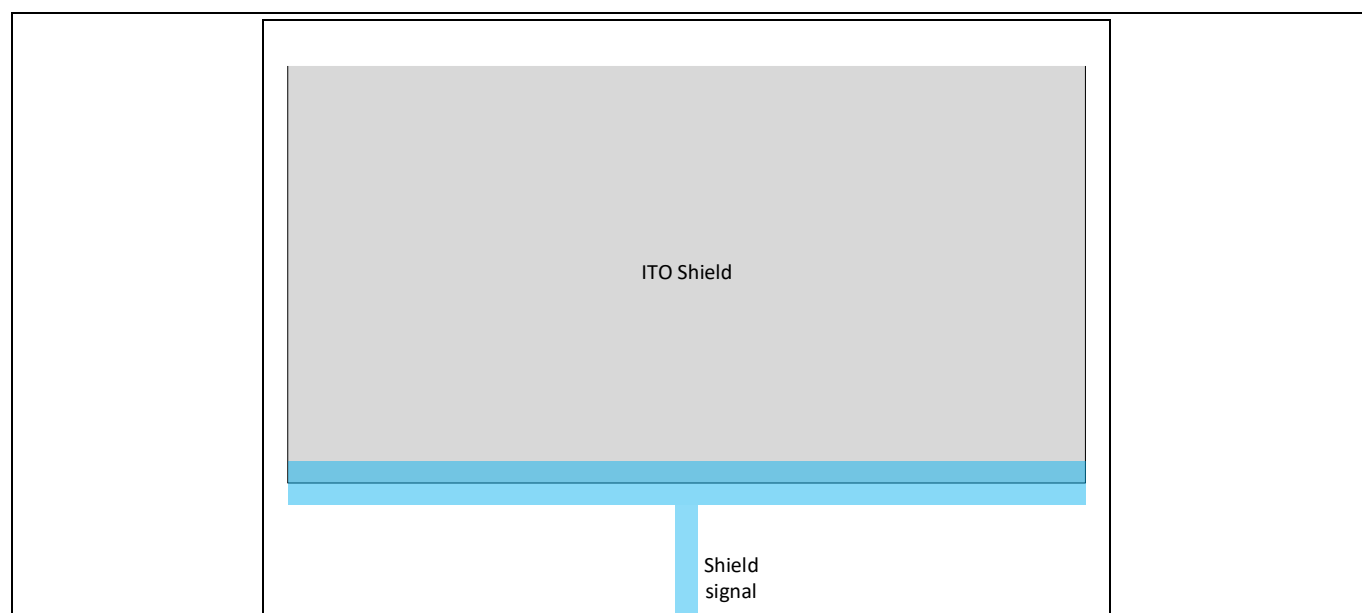


Figure 63 Connection to signal along long edge of shield

5.3 PCB design guidelines

The material presented here should be combined with other guidelines that may be available from other sources, such as component suppliers and PCB manufacturers.

Hover scanning requires that the touch controller be placed in an isolated ground domain. The key design requirements are:

- Component placement
- Ground isolation
- TX/RX signal shielding
- Digital line routing
- Boost converter layout
- Component selection

5.3.1 Component placement

Figure 64 shows the recommended components placement for hover designs, along with some layout recommendations discussed later. There are few things that need to be taken into consideration:

- Keep isolated area compact to reduce overall board emission and reduce floating driver capacitive load.
- Follow recommended RF suppression resistors placement.
- Place decoupling capacitors and ESD suppressor close to the power pins. The distance from chip to decoupling capacitor should be less than 5 mm.
- When using a parallel combination of capacitors for power decoupling, place the smallest capacitors closest to the power pin.
- Place two decoupling capacitors per each digital isolator IC (one for isolated and one for non-isolated domain). Locate them as close as possible to the corresponding digital isolator power pins.
- To reduce parasitic inductance, traces connected to decoupling capacitors should be wide (> 1 mm) and as short as possible.
- Place ESD suppressor close to V_{CCTX} pin, using a wide (> 1 mm) connection trace to minimize parasitic inductance.

Figure 65 shows an example of a complete four-layer board layout.

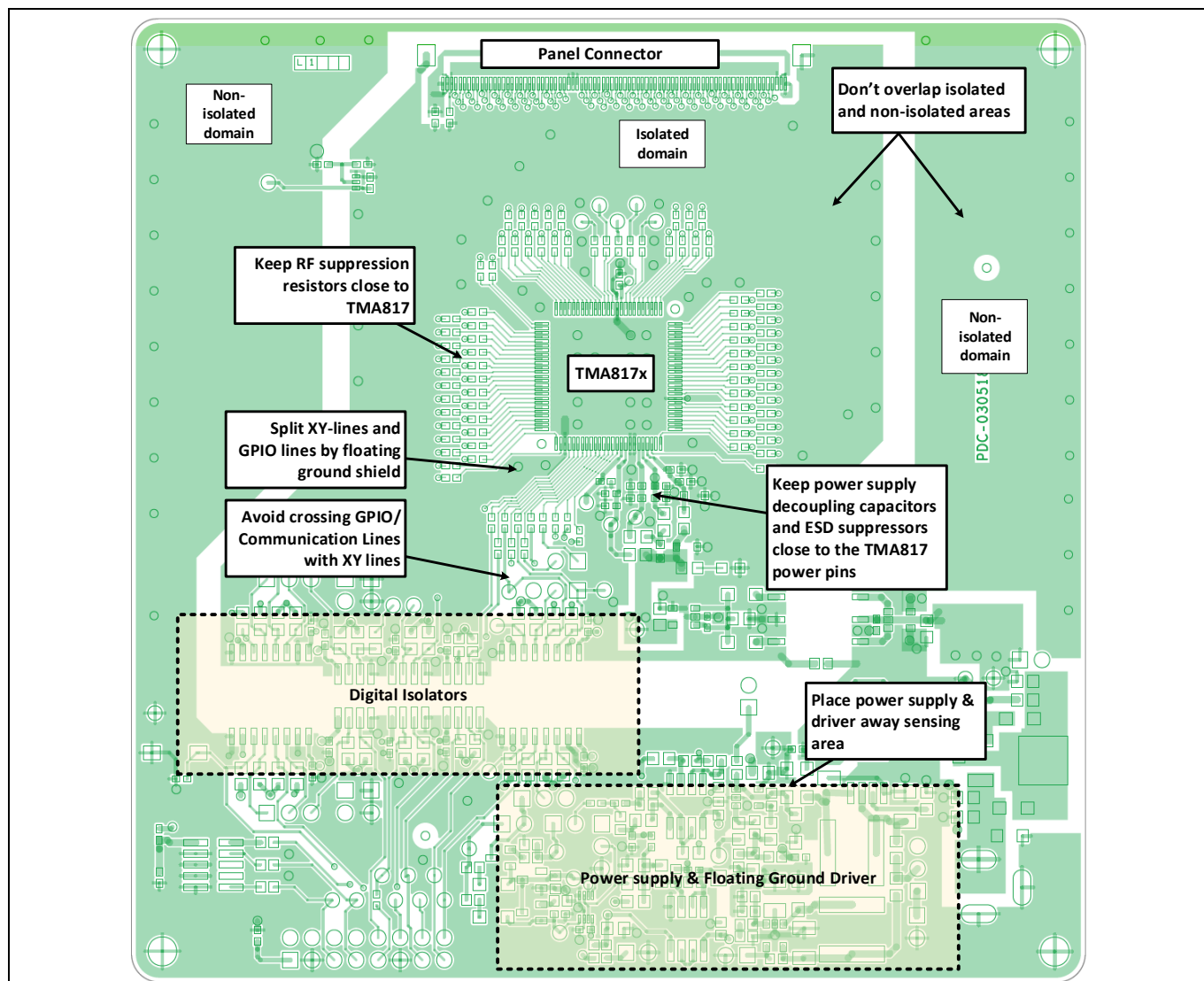


Figure 64 Recommended component placement and key layout tips

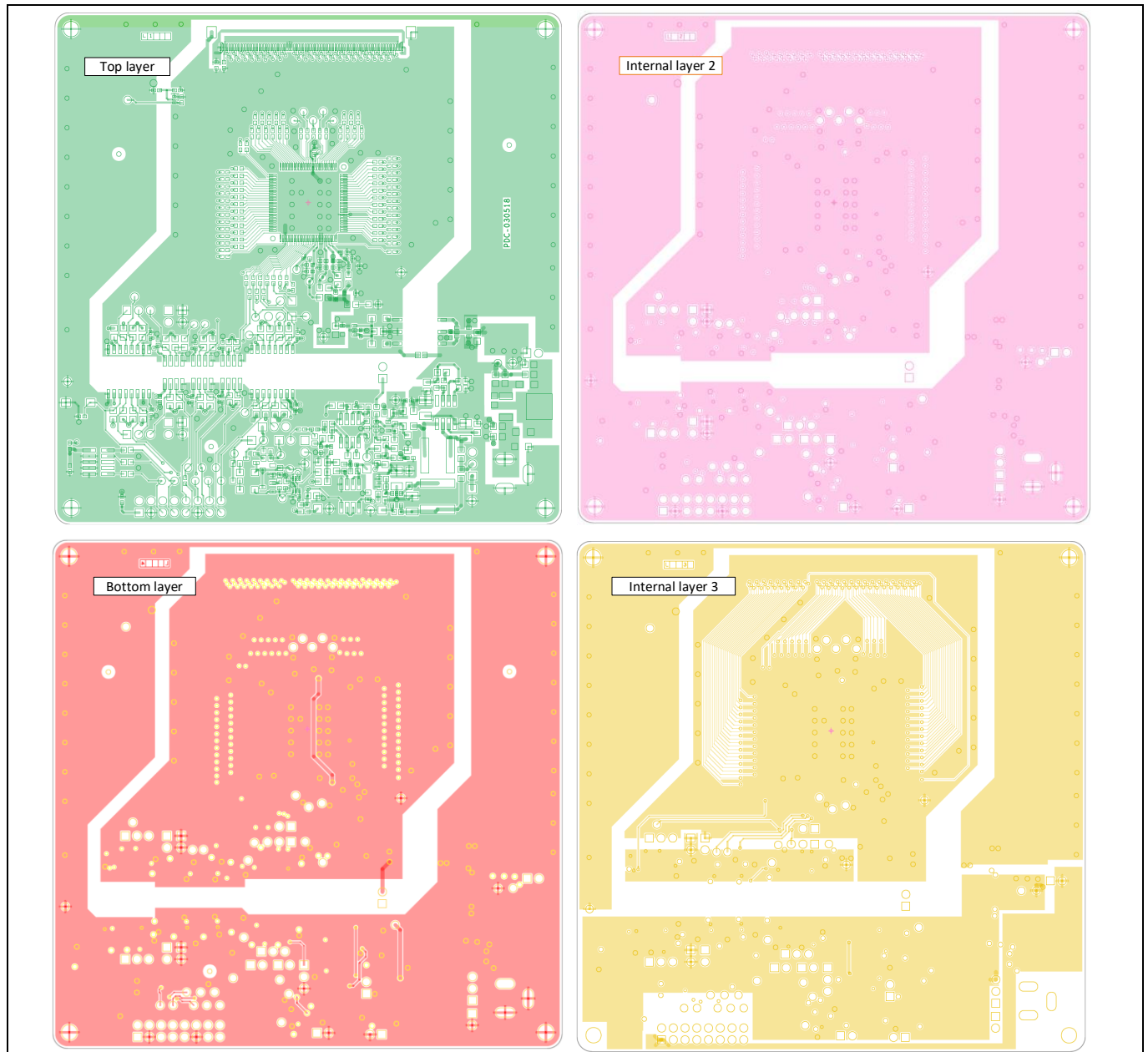


Figure 65 Design layout example

5.3.2 Ground isolation

The isolated domain must be separate from the regular domain. The layout example in [Figure 65](#) illustrates how to separate the isolated domain from the rest of design. Ensure that there are no traces passing between the two domains (except test point for the sensor shield connection).

The grounds are isolated from each other using digital isolators. Refer to section [Component selection](#) for component recommendations.

If the sensor includes an ESD ring, it should be connected to system ground.

5.3.3 TX/RX signal shielding

For best performance, the Tx and Rx routing traces should be shielded on both sides. This can be important when the board is subject to external RF fields in EMI testing, or when RF transmitters are nearby. Infineon recommends placing Tx and Rx traces in an internal layer (see example in [Figure 65](#), which uses layer 3), shielded from both sides by isolation ground planes. Those planes are in layer 2 and the bottom layer in the example.

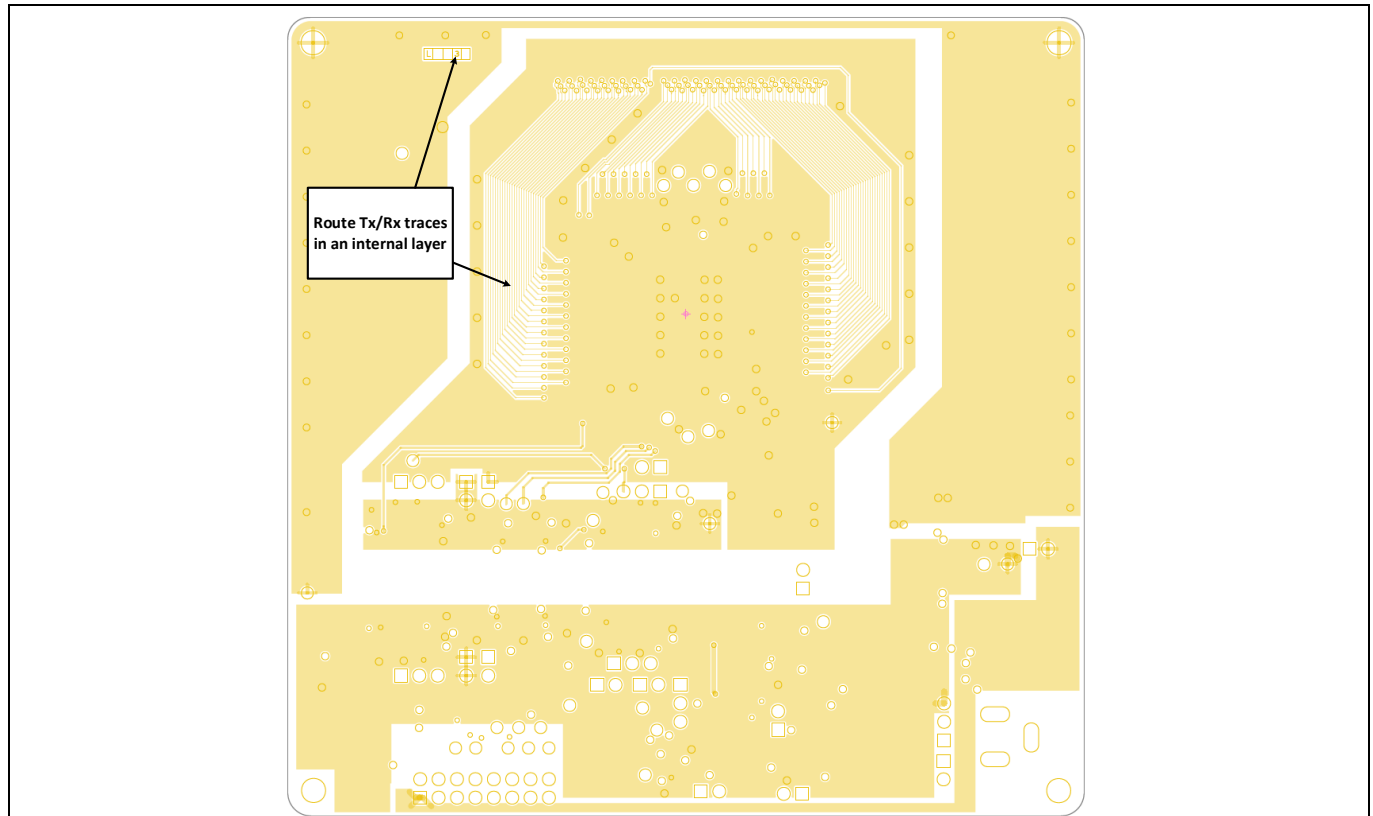


Figure 66 TX/RX trace routing in internal layer

5.3.4 Digital line routing

Keep TX/RX lines separated from communication and digital lines. Do not cross communication or digital lines over TX/RX lines.

5.3.5 Boost converter layout

The hover design uses a high-frequency switching boost regulator. The switching affects the overall RF emission of the system, potentially causing failures in EMI testing. It is critically important to follow recommended layout practices and to use recommended high-quality components, as described in the boost converter device datasheet.

5.3.6 Component selection

For best performance, component selection is important. Infineon has tested several components, and has the following recommendations. If you wish to use different suppliers or part numbers, please contact Infineon to review them.

- Boost regulator for low-voltage (down to 2.7 V) inputs:
 - Regulator: Texas Instruments LM27313
 - Inductor: Taiyo Yuden NRS6045T100MMGKV
- Boost regulator for high-voltage (12 V) inputs:
 - Regulator: Texas Instruments LM5001
 - Inductor: Murata Electronics MBH7045C-101MA=P3
- Digital isolators:
 - For I²C: Texas Instruments ISO1540
 - General purpose: Texas Instruments ISO7341CQDWRQ1
- Driver opamp: Texas Instruments LM7321QMF/NOPB
- Power isolation:
 - Transformer driver: Texas Instruments SN6501QDBVRQ1
 - Regulator: Texas Instruments TLV70033DDCR
 - Transformer: Würth Electronics 760390014
- Filter opamp: Texas Instruments OPA2365AQDRQ1

6 Multichip

This section provides extra requirements when designing a multichip module.

6.1 External oscillator

An external oscillator is required by multichip designs, see section [External oscillator](#) for guidelines.

6.2 Inter-chip communication

Inter-chip communication uses the SPI protocol. The highest supported SPI clock speed is 8 MHz. However, this may not be achievable depending on the inter-chip trace lengths. Keep the SPI communication lines as short as possible with the lowest achievable parasitic capacitance.

6.3 Sine wave drive

The multichip solution supports driving the TX electrodes with a sine wave signal. The ATT0_ATT_RX_INPUT and ATT1_SC_BUS_OUTPUT pins are used in sine wave generation. It is recommended to route the sensor pins (especially the RX electrodes) as far from these pins as possible to avoid parasitic coupling.

7 Checklists

This section provides checklists for all the key recommendations in the document. The specifications that deal with mutual-capacitance and self-capacitance also apply to CAPSENSE™ buttons.

The tables below describe Infineon’s “best practices” recommendations. Generally, a design that meets these requirements can be expected to perform to specifications. However, most real-world designs have some deviations from these values. It is strongly recommended that customers work directly with Infineon for simulation and consultation during the design phase.

Table 6 Touch panel parameters

Parameter	Section	Condition	Value	Remarks
Sense pins	N/A	TMA460A	48	8 RX channels
		TMA461A	48	8 RX channels
		TMA768A	56	8 RX channels
		TMA1036A	65	8 RX channels
		CYAT8165X	Up to 48	17 RX channels
		CYAT8168X	Up to 88	54 RX channels
		CYAT817X	Up to 103	64 RX channels
		CYAT837X and CYAT847X	Up to 294	64 RX channels per device
Intersections	N/A	TMA460A	576	
		TMA461A	576	
		TMA768A	768	
		TMA1036A	1056	
		CYAT8165X	560	
		CYAT8168X	1836	
		CYAT817X	2500	
		CYAT837X and CYAT847X	10320	
Electrode pitch	N/A	All	5 mm typical	Larger pitch allows larger panel
Panel size	N/A	TMA460A	7.0 inches	
		TMA461A	7.0 inches	
		TMA768A	8.1 inches	
		TMA1036A	9.4 inches	
		CYAT8165X	8.5 inches	
		CYAT8168X	12.6 inches	
		CYAT817X	15.0 inches	
		CYAT837X and CYAT847X	35 inches	

Checklists
Table 7 Key parameters

Parameter	Section	Condition	Value	Remarks
ITO sheet resistance	2.3	SLIM-E pattern, glass substrate	75 Ω /sq max	Assuming max routing Channel resistance is met
		SLIM-E pattern, PET substrate	100 Ω /sq max	
		Other patterns, glass substrate	120 Ω /sq max	
		Other patterns, PET substrate	270 Ω /sq max	
Electrode parasitic capacitance (C_P)	2.1	TMax devices	50 pF max	Along entire row or column and including routing. Specified value is the recommended maximum self-cap for optimal performance. Higher values may be successfully supported with tuning, but panel speed and tuning flexibility will be reduced. Recommend checking the panel speed with simulation.
		CYATx devices, single-routed panel	75 pF max	
		CYATx devices, double-routed panel	150 pF max	
Self-cap finger (C_{FS})	2.1	TMax devices	3 pF max	Change in self-cap due to finger touch
		CYATx devices	5 pF max	
Electrode C_P variation	2.1	All devices	25% max	Variation in C_P across all sense pins, including panel and routing.
Sensor mutual capacitance (C_M)	2.1	All devices	3 pF max	
Mutual-cap finger (C_{FM})	2.1	TMax devices	0.4 pF max	Change in mutual-cap due to finger touch
		CYATx devices	1 pF max	
Sensor C_M variation	2.1	All devices	25% max	Variation in C_M between all sense pins, including panel and routing.
Signal uniformity (SU)	2.1	All devices	7% max	Variation in CFS or CFM across any 3 adjacent sensors must be less than or equal to this value.
Leakage resistance	2.1	All devices	25 M Ω min	From any XY pin to V_{SS} . This may be due to ITO leakage between XY pins or flux leakage on circuit board.

Checklists

Table 8 Stack-up design parameters

Parameter	Section	Condition	Value	Remarks
Cover lens thickness	2.2.1	SLIM-E pattern, all devices	0.5 mm to 1.5 mm	Thicker lenses are often feasible. Infineon recommends simulation to check sensitivity if cover lens thickness is close to maximum value.
		SSD pattern, TMAx devices	1.2 mm to 1.5 mm	
		SSD pattern, CYATx devices	1.2 mm to 3.0 mm	
		DSD, MH3 pattern, TMAx devices	0.5 mm to 1.5 mm	
		DSD, MH3 pattern, CYATx devices	0.5 mm to 3.0 mm	
Dielectric constant	2.2	all	3.0 min	Higher values give stronger signal. Designs with lower values may be feasible but should be simulated.
Air gap	2 2.2.4 2.3.1 2.6	OLED and ACVCOM displays, all devices	0 mm min	
		DCVCOM displays, TMAx devices	0 mm min	RECOMMEND GG and GFF stack-ups. (SOL) direct lamination NOT RECOMMENDED without additional OCA layer to increase distance from display.
		DCVCOM displays, CYATx devices	0 mm min	Display surface noise voltage <1.5 V _{PP}

Table 9 Diamond pattern design parameters

Parameter	Section	Condition	Recommendation	Remarks
Diamond gap (G)	2.3.1	All devices, diamond patterns	30 µm to 100 µm typ	Glass substrate
			100 µm to 300 µm typ	Film substrate
Metal bridge width (W)			10 µm typ	
ITO bridge width (W)			100 µm min	On short axis, min can be reduced to 70 µm
Shield layer				Required for DCVCOM displays with no air gap

Checklists

Table 10 Manhattan pattern design parameters

Parameter	Section	Condition	Recommendation	Remarks
Bottom width (BW)	2.3.2	All devices, Manhattan patterns	(D - 0.3) to (D - 0.1) mm	D = electrode pitch
Number of stripes			3 typ	Optimal value determined in simulation
Top width (TW)			0.3 mm to 0.5 mm	Optimal value determined in simulation
TX/RX electrode separation			45 µm min	
TX/TX electrode gap			100 µm typ	Smaller gap is preferred
Shield layer				Not required

Table 11 SLIM-E pattern design parameters

Parameter	Section	Condition	Recommendation	Remarks
Routing channel trace space	2.3.3	All devices, SLIM-E pattern	10 µm to 50 µm	See Figure 8 , Figure 11 , Figure 12 , and Figure 13 .
Routing channel resistance			230 kΩ typ, 300 kΩ max	
Electrode pitch			5 mm typ, 5.5 mm max	See Figure 13 .
Routing channel width			35% max	Percent of electrode pitch size. (Figure 13)
S1/S2 gap			350 µm typ	Distance between S1/S2 electrodes. (Figure 13)
s1Prong			650 µm typ	Width of prong/finger of S1 electrode. (Figure 13)
s1BackBone			500 µm typ	Distance between S1 and routing channel. (Figure 13)
s2Prong			320 µm typ	Width of prong/finger of S2 electrode. (Figure 13)
s2BackBone			300 µm typ	Distance between S2 and routing channel. (Figure 13)
DelDummy			50 µm typ	Deletion (gap) between dummies. (Figure 13)

Checklists
Table 12 Routing parameters

Parameter	Section	Condition	Recommendation	Remarks
Sensor RC delay including pattern and routing	2.4.1.2	TMAx devices	1111 ns typ	3-tau of 95% rise time, for min TX frequency of 100 kHz and typical of 150 kHz. $((1 / 0.10 \text{ (MHz)}) / 2 \text{ (half-period)}) / 3 \text{ (tau)} = 1.67 \mu\text{s}$ $((1 / 0.15 \text{ (MHz)}) / 2 \text{ (half-period)}) / 3 \text{ (tau)} = 1.11 \mu\text{s}$
		CYATx devices	1667 ns max	
Max electrode resistance	2.4.1.2	TMAx devices	15 kΩ typ, 30 kΩ max	End-to-end. Panel speed simulation recommended. Higher resistance values are acceptable if simulation shows that the panel speed is OK.
		CYATx devices, single-routed panel	30 kΩ max	
		CYATx devices, double-routed panel	40 kΩ max	
Bond contact resistance	4.4	FPC to routing traces	100 Ω max	
	2.4.4.4	Metal to ITO, all devices	100 Ω max	
Bond contact width	2.4.4	Metal to ITO, all devices	0.3 mm min	PET substrates with silver ink routing traces
Bond contact height	2.4.4	Metal to ITO, all devices	90%	Percentage of electrode pitch
Metal routing trace width and space	2.3.3	SLIM-E patterns	30 μm to 100 μm	Min for Glass, Max for Film
Trace width and gap	2.4.1	Non-SLIM-E patterns	20 μm min	
		SLIM-E patterns		See section 2.3.3
Trace resistance	2.4.1	All devices	5%	Percentage of sensor resistance
TX/RX overlap	2.4.3	All devices	1 mm max	

Table 13 Button parameters

Parameter	Section	Condition	Recommendation	Remarks
Buttons	2.5	All devices		Buttons should be simulated for mutual-capacitance and self-capacitance.

Checklists

The FPC and PCB parameters below are provided in the form of a checklist. For CYAT6165x, CYAT8165x, and CYAT8168x devices, Infineon has created a detailed design checklist spreadsheet. It is available under NDA from Infineon as document number 002-25014.

Table 14 **Circuit, FPC, and PCB checklist**

ID	Device	Result (Pass/Fail)	Checklist requirement	Comment
1	All		Confirm row/col count and bonding pads are the same as sensor drawing.	
2	All		Confirm pin definition in the schematic and package in the layout. Confirm I2C, INT and programming pins (ISSP/SWD) are correctly defined.	
3	All		TX and RX assignment is following recommendation in section 3.1 and Table 6.	
4	All		No TX and RX traces routed in parallel on PCB or FPC, or TX and RX traces have been isolated by system ground traces or plane.	
5	All		No sensor traces and communication traces routed in parallel, or both sets of traces have been isolated by system ground traces or plane.	
6	All		For double routing traces, branch near the FPC pin bonding area to minimize trace length.	
7	All		Touch controller ground should be connected to the system ground through a single point (including the EPAD connection). Pads should be provided for a ferrite bead to connect the touch controller ground (VSS) to the system ground. System and device ground areas should be separated by 0.5 mm.	The ferrite can be replaced with a 0 Ω resistor if it is not required to pass ESD testing.
8	All		EPAD (if present) is connected to VSS pin and to the touch controller ground plane through multiple vias. This connection must not violate the single point ground connection.	
9	All		Pads should be provided for a unidirectional TVS from the TX supply pin to VSS. The TVS should be no more than 3 mm from the touch controller.	The TVS can be “Not Installed” if it is not required to pass ESD testing.

Checklists

ID	Device	Result (Pass/Fail)	Checklist requirement	Comment
10	All		Hatch fill 50% ground. Unused PCB/FPC space is filled with system ground. Ground area should have vias at regular intervals. Bending area of FPC can be without hatch if necessary.	The guideline is between 30% copper and 60% copper hatch fill.
11	All		If the PCB/FPC is routed through or near an opening in the chassis, route a solid system ground ring around the perimeter of the PCB/FPC to capture ESD. The ground ring should surround the entire PCB/FPC circuit.	RECOMMENDED ONLY. Can be waived if there is no space.
12	All		A plastic stiffener is recommended. If a conductive FPC stiffener is used, do not route TX/RX lines directly above it. Route TX/RX lines on the opposite side of FPC.	For FPC assemblies only.
13	All		TX/RX signals are not routed underneath CAPSENSE™ buttons (does not apply when CAPSENSE™ buttons are assigned dedicated sense pins).	
14	All		Ground and power traces from connector to chip are as short and wide as possible.	
15	All		Use an ESD guard ring on the FPC/PCB to shunt ESD energy away from the sensor routing. See section 2.7.1 .	RECOMMENDED ONLY. Can be waived if there is no space.
16	All		Connect ESD guard ring and FPC ground to the system ground, not the touch controller ground (V_{SS}).	
17	All		Add 2 kΩ series resistors to all TX/RX lines to the panel.	
18	All		All bypass capacitors should be within 3 mm of the pin they are connected to.	
19	All		Gap between routing traces on PCB and FPC is equal to or more than the width of the routing traces. Gap between routing traces and ground at least 1.5 times the width of the routing traces.	
20	All		All power and ground traces on PCB and FPC are at least 0.4 mm.	

Checklists

ID	Device	Result (Pass/Fail)	Checklist requirement	Comment
21	All		For PCB and FPC, Tx and Rx traces are consistent width, with no more than two vias on PCB or FPC.	
22	All		Ground trace and power trace from the connector to the chip's V_{SS} , V_{DDD} and V_{DDA} are as short and wide as possible to reduce inductance.	
23	All		V_{DDD} to chip can be controlled so that a power cycle can be used to clear a latch-up condition.	
24	TMA460A TMA768A TMA1036A		Power supply circuit should match one of reference circuits in device datasheet 001-86988.	
	TMA461A		Power supply circuit should match one of reference circuits in device datasheet 001-96264.	
	TMA616A TMA884A		Power supply circuit should match one of reference circuits in device datasheet 001-68214.	
	CYAT8168X		Power supply circuit should match one of reference circuits in device datasheet 001-97033.	
	CYAT8165X		Power supply circuit should match one of reference circuits in device datasheet 002-10076.	
	CYAT817X		Power supply circuit should match one of reference circuits in device datasheet 002-18788.	
	CYAT837X and CYAT847X		Power supply circuit should match one of reference circuits in device datasheet 002-26635.	
25	CYTMAx devices		Capacitor Value Check: $C_{VCCD} = 1 \mu F$, $C_{VDDD} = 0.1 \mu F$, $C_{VDDA} = 1 \mu F + 0.1 \mu F$ Check capacitor datasheet to ensure that capacitance is OK at operating DC voltage.	
	CYATx devices		Capacitor Value Check: Requirements vary; refer to the device datasheet for capacitor details. Check capacitor datasheet to ensure that capacitance is OK at operating DC voltage.	

Checklists

ID	Device	Result (Pass/Fail)	Checklist requirement	Comment
26	CYTMAx devices only		No Signal trace pass through the Ref Cap.	
27	CYTMAx devices only		V _{DDA} should be ≥ 3.0 V to have the highest TX drive strength.	
28	CYATx devices only		Series resistors at V _{DDA} and V _{DDA_Q} = 1.0 Ω , 5%.	
29	CYATx devices only		Place caps C _{VCCD} , C _{VDDD} , C _{VDDA} , and C _{VCCTX} as close to chip as possible. All traces connecting to capacitors should have length < 3 mm and width >1.5x of signal traces. Lower value caps should be closer to package than higher-value caps.	
30	CYATx devices only		Uni-directional TVS diodes at VCCTX, VDDD, and VDDA, within 3 mm of the device package.	
31	CYATx devices only		Single-point ground connection from V _{SS} to system ground, through ferrite bead.	
32	CYATx devices only		Pi filters on V _{DDD} and V _{DDA} .	
33	CYATx devices only		Check for single-slot sensor assignment. Do this by entering Rx and Tx pins into Infineon's Touch Tuning Host Emulator software, then checking Rx slot assignment.	
34	CYATx devices only		Check for 10 k Ω pull-up resistor on XRES pin.	
35	CYATx devices only		Check interface pins – COMM_INT and I ² C or SPI.	
36	CYATx devices only		Confirm I ² C SDA and SCL lines have pull-up resistors.	
37	CYATx devices only		Confirm that unused GPIO pins, unused sensor pins, and all DNU pins are left open.	
38	CYATx devices only		Check that SWD pins are connected.	
39	CYATx devices only		Check PCB and FPC stack-up.	
40	All		Interrupt line should have an external pull-up/down.	Recommended only – see section "Communication line guidelines"

Checklists

ID	Device	Result (Pass/Fail)	Checklist requirement	Comment
41	All		Error pin should be connected to the host controller.	Recommended only

Table 15 Checklist for CYAT817X hover sensing

Checklist ID	Result (Pass/Fail)	Checklist requirement	Comment
1		Ensure that panel shield layer is connected to the isolated ground, not to non-isolated ground.	
2		The lowpass filter uses resistors with 1% tolerance and capacitors with 5% tolerance.	
3		There is no overlapping between routing in isolated and non-isolated domain routing.	
4		The RF suppression resistor connection traces are not more than 10 mm in length.	
5		Panel TX/RX connection routing traces are shielded on both sides by using individual planes, connected to isolated ground net.	
6		Decoupling capacitors and ESD suppressor are located close to the corresponding pins power pins. The smaller capacitor is placed closer to corresponding power pin.	
7		The communication and sense lines are separated by shield area, connected to isolated ground.	
8		There is no crossing between TX/RX lines and communication/GPIO lines.	
9		Power supply and communication lines do not overlap and are not placed in parallel.	

8 Appendix 1 - ESD considerations

This section provides background information about ESD. The design guidelines are already described in the sections covering the touch panel and PCB.

8.1 IEC ESD test standard

The International Electrotechnical Commission (IEC) developed a series of standards for Electromagnetic Compatibility (EMC). IEC standard 61000-4-2 is specifically focused on testing and immunity requirements for electrical and electronic equipment to system-level ESD (electrostatic discharge) events. The IEC 61000-4-2 standard documents the test methods, equipment, and measurement techniques to verify that a product design is sufficiently robust to survive in the real world.

8.1.1 EMC standard definitions

The intent of the IEC 61000-4-2 ESD standard is to realistically emulate worst-case electrostatic discharge that might occur with the product in a real-world end-user environment. Since this is a general test standard, adherence to the test method in IEC 61000-4-2 is a must in order to evaluate the Equipment Under Test (EUT) correctly and ensure the test is repeatable.

The IEC standard defines:

- Waveform of the discharge current into a specified test load
- Range of test levels
- Test equipment
- Test setup
- Test procedure
- Calibration procedure
- Measurement uncertainty

8.1.2 EMC standard test levels

Selection of a test level for a product is normally determined by the specific market that the product serves. The IEC standard supports four different test levels for ESD discharge, as documented in [Table 16](#). Most consumer equipment (cellular phones, personal media players, GPS units, etc.) are required to pass the most stringent (level-4) test levels.

Table 16 IEC 61000-4-2 ESD test levels

Contact discharge		Air discharge	
Level	Test voltage (kV)	Level	Test voltage (kV)
1	2	1	2
2	4	2	4
3	6	3	8
4	8	4	15

Care must be exercised to properly understand what these test levels actually mean relative to testing of a device or piece of equipment. For example, a level-4 air discharge test does not mean that a product must withstand 15 kV across its terminals and operate forever.

8.1.3 ESD simulator

The test levels in [Table 16](#) actually document the voltage to which a reference capacitor within the ESD simulator is charged. This ESD simulator is a piece of hand-held equipment (often referred to as an ESD gun) that generates calibrated ESD pulses. This ESD simulator is implemented to mimic the discharge profile that would occur if a human discharged themselves through a ring or other small piece of metal into the equipment. This is considered equivalent to the HMM (Human Metal Model). A simple model for that ESD simulator is shown in [Figure 67](#).

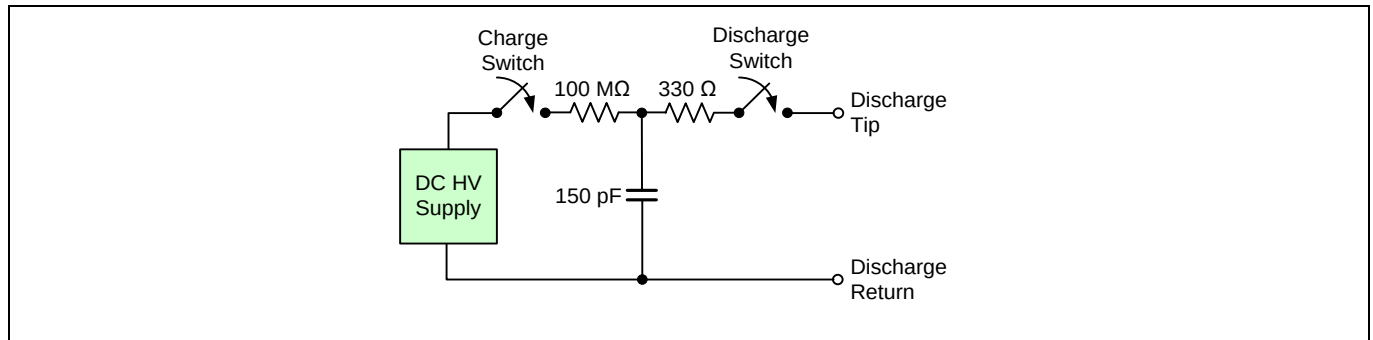


Figure 67 HMM ESD simulator

Note the size of capacitor within the ESD simulator. A current flow of 1 A flowing for 1 s provides 1 C of charge. This capacitor, when charged to the full 15 kV potential for a level-4 air discharge, only stores 2.25 μC of charge. Thus the pulses generated can potentially damage the microscopic circuits found on complex integrated circuits, but are too weak to harm a person.

This 2.25 μC of charge also is important in understanding where these ESD events can potentially cause trouble. If fully discharged into a power rail with 10 μF of supply filter capacitance present, that rail could move at most 225 mV. But if discharged into an individual pin (with a capacitance of ~ 10 pF), the charge sharing would equalize with the pin charged to 14 kV. This means that low-current device-generated supplies such as V_{DD} or V_{CCTX} in the CYAT81X family generally require more stringent ESD protection than externally-supplied power supply rails.

8.1.4 ESD pulse waveform

ESD contact and air discharges are used to test the equipment. A contact discharge allows the ESD pulse to be explicitly injected into one or more specific conductive-points on the equipment. An air discharge allows the ESD pulse itself to find the path of least resistance into the EUT.

When applying either a contact or air discharge, the Charge Switch in the ESD simulator is first closed to charge the capacitor, following which the Charge Switch is opened. For a contact discharge, the discharge tip is then placed in direct contact with a conductive part of the EUT (such as a connector shell), the discharge switch is closed, and the capacitor is discharged into the EUT. For an air discharge, the discharge switch is closed and then the discharge tip is moved close to the EUT. At some point, the potential at the discharge tip is enough to cause a spark to jump that gap and discharge the capacitor into the EUT.

Regardless of discharge type, what is seen by the equipment is a discharge of current, not of voltage. How fast that current is transferred is determined by the complex impedance of the EUT and how it couples to the Discharge Return of the ESD gun. The IEC standard documents the current transfer waveform that a properly designed and calibrated contact discharge would produce, as shown in [Figure 68](#), when coupled to a specific test load. The peak current in that waveform for a 15 kV discharge (assuming only the test load resistance in the discharge path) can exceed 40 A.

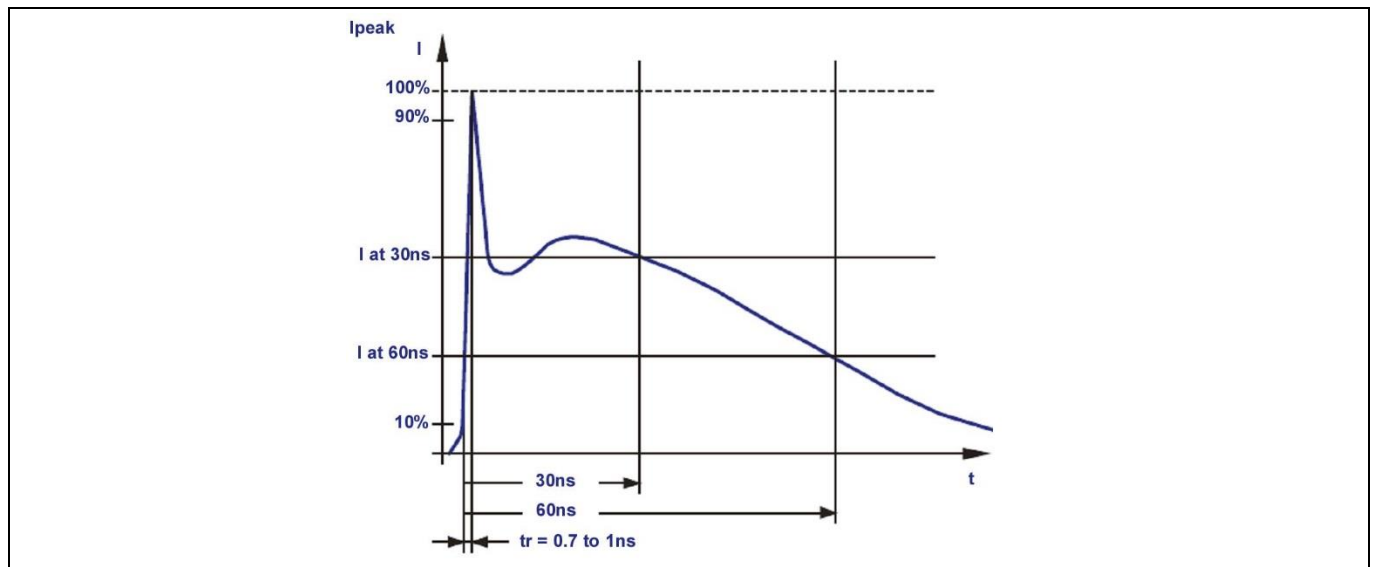


Figure 68 IEC contact discharge waveform

Within this waveform, the initial peak current represents the discharge from the tool (or hand) while the secondary peak represents the balance of the discharge from the body.

This IEC discharge waveform is significantly more stressful for semiconductors than the MIL-STD-883 or JEDEC JS-001-2012 human body model (HBM) tests. There the discharge resistance in the tool is 1500 Ω , and the storage capacitor is only 100 pF.

8.1.5 Classification of test results

The IEC standard documents four different classifications of results from ESD testing, designated by the letters **a** through **d**. These are listed in [Table 17](#), and document the functionality of the equipment following ESD testing.

Table 17 IEC ESD immunity test result classes

Class	Description
a	normal performance within limits specified by the manufacturer, requestor, or purchaser
b	temporary loss of function or degradation of performance which ceases after the disturbance ceases, and from which the equipment under test recovers its normal performance, without operator intervention
c	temporary loss of function or degradation of performance, the correction of which requires operator intervention
d	loss of function or degradation of performance which is not recoverable, owing to damage to hardware or software, or loss of data

Depending on the specific product, any of immunity class **a** through **c** may be acceptable, while class **d** results are generally considered unacceptable. Likewise an ESD immunity test result classification of **b** through **d**, when applied to an individual component, may equate to a classification of **a** at the product level, if the equipment is designed with the necessary recovery and redundancy mechanisms to handle the specific ESD induced failures.

8.2 ESD in touchscreen devices

Touchscreens are now found in a wide range of products, including items such as ticketing kiosks, printers, car dashboards, cell phones, cameras, and watches. The IEC standard places these products into three different categories, each with different test setups and procedures:

- Table-top equipment (grounded)
- Floor-standing equipment (grounded)
- Ungrounded equipment

For automotive applications, there will generally be a connection from the touchscreen ground to the vehicle chassis, which serves as the “earth” ground for the system. For this reason, the “grounded equipment” tests apply.

Grounded equipment has an inherent benefit in both design and testing for ESD tolerance. This equipment always has an integrated low-impedance path to chassis ground. Although it is possible to electrically isolate the touchscreen, this is not - and should not - typically be done in practice.

In most cases, the display’s cover lens does not provide adequate protection for the device. The challenge for touchscreen systems is that the touch surface is by definition exposed to ESD touches, and cannot be shielded. Therefore, ESD management in touchscreens is achieved via two design elements:

- Adding a grounded ESD ring (“guard ring”) around the periphery of the touchscreen, to shunt the ESD charge to chassis ground, away from the touch controller.
- Add circuit protection to the touchscreen controller board to protect against ESD charge that is not captured by the guard ring.

Design guidelines for the guard ring are given in section [Panel guard ring guidelines](#), and circuit protection guidelines are covered in the schematic and layout guidelines for the PCB (section [External oscillator](#) and section [Debug connector/adapter PCB](#)).

9 Appendix 2 - EMC considerations

This section provides background on issues related to electromagnetic compatibility (EMC). Circuit design recommendations for electrical fast transient (EFT) and immunity to radiated electromagnetic interference (EMI) are already described in section [External oscillator](#). The information here is for reference.

9.1 EFT/burst immunity test standard

The International Electrotechnical Commission (IEC) developed a series of standards for Electromagnetic Compatibility (EMC). IEC standard 61000-4-4 is specifically focused on system-level EFT (Electrical Fast Transient) and burst immunity to validate the end-product immunity required for electrical and electronic equipment. EFT and burst are high frequency noise on the AC lines of a device (in a hand-held device it is noise to the charger). The IEC 61000-4-4 standard documents the test methods, equipment, and measurement techniques to verify that a product design is sufficiently robust to survive in the real world.

9.2 EMI immunity test standards

Touchscreen systems are often subject to strong electromagnetic fields from devices such as cellular phones and other mobile devices. The IEC standard 61000-4-3 covers testing and measurement techniques for radiated, radio frequency, electromagnetic field immunity. Other standards, such as IEC 11452-2, and CISPR (International Special Committee on Radio Interference) standard 25 are focused on automotive applications.

By design, a touchscreen system has large exposed conductors that connect to sensitive electronics. Therefore, the designer must consider techniques to prevent applied fields from affecting operation.

10 Appendix 3 - Display guidelines

The primary impact of the display is electrical noise coupled into the touch system. If not considered, display noise can render a system useless. Less display noise means there is a larger budget for other sources of noise. To maximize noise handling, display noise should be reduced by adding an air gap between the panel and display, shield layer (section [Shielding](#)), choosing a low-noise display such as AMOLED, or by using an MH3 pattern to help shield the noise.

10.1 Measuring display noise

Displays emit noise that couples into the touchscreen electrodes. A good practice is to measure the magnitude and spectral content of display noise before going forward with a project. Use the following procedure to measure display noise:

- Use a display module that is not laminated to the touchscreen so that you can directly access the surface of the display.
- Turn on the display module and display an image that emits the most amount of noise (worst case). Typically the worst-case image is a black/white checkerboard image (alternating at the pixel level), but check with your display manufacturer on possible worst case patterns. The worst-case pattern is normally the one that switches a maximum amount of transistors on this display at a time.
- Cover the display with a copper plate to capacitively couple the noise waveform to the copper plate (see [Figure 69](#)).

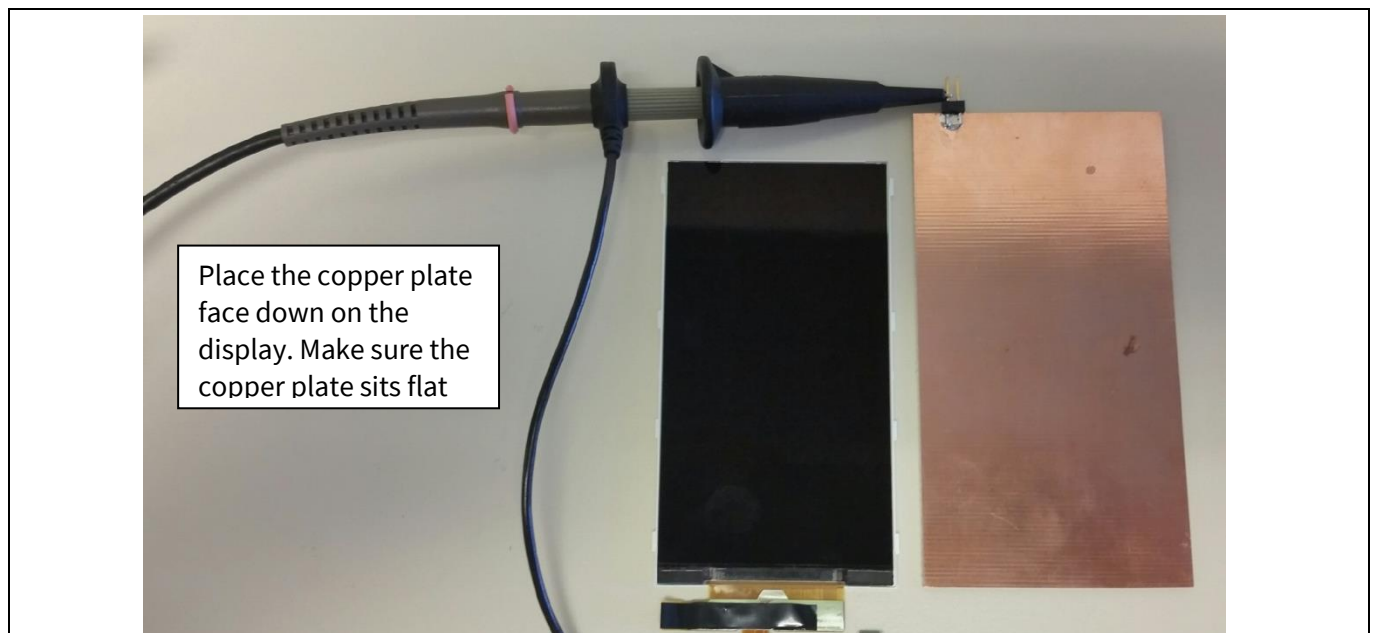


Figure 69 Display noise measurement setup with copper plate

- Connect an oscilloscope (10x probe) to the copper plate and capture the noise generated by the display. The oscilloscope ground should be connected to the display ground.
- Note the peak-to-peak amplitude of the noise (see [Figure 70](#)), and use the DSP capabilities of the oscilloscope to extract the spectral characteristics (primary switching frequencies) of this noise.



Figure 70 Measuring display noise amplitude on oscilloscope

10.2 High, medium, and low-noise displays

Use [Table 18](#) to help classify low- versus high-noise displays.

Table 18 Classifying displays for low, medium, and high noise

Classification	Min	Max	Units
Low Noise	0	30	mV _{PP}
Medium Noise	30	300	mV _{PP}
High Noise	300	3000	mV _{PP}

10.3 Minimizing the effect of display noise

In addition to choosing a low-noise display, there are several ways to minimize the effect of display noise:

- Use a Manhattan electrode pattern (section [Manhattan patterns](#)). The large TX electrodes of this pattern type help shield the RX electrodes from noise.
- Use an ITO shield in the stack-up (section [Shielding](#)).
- Use a “fast” panel with low ITO resistance and low CP (section [Key parameters](#)). A fast panel allows greater flexibility in the tuning stage, and increases the probability of finding a TX frequency that is insensitive to the display noise. It also allows more averaging in the same amount of time, and more flexibility in tuning to avoid noise generated by external sources.

Revision history

Document revision	Date	Description of changes
**	2009-11-12	New document.
*A	2010-02-25	Reformatted for consistency among touch documents.
*B	2010-05-20	Removed “Definitions” and “Key Concepts and Terminologies” Added Section 1.3 - Document Conventions Added Section 1.3.1 - Acronyms Added Section 2 - Touchscreen Sensor Patterns Added Section 8 - Pin Assignments Added Section 9 - Miscellaneous
*C	2010-09-02	Recreated graphics to meet current standards. Removed Section 8 - Pin Assignments Numbered recommendations and requirements
*D	2011-03-22	Added Section 5.5 Package Recommendations Added Section 6.3 and 6.4 on FPC Layout Errors and Review Checklist Updated Table 2-1; added requirements to Sections 2.1.4, 2.3.1, and 2.2.6 Removed Section 8 Miscellaneous
*E	2011-04-29	Added Touchscreen Buttons section Added Section 3.7 and Figure 6-1; added pattern summaries - Sections 2.1.6, 2.2.8, and 2.3.3 Added Bezel and Bezel Recommendations sections in the Mechanical Designs section In Section 6.2, changed part number from CY8CTMA3xxE to CY8CTMA3xx; added table Updates to Section 2.1.4, 2.2.6, and 2.3.1
*F	2012-02-20	Major revision. Reformatted entire document into single column. Added reference to device Technical Reference Manuals. Added sensor design guidance for each part family. Added requirement that split routing not be used for designs that require stylus, hover, or proximity gesturing functionality. Added periphery ground trace to sensor routing guidelines. Revised capacitive button guidance to match current best practice. Added ESD protection considerations to mechanical design guidelines. Reorganized/rewrote FPC layout section.
*G	2012-04-23	Reworded introduction. Added detailed SLIM requirements Added PMMA shield requirement

Revision history

Document revision	Date	Description of changes
		Added advantages and disadvantages to Manhattan pattern description. Corrected incorrect units for sheet resistance Added SLIM sensor/trace design guidelines to Sections 2 and 3. Consolidate Figure 3-9 and Figure 3-10, also enlarged it to make the “A”, “B” dimensions more readable. Corrected incorrect references to Figure 3-7 in Section 3.5. Added Section 3.2.4. Corrected erroneous description of hybrid mode in Section 4.2. Added statement the mutual capacitance scanning reduced performance loss due to increasing Cp in Section 4.2. Inserted bullets 1 and 2 and Figure 3-5 into Section 3.3. Added FPC tail length parameter to Table 7-1. Inserted Section 7.2.1 and Figure 7-8.
*H	2012-09-25	Added definitions to Section 1.4 Acronym table. Update Section 2.3 SLIM Patterns specifications Updated Figure 4 and Figure 6 for readability Update Panel Design Guidelines in Table 2-3 Table 2-2 and Table 2-4. Update description in Section 3.3 Sensor Routing Guidelines Update Figure 16 Metal to ITO contacts Add Section 4 Optimal Pinout Selection Added Section 8.3 Unused Pins in FPC design Completing Sunset Review.
*I	2013-02-20	Major Revision. See IOA-347 for all changes and justifications. Added Totem Pole (TOT) Patterns section. Added TMA44x/TMA5xx Design Guidelines Section Added Chip-on-Board Section Added Signal Disparity Section Added Pattern Corner Cutting Figure Updated Metal-to-ITO Contacts Figures for MH3 and SSD
*J	2013-03-15	Minor Revision. See IOA-351 for all changes and justifications. Added stylus spec for G4.BASE.V2 Stylus. No changes needed for Glove.
*K	2013-06-12	Minor Revision. Updated for TMA568.
*L	2013-08-05	Major Revision. See IOA-368 for detailed changes and justifications. Added Charger Noise Guidelines section to show how to get to 40Vpp Added max self/mutual signal in device design guidelines tables Added max RC time constant to meet charger noise performance levels Added max RX and TX sensors in device guidelines tables Added link to web-based Cp calculator Added ESD ring gap size Added section on extending ITO to match metal contact area Updated CAPSENSE™ button design style to star shape

Revision history

Document revision	Date	Description of changes
		Added “sensor routing under button” error in common layout errors section
*M	2013-12-16	New expanded ESD section. Updated CAPSENSE™ button recommendations New Hover Guidelines Fixed several minor errors and inconsistencies.
*N	2014-01-20	Added Industry Stack-Ups Added 2-prong LCS to Sensor Guidelines Updated MH3 TX/RX Separation Guideline Added new Small Screen Guidelines section Added definitions of “active” and “viewing” areas Added new Antenna Guidelines section Removed guideline to leave EPAD floating in ESD section Added new Manufacturing Guidelines section Fixed several minor errors
*O	2014-03-10	Added ESD acronyms to acronym table Fixed defect in GF1 drawing Added new Totem Pole table to show improvements over Diamond pattern Added interlaced routing drawing and description Fixed defect in number of RX sensors supported for TMA44x/54x parts Added definition of signal disparity with example Added recommendation for $V_{DDA} \geq 3V$ for TMA46x devices to get fast TX Reverted mutual cap button design guidelines to *L version (no shared RX) Clarified that shield layer should connect to ground Removed recommendation for RX sensors to be rows for antenna noise immunity. Clarified the antenna overlap with CAPSENSE™ ground. Increased resolution of ESD table setup drawing and emphasized with red box that the table must be non-conductive Updated ESD path drawings and associated text Removed generic ESD clearance drawing Added new Display Guidelines section which contains method for measuring display noise. Added general and device-specific checklist tables with specific ESD checklist items
*P	2014-05-28	Fixed error in totem pole drawing TX_CW Make MH3 suggestions more clear and included GND trace guideline on MH3 bezel. Complete overhaul of SLIM section Added Sensor max resistance into tables (end-to-end) Added max bond resistance into tables

Revision history

Document revision	Date	Description of changes
		Added new display COB layout guideline Added WLCSP guideline reference
*Q	2014-07-17	Added CY8CTMA460A, CY8CTMA768A, CY8CTMA1036A parts
*R	2014-08-07	Added recommendation that 90-degree corners are not allowed with metal traces on the panel. Updated panel drawings to have 45-degree corners on traces. Added recommendation for SLIM designs to put the outside routing channel outside of the viewable area. Added FPC and PCB component placement recommendations.
*S	2014-10-10	Added device CYTMA445A Changed device CY8CTMA44x to CYTMA445 Changed device CY8CTMA54x to CYTMA545 Section 2.6.4: Updated LCS sensor design description Section 4.2: Removed CY8CTMA768A and CY8CTMA1036A from restricted pinouts
*T	2014-11-05	Corrected typos Added changes to support SLIM-D Added information about single point ground in Ground Pins section in the Circuit Layout section. Added minimum clearance for an ESD ground ring around buttons.
*U	2015-01-16	Revised Section 9, ESD Protection Guidelines. Updated Circuit Design Checklist Changed Figures from “Figure X” to “Figure X-Y” where X is the section #. Cleaned up dimension information
*V	2015-01-28	Revised Section 9, ESD Protection Guidelines General document format alignment Removed future tense references
*W	2015-02-18	Added Button Routing section (2.5.5) Updated Button Layout figure to include keep out area (Figure 2 28) Replaced Figure 2 27
*X	2015-04-23	Added CYTT21X/31X (40, 44, 48 IOs) Panel Design Guidelines to Section 2.7.5. Added checklist for CYTT21X/31X devices. Updated ESD guidelines to only have 2 required components (ferrite bead in the ground connection, TVS diode on VCCTX). The other components (pi filters, additional TVSs) are only used to rework a problem that is discovered. Updated checklist with new ESD information. Added figure for tail effect. Improved the description. Added figure for putting the edge routing channel under the bezel, Improved the description. Added definition of “ceiling()” in Section 3.1.1.

Revision history

Document revision	Date	Description of changes
		Improved the figure and modified the text for ITO shield and metal ground ring. Removed support for SLIM-D.
*Y	2015-06-22	Table 2-10. CYTMA568, CYTT21X, CYTT31X, CYTT41X, and CYAT81X Panel Design Guidelines Changed panel size for stylus with TMA568, TT21xxx and TT31xxx to 5.5 inch based on maximum pitch for stylus to 4mm. Figure 9-4, Figure 9-5, Figure 9-6, and Figure 9-7 changed to properly show contact discharge for to exposed metal in ESD testing. Table 2-7, Table 2-8, Table 2-9, and Table 2-10: Sensor Mutual Capacitance (Cm) minimum of 0 pF removed from This is mathematical true, but is not useful information and has caused confusion. Improved the quality of Figure 10 1. Changed to directional descriptions to be consistent so that there should be no top-bottom/front-back confusion. Added TT41xxx to Table 2-10's panel design guidelines. Corrected the maximum number of intersections for TMA460A in Table 2-8. Unified the terms for partial ground fill to "hatch fill". Clarified what % fill meant. Updated Figure 4-1 and Figure 4-2 to show stack-up with non-component side towards the Cover Lens. Checklist updated for CYTT4X device, to include additional bypass requirements for CVDDA_Q and 1-Ω instead of 1.2-Ω resistors. Added checklist for additional automotive requirement with two CVDDA_Q pins Added guidance to set XRES low when powering the device down.
*Z	2015-08-14	Added TT41xxx to Table 2-10's panel design guidelines. Added AT81xxx to Section 13.8.6 Specific Checklist for TT21xxx, TT31xxx, TT41xxx and AT81xxx Corrected part number AT81xxx to Section 13.8.7 Specific Checklist for Automotive AT81xxx
AA	2015-09-10	Reformatted equations for proper referencing Replaced "sensor pitch" with "electrode pitch" Replaced all instances of Overlay with Cover Lens Changed LCS electrode shape description from triangle to trapezoid Added totem-pole subsections Updated figures for common definition of pitch. Revised SLIM section for only SLIM-E support. Replaced "chapter" references with "section" Replaced IC GND references with V _{SS}
AB	2016-03-31	Remove or replace "phone" throughout document so content is relevant to all applications. Fix minor grammar errors throughout document. Section 9: update text and figures to clarify system vs. device ground. Section 9.1: change heading to "ESD". Section 9.1.3: update to

Revision history

Document revision	Date	Description of changes
		clarify that VCCTX is CYAT81x family only, rewrite to remove comment about specialized ESD protection on external rails. Section 9.3.1: add note about electrode shape, remove “phone” so content is more relevant to other applications. Section 9.3.2: add comment about thin overlay or higher discharge voltage. Figure 9-8: change RX to TX/RX. Figure 9-9: add label for guard ring, delete VSS text. Section 9.6.1: recommend larger cap for designs not requiring deep sleep. Section 9.6.2: add TVS example for clarity. Delete Section 9.7.2, move content to Section 10. Section 10: rename, add section for EMI immunity. Section 2.7.5: add note about larger sensor pitch. Section 14: clarify system vs. chip ground throughout. Section 14.8.6: remove CYAT81x and replace supplemental checklist with separate checklist in new Section 14.8.7.
AC	2017-01-01	Throughout document: Update Cypress logo, move logo to page header, minor typo and style fixes, use subscripts for signal names and all upper-case for device pin names (CDT 238068). Figure 5-2: add outer ring to detail drawing, change picture to embedded Visio object (CDT 238068). Sections 2.2.1 and 2.4.1: add paragraph recommending Tx electrode count that is multiple of 7 or 9 for low-SNR designs (CDT 248769). Remove Section 9.7.2 (CDT 249323). Section 10.1.2: clarify requirement on ferrite; add second paragraph noting tradeoff with ESD recommendations (CDT 237419). Section 10.2.2: add paragraph about series ferrites for communication lines (CDT 237419). Add Section 9.6.8 to recommend gap for cover films (CDT 257881). Completing Sunset Review.
AD	2017-05-03	Update Cypress logo. Update recommended ESD diode part number (CDT 274548). Update FPC ground fill requirements in Section 13.7.5 (CDT 265418). Add requirement to check voltage coefficient for capacitors in Section 9.7.1, Table 13-1, and checklists 13.8.2-13.8.7 (CDT 268666). Clarify signal uniformity requirement in Table 2-10 (CDT 274352). Clarify sensor RC delay requirement in Table 2-10 (CDT 274349). Update button section with recommendations for simulation and dual scan, add two-layer example (CDT 238530). Change series resistor recommendation to 2 kΩ for all controllers in Sections 10.2.2, 13.8.1 and 13.8.7 (CDT 235489). Add button simulation to checklists in Tables 2-5 through 2-10.
AE	2017-09-29	Update supported Cm and Cp range for CYAT8 and CYTMA4 devices (CDT 279286 and 281616). Update sensor pitch, cover lens, and separation requirements in Table 2-10 (CDT 290292).
AF	2017-12-11	Update and clarify resistance and capacitance requirements in Table 2-10. (CDT 294761). Update and clarify resistance requirements for ESD ring in Section 9.6.6 (CDT290480). Completing Sunset Review.
AG	2018-06-20	Added best practices for the hover design guidelines and CYAT817x devices. Fixed miscellaneous small errors.
AH	2019-02-01	Substantial update to simplify and include only automotive devices and solutions. Remove information from one type of hover scan. Remove

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Document revision	Date	Description of changes
		totem pole patterns. Remove references to old TST, TMG, CY8CTMA14x, CY8CTMA3xx, TMA5x, TMA44x, TT21x, TT31x, and TT41x devices. Remove low-cost sensor (LCS) pattern. Change Charger Noise Guidelines to Noise Guidelines and remove content and figures related to TMA5. Revise ESD section to remove non-automotive conditions such as face-down discharge and ungrounded equipment. Remove reference to AN96096. Remove mention of stylus. Simplify hover design section. Update stack-up table. Remove section on antenna guidelines. Update Table 2-4 and 2-5. Add table for routing requirements. Update tables to match checklist 002-25014, and add reference to checklist. Change from assumed active to passive FPC to match automotive applications. Update reference schematic. Remove diagonal dimension specs from tables. Update drawing of single-layer buttons using automotive example. (CDT 318276)
AI	2020-01-31	Added multichip section and references to double split TX routing. Completing Sunset Review.
AJ	2022-06-07	Rebranding to PSoC™ Automotive Multitouch.
AK	2023-02-23	TSG4-non-auto delete throughout, except one "not for new designs" TSG4-auto throughout, either delete or change focus to TSG6/7 Note in introduction (section 1) display is "optional" Update #TX/#RX recommendations in pitch sections (2.3.1.1 and 2.3.2.1) Restore Figure 2-19 (was missing) Add ground plane recommendations (section 2.4.3) Add that shield improves Cp uniformity (section 2.6), driven shield more so Routing recommendations added for CYAT81x devices (section 3.1.1) Restricted pinout (3.1.2), remove non-auto TMA4x, add 103-pin CYAT81x Recommend interrupt line external pull-up (3.3.5), also add checklist item#40 Recommend error pin connection (3.3.5), also add checklist item#41 Comms TVS diodes in Fig 3-12 (section 3.3.6) go to system ground Add section for "Debug Connector / Adapter PCB" (3.3.11) Note in Hover (5.3.2) that an ESD ring connects to system ground Checklist table 7-1, add TSG6L rows, TSG6XL (CYAT816X -> CYAT8168X) Checklist table 7-7, fix rise time (1167->1667), merge rows, add equation info
AL	2024-03-04	Updated Introduction: Updated description. Updated Touch Panels: Updated description. Updated Stack-up: Updated Common Industry Stack-ups:

Document revision	Date	Description of changes
		Updated Table 2 Industry Stack-ups. Updated Air Gap: Updated description. Updated Signal Disparity guidelines: Updated description. Updated Electrode patterns: Updated Diamond patterns: Updated Electrode Pitch (DX, DY): Updated description. Updated Manhattan patterns: Updated Electrode Pitch (DX, DY): Updated description. Updated Routing: Updated Routing Configurations: Updated Split Routing: Updated description. Updated Double Routing: Updated description. Updated Double Split TX Routing: Updated description. Updated Capacitive Buttons: Updated description. Updated PCB Design: Updated Pin Assignment: Updated Unrestricted Pinout Parts: Updated Single Routing: Updated description. Updated Double Split TX Routing: Updated description. Updated Schematic guidelines: Updated Power Bus guidelines: Added VDDD ramp-up and ramp-down. Updated Additional ESD options: Replaced “Power Bus guidelines” with “ESD options” in heading. Updated Checklists: Updated Table 14 Circuit, FPC, and PCB Checklist. Migrated to Infineon template. Completing Sunset Review.
AM	2024-04-19	Updated metadata.
AN	2024-07-30	No content updates. Only ECN Metadata updated.

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